

Carrot & Stick Policies and the Housing Affordability Puzzle^{*}

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Abstract

Housing supply in cities operates through replacement and upgrading of existing structures rather than through aggregate expansion of the stock. In this setting, supply reflects the replacement of the housing stock through parcel-level redevelopment decisions. We analyze this replacement margin by developing a forward-looking spatial model in which developers choose when to replace existing structures and how intensively to build. Policy affects supply by shifting parcel-level redevelopment thresholds rather than moving an aggregate supply curve. As a result, more construction need not lower average rents once changes in composition are taken into account, because new development replaces lower-rent units with higher-rent housing. The model, calibrated to New York City data, shows that tax incentives accelerate development, while density bonuses increase project scale but often delay construction. Affordability mandates primarily crowd out market-rate construction and generate non-trivial costs, whose incidence depends on the policy instrument.

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Despite extensive policy intervention, housing affordability pressures persist across cities, even where housing production has remained strong. A central tension is that in mature housing markets, supply arises from the redevelopment of the existing stock, replacing lower-rent units with higher-rent construction. To address this tension, policymakers have often combined fiscal and regulatory incentives with affordability mandates. Because real estate investment is infrequent, irreversible, and forward-looking, these policies primarily shift the conditions under which individual parcels are redeveloped. In this paper, we show that housing supply arises from these parcel-level redevelopment decisions, so that aggregate supply reflects developers' individual choices over the existing housing stock. As a result, policies that expand supply alter the composition of the housing stock and need not lower average rents once these composition effects are taken into account. We also show that affordability mandates reallocate access to new units and generate non-trivial costs, whose incidence depends on the policy instrument.

We study how these redevelopment decisions shape housing production, rents, and the cost of affordable housing mandates in a spatial model of real estate redevelopment. In this environment, construction primarily occurs through the replacement of existing structures rather than through the expansion of an abstract housing stock. Developers choose whether and when to redevelop and how intensively to build. We refer to these parcel-level choices as the *replacement margin*. This margin underlies the main results. It determines which parcels are redeveloped, how intensively they are built, and thus what replaces the existing stock. It also explains why tax incentives and density bonuses generate distinct redevelopment patterns, and why the incidence of affordable mandates differs across policy instruments.

Understanding this replacement mechanism is essential because the durable nature of real estate implies long-lasting effects on residents (Rosenthal, 2014), neighborhoods (Brueckner and Rosenthal, 2009), the urban environment (Glaeser and Gyourko, 2018), and the broader economy (Duranton and Puga, 2023; Hsieh and Moretti, 2019). We provide this framework, which, through individual redevelopment decisions, links policy design to housing production, rents, and their distributional consequences. We thus contribute to the growing literature on the determinants and deterrents of housing supply (Baum-Snow and Duranton, 2025; Glaeser and Gyourko, 2018). Our approach specifically complements quantitative urban models à la Ahlfeldt et al. (2015) by showing that supply and rent responses arise from heterogeneous parcel-level replacement decisions rather than from aggregate equilibrium responses.¹ We show that redevelopment is inher-

¹Recent applications of quantitative urban models particularly focus on the importance of including dynamics to understand long-term urban outcomes (Greaney et al., 2025; Hsiao, 2025; Peng, 2023; Rollet, 2025), the endogenous quality of the supply (Koster and Loumeau, 2026; Ma, 2026), endogenous urban amenities (Almagro and Domínguez-lino, 2025), and the distributional consequences of housing supply (Giacoletti et al., 2025; Hong and Macek, 2025; Rollet and Weiwu, 2025; Tsivanidis, 2026). Our model incorporates those elements.

ently nonlinear because it is governed by threshold decisions. Small policy changes can have no effect on most parcels but trigger discrete redevelopment on a subset of marginal parcels. This micro-founded mechanism links policy directly to the composition of the housing stock and to the incidence of policy across landlords, tenants, and the public sector.

This paper makes three contributions. First, we develop a forward-looking spatial model in which housing supply arises from parcel-level redevelopment decisions. Supply is determined by policy-driven shifts in redevelopment thresholds rather than by aggregate quantities. Second, we show that this mechanism has important implications for housing outcomes. Supply expansion alters the composition of the housing stock, so average rents often increase with greater housing production once these effects are taken into account. Affordability mandates primarily crowd out market-rate construction and do not substantially expand the lowest segment of the rent distribution. Third, we quantify the cost and incidence of affordable housing policies. The cost per affordable unit is substantial and varies by the policy instrument. Fiscal incentives shift costs to the public sector. Regulatory instruments operate through land value capitalization.

To study this replacement margin, we develop a forward-looking model of city redevelopment in which developers choose when to undertake irreversible investments in durable housing and how intensively to build under alternative policy regimes. The city contains heterogeneous parcels defined by location, built density, and depreciation. Developers form expectations about future rents that depend on aggregate and local construction activity. They choose the timing and scale of redevelopment to maximize asset values subject to zoning, construction costs, and policy parameters.² We model investment as an irreversible option under uncertainty, building on classic frameworks (Capozza and Li, 1994; McDonald and Siegel, 1986; Paddock et al., 1988). We embed these decisions in a spatial setting where redevelopment on one parcel affects others. Higher expected construction reduces citywide rent growth through a supply effect (Mast, 2023; Rosenthal, 2014). New market-rate development increases local rent growth through endogenous amenities and spatial spillovers (Almagro and Domínguez-Iino, 2025; Diamond, 2016).

In our model, governments can combine incentives (*Carrots*), such as property tax exemptions and density bonuses, with obligations (*Sticks*), such as affordable housing mandates.³ These policies affect supply along three margins. First, they alter timing by shifting the rent level at

²We also provide an extension of the model that allows for renovation as an alternative to redevelopment. The outcomes from this extended model show that the results are mostly driven by redevelopment decisions that replace older structures with new construction.

³The ability of governments to recapture value increases resulting from public actions, often referred to as “land value capture” (OECD and Lincoln Institute of Land Policy, 2022), plays a central role in contemporary urban policy debates. Incentive zoning and related mechanisms are widely used in the United States (Homsy and Kang, 2023), Canada (Moore, 2013), and Europe (Druckendroff and Beckmann, 2018).

which redevelopment becomes optimal. Second, they change scale by affecting the profitability of building more intensively. Third, they shape the spatial distribution of redevelopment through interactions across parcels. The replacement margin bundles these parcel-level responses, while supply effects and amenity spillovers operate alongside it.

We calibrate the model to parcel-level data from New York City and evaluate counterfactual policies. The calibration combines parcel-level data on density, building age, and location with rental data from the American Community Survey and neighborhood-level rent dynamics from CoStar. Rental income decreases with building age and the share of affordable units, and increases with height (Ahlfeldt and Barr, 2022; Danton and Himbert, 2018; Liu et al., 2018). Construction costs are convex in height (Ahlfeldt et al., 2026; Ahlfeldt and McMillen, 2018; Eriksen and Orlando, 2022). The model matches the observed cross-neighborhood property values and predicts out-of-sample construction patterns when calibrated to pre-policy data.

We then compare affordability mandates, density bonuses, and tax exemptions both individually and in combination, focusing on their effects on citywide housing production, the rent distribution, and the cost and incidence of affordable units. Both density bonuses and tax exemptions expand housing supply relative to the baseline but operate on different margins of the replacement decision. We find that tax exemptions primarily raise redevelopment probabilities by lowering post-construction tax liabilities and bringing forward replacement on parcels near their investment threshold, whereas density bonuses increase optimal density and thus the scope for replacement. A higher allowable density can thus delay construction, as developers optimally wait to build taller structures. Because of heterogeneity in the built environment, policy impacts are spatially heterogeneous. Parcels closer to their redevelopment thresholds respond more strongly to fiscal incentives, while high-quality locations benefit more from density bonuses. Introducing an affordability mandate to either of these policies reduces the number of market-rate units, although substitution toward affordable units partly offsets this effect.

We also evaluate how each policy counterfactual affects expected rents. Rent changes reflect the interaction of three channels: the discount associated with the share of affordable units in new buildings, the replacement premium arising when redevelopment substitutes older, depreciated units with new housing that enters without accumulated depreciation, and the downward (upward) effect of additional citywide (neighborhood market-rate) supply. We find that policies that expand supply increase the number of units in both the bottom and the top quintiles of the rent distribution, reflecting the competing supply and replacement effects. For any plausible demand elasticity, the replacement effect always dominates, so additional construction does not reduce average rents. Affordability mandates expand access to below-market-rate units in new

construction, but these units tend not to be in the bottom quintile of the distribution, because more stringent rent discounts otherwise hinder redevelopment.

Because below-market housing production is not privately profitable, affordability mandates introduce a wedge between private returns and social objectives. We quantify this cost in terms of changes in land values and foregone tax revenues, making explicit how the cost of affordability is distributed across agents. The incidence of affordability mandates is thus shared between landowners, through land value capitalization, and the public sector, through foregone tax revenues. We estimate that each mandated affordable unit costs approximately \$60,000. The incidence depends on the policy instrument. Under tax exemptions, the cost falls almost entirely on the public sector through foregone tax revenues. Under density bonuses, the cost is shared between landowners, through lower land values, and the public sector. This result reflects how different policy instruments allocate costs through redevelopment, providing a micro-founded mechanism for the incidence patterns studied in the land-use regulation literature ([Diamond and McQuade, 2019](#); [Fischel, 2002](#); [Glaeser et al., 2005](#)).

Our paper contributes to two strands of literature. The first concerns the determinants of real estate supply ([Baum-Snow and Duranton, 2025](#); [Glaeser and Gyourko, 2018](#)). Prior work emphasizes the role of market structure and productivity in constraining housing growth ([D’Amico et al., 2024](#); [Goolsbee and Syverson, 2023](#)), and the role of construction technology in enabling the rise of high-rise buildings ([Ahlfeldt and Barr, 2022](#); [Ahlfeldt et al., 2026](#); [Ahlfeldt and McMillen, 2018](#)). A parallel line of research shows how physical and regulatory constraints generate spatial variation in supply elasticities ([Albouy and Ehrlich, 2018](#); [Baum-Snow and Han, 2024](#); [Brueckner and Singh, 2020](#); [Büchler and v. Ehrlich, 2023](#); [Gyourko et al., 2021](#); [Saiz, 2010](#)). By modeling redevelopment decisions at the parcel level, our study contributes to this literature by making explicit how the current built environment and policy instruments jointly determine the timing and intensity of redevelopment.

Our study is most closely related to work that models forward-looking developers in dynamic settings ([Cunningham, 2006](#); [Hsiao, 2025](#); [Murphy, 2018](#); [Peng, 2023](#); [Rollet, 2025](#)), where land embodies an option to redevelop. This approach builds on the real options tradition ([McDonald and Siegel, 1986](#); [Myers, 1977](#)) applied to real estate development ([Arnott and Lewis, 1979](#); [Capozza and Li, 1994](#); [Grenadier, 1996](#); [Lange and Teulings, 2024](#); [Quigg, 1993](#); [Titman, 1985](#)). Building on the development framework of [Lebret and Liu \(2023\)](#), we extend this class of models by introducing endogenous interactions across parcels. Redevelopment decisions affect both aggregate and local rent dynamics, which in turn feed back into future redevelopment incentives. This feature addresses a key limitation of asset-pricing-based development models

that treat prices as exogenous. Embedding parcel-level development decisions in a spatial equilibrium allows us to evaluate policy counterfactuals.

Second, our study contributes to the literature on the effects of urban regulatory and fiscal policies (Gyourko and Molloy, 2015; Quigley and Rosenthal, 2005). Existing literature examines the impact of supply-side policies such as tax incentives in opportunity zones (Chen et al., 2023; Corinth and Feldman, 2024), upzoning (Büchler and Lutz, 2024; Freemark, 2020, 2023; Greenaway-McGrevy and Phillips, 2023; Liao, 2025; Rollet, 2025), and affordable housing mandates (Favilukis et al., 2023). Other studies analyze inclusionary housing mandates coupled with density bonuses (Schuetz et al., 2011) or fiscal incentives (Baum-Snow and Marion, 2009; Eriksen and Rosenthal, 2010; Soltas, 2024, 2022). Krimmel and Wang (2026) show that attaching affordability constraints to upzoning can shift development across locations, highlighting the tension between incentives and constraints. Although these incentive-based and mandate-based policies have also been studied by urban planners and political scientists (Freemark, 2020; Homsy and Kang, 2023; Kim, 2020; Korngold, 2022), our paper provides a structural framework that evaluates these instruments jointly and quantifies how they affect redevelopment decisions, rent outcomes, and policy incidence.

The paper proceeds as follows. Section 1 presents the model, describing developers' decision rules in a spatial economy. Section 2 illustrates the mechanics of the model on a representative parcel and describes how spatial spillovers operate. Section 3 outlines the empirical setting and calibration. Section 4 reports the outcomes of the status-quo calibration and validation exercises. Section 5 analyzes the effects of policies on housing production and rents. Section 6 describes the incidence of each policy and quantifies the cost of mandated affordable units. Section 7 concludes.

1 A Forward-looking Spatial Model of Real Estate Development

We develop a forward-looking model of city redevelopment in which developers decide both when to redevelop a parcel and how intensively to build. We consider a city composed of parcels $i \in I$, each owned by a price-taking developer. Parcels of size L_i may contain an existing building characterized by its built density $F_{B,i}$ and its rental depreciation index $z_{B,i}$. The location of parcel i determines the market rent per unit of built density, denoted R_{it} , which reflects both local and citywide housing market conditions. Developers form expectations about future rents when making redevelopment decisions. In our setting, the government regulates redevelopment through a policy environment that may change from a baseline regime $\mathcal{P}_0$ to an alternative regime $\mathcal{P}$. Policy affects redevelopment incentives directly through regulatory and fiscal instruments and indirectly by altering future housing supply and, thus, future rent dynamics across the city.

1.1 The Real Estate Developer Decision Rule Under Status Quo Regulation

To characterize the redevelopment decision, we first analyze the problem of a single parcel and suppress the parcel index i for notational simplicity. The market rent per unit of space, R_t , is our fundamental state variable. Under the risk-neutral measure $\mathbb{Q}$, rent evolves according to a geometric Brownian motion:

$$dR_t = \delta R_t dt + \sigma R_t dW_t \quad (1)$$

where δ is the risk-neutral expected growth rate of rent used for valuation that adjusts with future supply⁴, σ is the volatility, and W_t is a standard Brownian motion on the probability space $(\Omega, \mathcal{F}, \mathbb{Q})$. We describe how δ adjusts endogenously with respect to unexpected future supply in Section 1.3. This process is log-normal, implying that both expected growth and volatility scale proportionally with the current rent level.

Valuation relies on standard assumptions that ensure the existence of a risk-neutral measure. Under these conditions, the stochastic process for rents can serve as the fundamental state variable for arbitrage-free pricing, and development options can be valued as contingent claims on future rents.⁵ Denoting R as the market rent per unit of space at $t = 0$, we can rewrite the potential rent at $t \geq 0$ as $R_t = Re^{(\delta - \sigma^2/2)t + \sigma W_t} = Re^{\sigma Z_t}$, where $Z_t = bt + W_t$, and $b = 1/\sigma(\delta - \sigma^2/2)$.

To obtain the rental income of a parcel, we introduce two elements. First, we define $\Gamma(F, z, \alpha)$ as the function that scales the market rent of a unit of density R_t , based on the characteristics of the built property, to yield the stochastic rental income of a developed property.⁶ In particular, the rent increases with the built density F^7 , decreases with the accumulated rental depreciation index ($z \in [0, 1]$), and decreases with the portion of built space allocated to affordable housing (α). Because market rent R_t is defined on a per unit basis of newly built property, we let $\Gamma(F = 1, z = 1, \alpha = 0) = 1$ correspond to a property with a built density of 1 ($F = 1$), with no operational efficiency losses due to rental depreciation ($z = 1$), and no affordable housing ($\alpha = 0$) that would alter the rental incomes. We revert to the full parametrization of $\Gamma(F, z, \alpha)$ in Section 3. Second, we assume the government imposes a tax on developed properties proportional to the rental incomes,

⁴Under standard asset-pricing conditions, the risk-neutral rent growth rate is linked to the observed cap rate c_i through the relationship $\delta_i = r - c_i$, where r denotes the risk-free rate.

⁵In practice, rents are not continuously traded and markets are incomplete. The assumption that rents are spanned by traded assets should be interpreted as an analytical convenience supported by the existence of correlated securities (e.g., REIT shares, lease-backed instruments) that allows developers to partially hedge rent risk. We discuss these assumptions in more detail in Appendix A.1.

⁶In this theoretical section, we assume that parcels are of unit size one ($L = 1$). In the empirical application, parcel size scales both rents and costs proportionally.

⁷Throughout the paper, density F should be interpreted as effective built volume per unit of land rather than purely floor-area density. In addition to total floor area, F captures building height that affects the productive space available to tenants.

reducing the developer's effective payout by ψ . Consequently, the value of the developed parcel without its development option is

$$\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{\infty} e^{-ru} \underbrace{(1 - \psi)}_{\text{Net of tax}} \underbrace{\Gamma(F_B, z_B, \alpha_B) R_u}_{\text{Gross rental income}} du \right\},$$

where the subscript B describes the characteristics of the current building.

Private developers have the option to replace the current building with a new one constructed at density $F \leq \bar{F}_0$, where $\bar{F}_0$ denotes the maximum allowable built density (volume) under the regulatory environment $\mathcal{P}_0$.⁸ Exercising this option requires paying an instantaneous demolition cost $\gamma(F_B)$ to remove the existing structure, in addition to incurring an annual construction cost during the construction period of d years, denoted as $\kappa(F, \alpha)$. Construction costs increase with the density of the new construction (F) and vary with the portion of built-up area dedicated to affordable housing (α).

Developers exercise the option to redevelop under $\mathcal{P}_0$ only if favorable market conditions prevail. Because developers already own the parcel and the existing structure, the redevelopment decision corresponds to exercising an embedded option to replace the current building. We denote by R^e the predetermined rent level (the expansion threshold) at which developers initiate construction at density F , and by T_{R^e} the first time this threshold is reached. We assume deterministic time-to-build: completion occurs exactly d years after initiation. Let H_{R^e} denote the time at which the property is fully developed.⁹ Consequently, the value of the parcel under $\mathcal{P}_0$ inclusive of the development option is

$$\begin{aligned} \pi_0(R, F_B, z_B, \alpha_B, F; \mathcal{P}_0, \Theta) = & \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{T_{R^e}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right\}}_{\text{Value of existing property}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ e^{-rT_{R^e}} \gamma(F_B) \right\}}_{\text{Demolition costs}} \\ & - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{T_{R^e}}^{H_{R^e}} e^{-ru} \kappa(F, \alpha) du \right\}}_{\text{Construction costs}} + \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{R^e}}^{\infty} e^{-ru} (1 - \psi) \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Value of new property}} \quad (2) \end{aligned}$$

where Θ is the vector of parameters. The value of a parcel thus consists of four discounted terms: net-of-tax rents from the current property until demolition, demolition costs at the stopping time,

⁸In Appendix E, we extend the model by adding the option for developers to renovate their properties in addition to the option to undergo a demolition and rebuilding and show how the baseline outcomes and the counterfactual results differ under this extended model.

⁹Formally, $T_{R^e}(R) = \inf\{t \geq 0 : R_t = R^e\}$ and $H_{R^e}(R) = T_{R^e}(R) + d$.

expenditures during construction, and the net-of-tax rental income from the completed asset.

1.2 Adding Policy Instruments into the Development Decision Rule

We summarize the policy environment by the vector $\mathcal{P} = [\alpha, \phi, \tau]$, where α denotes a mandate requiring that a share of the project be allocated to affordable housing units, ϕ represents additional allowable density, and τ denotes a temporary tax exemption following redevelopment. The status quo corresponds to $\mathcal{P}_0 = [\alpha = 0, \phi = 0, \tau = 0]$.¹⁰ The three policy instruments affect redevelopment decisions through different channels. The mandate α reduces the market rental income generated by the property and alters construction costs through the functions $\Gamma(\cdot)$ and $\kappa(\cdot)$, respectively. The density bonus ϕ relaxes the density constraint by allowing developers to choose $F \leq \bar{F}$, where $\bar{F} = \bar{F}_0(1 + \phi)$. Finally, the tax exemption τ postpones property taxation for τ years after redevelopment is completed.

We can thus rewrite the value of a parcel under the new regulatory environment as:

$$\begin{aligned} \pi(R, F_B, z_B, \alpha_B, F; \mathcal{P}, \Theta) = & \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{T_{Re}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right\}}_{\text{Value of existing property}} \\ & - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ e^{-rT_{Re}} \gamma(F_B) \right\}}_{\text{Demolition costs}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{T_{Re}}^{H_{Re}} e^{-ru} \kappa(F, \alpha) du \right\}}_{\text{Construction costs}} \\ & + \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{Re}}^{\infty} e^{-ru} \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Value of new property gross of tax}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{Re} + \tau}^{\infty} e^{-ru} \psi \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Property taxes after } \tau \text{ years of exemption}} \quad (3) \end{aligned}$$

Similar to Equation (2), the property value consists of the sum of present values with an additional final term that captures the impact of postponing property taxes for τ years after the development has been completed.

We solve the model analytically by first characterizing the five integrals that define parcel value conditional on a given expansion threshold and density. Rational developers maximize their parcel values, which yields a closed-form solution for the optimal expansion threshold and a numerical solution for the optimal density. These expressions allow us to derive the parcel-level

¹⁰Some parcels may already contain units subject to rent stabilization, rent control, or existing affordability requirements. Accordingly, the policy parameter α should be interpreted as an incremental mandate beyond existing regulations. If the current share of regulated or affordable units on a parcel (denoted α_B) already exceeds the policy requirement, the mandate does not bind, and the existing constraint remains in place.

outcomes: the probability that a parcel is redeveloped at a future date, and its optimal redevelopment density.

1.2.1 Parcel value

Proposition 1 *In the presence of the expansion threshold R^e and given a construction density $F \leq \bar{F}$, the property value $\pi(\cdot)$ satisfies*

$$\begin{aligned} \pi(R, R^e, F, F_B, z_B, \alpha_B; \mathcal{P}, \Theta) = & \underbrace{\Gamma(F_B, z_B, \alpha_B) \frac{1-\psi}{r-\delta} \left[R - R^e \left(\frac{R^e}{R} \right)^\xi \right]}_{\text{Value of existing property}} - \underbrace{\gamma(F_B) \left(\frac{R^e}{R} \right)^\xi}_{\text{Demolition costs}} \\ & - \underbrace{\frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}) \left(\frac{R^e}{R} \right)^\xi}_{\text{Construction costs}} + \underbrace{\Gamma(F, 1, \alpha) \frac{R^e}{r-\delta} e^{-(r-\delta)d} \left(\frac{R^e}{R} \right)^\xi}_{\text{Value of new property gross of tax}} \\ & - \underbrace{\psi \Gamma(F, 1, \alpha) \frac{R^e}{r-\delta} e^{-(r-\delta)(d+\tau)} \left(\frac{R^e}{R} \right)^\xi}_{\text{Property taxes after } \tau \text{ years of exemption}}, \quad (4) \end{aligned}$$

with $b = (1/\sigma)(\delta - \sigma^2/2)$, $\lambda = \sqrt{2r + b^2}$, and $\xi = (b - \lambda)/\sigma$.

The proof is in Appendix A.2.

1.2.2 Probability of development

Conditional on current market conditions and a given regulatory regime, developers optimally initiate construction when rent reaches a critical expansion threshold R^{e*} . At this threshold, the expected value of redevelopment exactly equals the continuation value of the existing structure, accounting for construction costs, policy distortions, and the stochastic timing of redevelopment.

Proposition 2 *Given a construction density $F \leq \bar{F}$ and a regulatory regime $\mathcal{P}$, the optimal expansion threshold is*

$$R^{e*} = \frac{\xi}{1 + \xi} \frac{\gamma(F_B) + \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd})}{\vartheta}, \quad (5)$$

where

$$\vartheta = \Gamma(F, 1, \alpha) \frac{1}{r-\delta} (e^{-(r-\delta)d} - \psi e^{-(r-\delta)(d+\tau)}) - \Gamma(F_B, z_B, \alpha_B) \frac{1-\psi}{r-\delta}.$$

The proof is in Appendix A.3.

Proposition 2 characterizes the rent level at which redevelopment becomes optimal. Because market rents evolve stochastically, a higher expansion threshold does not simply delay development deterministically. Instead, it lowers the probability that redevelopment occurs within any finite horizon. Conditional on the current rent level R , the timing of (re)development is governed by the stopping time $T_{R^{e*}}(R) = \inf\{t \geq 0 \mid R_t = R^{e*}\}$, the first time the rent process reaches the optimal expansion threshold.

This formulation allows us to characterize the redevelopment start in probabilistic terms. Policies that raise construction or regulatory costs increase R^{e*} and therefore reduce the likelihood that development is initiated by a given date, even holding fixed the underlying rent dynamics. The next proposition provides a closed-form expression for this probability.

Proposition 3 *Given the expansion threshold R^{e*} and a construction density $F \leq \bar{F}$, the probability that development is completed by horizon h satisfies*

$$\mathbf{P}_{\mathbb{P}}(H_{R^{e*}} \leq h) = \mathbf{P}_{\mathbb{P}}(T_{R^{e*}} \leq h - d), \quad (6)$$

where $H_{R^{e*}} \equiv T_{R^{e*}} + d$. The probability is computed under the physical measure $\mathbb{P}$, under which rent evolves as $dR_t = gR_t dt + \sigma R_t dW_t^{\mathbb{P}}$. Here g denotes the expected rent growth rate under the physical measure. Section 1.3 allows this growth rate to adjust with policy-induced supply changes.

For $h > d$, letting $a = \ln(R^{e*}/R)/\sigma > 0$ and $m = (g - \sigma^2/2)/\sigma$, the redevelopment probability is

$$\mathbf{P}_{\mathbb{P}}(T_{R^{e*}} \leq h - d) = \mathcal{N}\left(\frac{m(h - d) - a}{\sqrt{h - d}}\right) + e^{2ma} \mathcal{N}\left(-\frac{m(h - d) + a}{\sqrt{h - d}}\right), \quad (7)$$

while $\mathbf{P}_{\mathbb{P}}(H_{R^{e*}} \leq h) = 0$ for $h \leq d$.

The proof is in Appendix A.4.

1.2.3 Optimal density

The preceding analytical results characterize optimal redevelopment timing conditional on a chosen construction density $F \leq \bar{F}$. We now turn to the developer's choice of density. For a fixed expansion threshold R^e , the tradeoff between higher post-redevelopment rental income and higher construction costs admits a closed-form solution for the optimal density (see Appendix A.5). Higher density increases post-redevelopment rental income but also raises construction costs, so the optimal choice of F^* balances these competing forces. Once the expansion threshold is chosen optimally as a function of density, the joint problem of selecting both redevelopment timing and density no longer admits a closed-form solution.

As a result, the developer’s optimal density must be determined numerically. Formally, we solve

$$\begin{aligned} F^* &= \arg \max_F \pi(R, R^{e*}(F), F, F_B, z_B, \alpha_B) \\ \text{s.t. } & 0 \leq F \leq \bar{F}, \end{aligned} \quad (8)$$

where $R^{e*}(F)$ is given by Proposition 2. This numerical optimization yields the optimal density F^* , the associated expansion threshold $R^{e*}(F^*)$, and the associated probability of development $\mathbf{P}^*$. Together, these objects characterize the developer’s decision rule, incorporating both the optimal timing and intensity of redevelopment, conditional on replacement construction.¹¹ Given this optimal decision rule, for any horizon h and any policy $\mathcal{P}$, the expected floor-space supply on a parcel of size L is

$$s(h; \mathcal{P}) = \underbrace{(1 - \mathbf{P}^*(h; \mathcal{P}))F_B L}_{\text{Supply with no development}} + \underbrace{\mathbf{P}^*(h; \mathcal{P})F^*(\mathcal{P}; \Theta) L}_{\text{Supply with development}}. \quad (9)$$

This expression makes the replacement margin explicit. With probability $(1 - \mathbf{P}^*)$, the parcel keeps the existing structure. With probability $\mathbf{P}^*$, the parcel is redeveloped at density F^* .

1.3 Accounting for Aggregate and Local Spillovers in Development Decisions

The parcel-level decision rule described above characterizes redevelopment incentives for a given rent process. However, large-scale policy reforms may alter rents by changing construction across the city. To capture this feedback, we embed the parcel-level model in a citywide environment in which redevelopment decisions determine future housing supply and, in turn, expected rental growth. Policy-driven changes in supply affect rent growth through both aggregate supply effects and local neighborhood spillovers.

In equilibrium, the pricing drift δ_i used in the valuation problem and the physical rent growth rate g_i introduced in the redevelopment probability depend on the policy environment and the evaluation horizon h . We denote these policy-adjusted parameters by $\delta_i(h; \mathcal{P})$ and $g_i(h; \mathcal{P})$. Consistent with the literature on the effects of new supply on rents (Asquith et al., 2023; Li, 2022; Pennington, 2025; Rollet, 2025), we expect rents to respond negatively to *citywide* aggregate rises in housing stock (supply effect) and positively to *neighborhood-level* increases in market-rate units (amenity effect).

¹¹In this framework, developers choose *when* to redevelop and *how much* to build, but not the quality of construction conditional on redevelopment. New construction is assumed to enter without accumulated rental depreciation, reflecting the fact that redevelopment typically replaces aging structures with buildings that must meet current building codes, safety standards, and regulatory requirements. To the extent that developers in high-demand locations could choose to upscale quality, allowing for such choices would tend to amplify replacement-driven rent increases.

Negative supply-induced adjustments. Let $S(h; \mathcal{P})$ denote the expected citywide housing stock at horizon h under policy $\mathcal{P}$, and let $S(h; \mathcal{P}_0)$ denote the corresponding expectation under the status-quo. We measure the policy-induced change in expected total housing stock as the log difference:

$$\Delta S(h; \mathcal{P}) = \ln S(h; \mathcal{P}) - \ln S(h; \mathcal{P}_0).$$

We assume that rental growth responds to this aggregate supply shock with elasticity of rents with respect to housing supply $\zeta_S < 0$, so that larger citywide increases in housing supply reduce expected rent growth.

Positive amenity-induced adjustments. Let $N_i(h; \mathcal{P})$ denote the expected housing supply in the neighborhood of parcel i under policy $\mathcal{P}$, and define the associated policy-induced change as the log difference

$$\Delta N_i(h; \mathcal{P}) = \ln N_i(h; \mathcal{P}) - \ln N_i(h; \mathcal{P}_0).$$

The elasticity ζ_A captures how an increase in neighborhood housing units through redevelopment affects rent growth through local amenity spillovers. We assume that $\zeta_A > 0$ to reflect that new nearby development enhances local amenities and spurs gentrification (Couture et al., 2024), although negative values remain possible when congestion or local opposition dominate.

Combining the aggregate supply and local amenity channels yields a common adjustment summarizing the effect of policy-induced construction on rent growth. For parcel i , define the annualized supply-induced adjustment

$$\Delta \delta_i(h; \mathcal{P}) = \frac{1}{h} \left(\zeta_S \Delta S(h; \mathcal{P}) + \zeta_A \Delta N_i(h; \mathcal{P}) \right).$$

This term captures the effect of policy-driven supply changes on rent dynamics.¹²

Mapping supply adjustments into rents. We incorporate this supply adjustment into the two drift parameters governing rent dynamics. The *risk-neutral drift* governs valuation under the pricing measure $\mathbb{Q}$. For parcel i , the baseline pricing drift implied by the cap rate c_i is $\delta'_i \equiv r - c_i$. Rent growth, however, evolves under the physical measure $\mathbb{P}$. We denote the baseline physical rent growth rate by g'_i , which reflects the expected rent growth consistent with the total return on real estate net of the cap rate absent policy-induced supply shocks.

¹²Under the risk neutral measure, the expected potential rent satisfies $\mathbf{E}_{\mathbb{Q}}[R_h] = Re^{\delta h}$. At horizon h under policy $\mathcal{P}$, an exogenous supply shock $\Delta S(h; \mathcal{P})$ scales expected rents by $e^{\zeta_S \Delta S(h; \mathcal{P})}$ and implies $Re^{(\delta + \Delta \delta)h} = Re^{\delta h + \zeta_S \Delta S(h; \mathcal{P})}$, which yields this drift adjustment.

For a given policy $\mathcal{P}$ and evaluation horizon h , the policy-adjusted parameters are

$$\delta_i(h; \mathcal{P}) = \delta'_i + \Delta\delta_i(h; \mathcal{P}), \quad g_i(h; \mathcal{P}) = g'_i + \Delta g_i(h; \mathcal{P}).$$

Within each policy counterfactual, both parameters remain constant over time in the geometric Brownian motion dynamics. The dependence on h reflects the annualization of the horizon-specific supply adjustment rather than a time-varying drift process.

Fixed point and convergence. In equilibrium, policy-induced supply adjustments must be consistent with developers' redevelopment responses. The policy-induced changes in citywide and neighborhood supply determine the drift adjustment $\Delta\delta_i(h; \mathcal{P})$, which updates both the pricing drift $\delta_i(h; \mathcal{P})$ and the physical rent growth rate $g_i(h; \mathcal{P})$. These updated parameters affect redevelopment probabilities and densities and, therefore, expected supply. The resulting equilibrium corresponds to a fixed point in which supply adjustments and redevelopment responses are mutually consistent. Appendix A.6 describes the numerical procedure used to compute this fixed point.

1.4 Expected Rental Incomes

Using the completion probability from Proposition 3 and the optimal density F^* , we derive the expected parcel-level rental income at horizon h incorporating the redevelopment option and spatial adjustments. These adjustments account for the effects of citywide supply changes and neighborhood spillovers on expected rent levels.

Corollary 1 *For parcel i , the expected rental income at horizon h is*

$$\begin{aligned} \text{RentInc}_i(h, \mathcal{P}) = & \underbrace{\left[1 - \mathbf{P}_i^*(H_{Re^*} \leq h; \mathcal{P})\right]}_{\text{Prob no development}} \times \underbrace{\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}) \times \mathbf{E}_{\mathbb{P}}\{R_{ih}\}}_{\text{Rental income of existing property}} \\ & + \underbrace{\mathbf{P}_i^*(H_{Re^*} \leq h; \mathcal{P})}_{\text{Prob development}} \times \underbrace{\Gamma(F_i^*, 1, \alpha) \times \mathbf{E}_{\mathbb{P}}\{R_{ih}\}}_{\text{Rental income of new property}}. \end{aligned} \quad (10)$$

where $\mathbf{P}_i^*$ and F_i^* are given in Equations (6) and (8), and $z_B(h)$ is the cumulative rental depreciation index of the existing structure after additional h years. The spatially-adjusted physical expectation of rents is

$$\mathbf{E}_{\mathbb{P}}\{R_{ih}\} = R_i \exp(g_i(h; \mathcal{P}) h), \quad (11)$$

Expression (11) reduces to $R_i e^{g'_i h}$ under the status quo.

The proof is in Appendix A.7.

Expression (10) shows that expected rental income is a weighted average of the income generated by the existing structure and the redeveloped property. Spatial effects enter through two channels. First, the drift adjustment $\Delta\delta_i(h; \mathcal{P})$ modifies the pricing drift $\delta_i(h; \mathcal{P})$ and therefore affects the redevelopment probability $P_i^*(h, \mathcal{P})$ and the optimal density $F_i^*(\mathcal{P})$. The weights in (10) therefore incorporate spatial feedback induced by policy. Second, spatial supply responses also affect the level of expected rents through the physical growth rate $g_i(h; \mathcal{P})$ in (11). These adjustments apply even when redevelopment does not occur, so spatial feedback affects both new and existing structures.

2 Impact of Public Policies on Parcel-level Outcomes

This section shows how policy changes redevelopment on a parcel. We focus on two objects: the value of the parcel and the redevelopment probability. We specifically highlight the parcel-level trade-off between *Carrots* (ϕ and τ) and *Sticks* (α). Appendix B develops these effects further, reports comparative statics for the remaining parameters, shows why parcel heterogeneity matters for redevelopment, and numerically describes how exogenous supply propagates through the spatial spillovers.

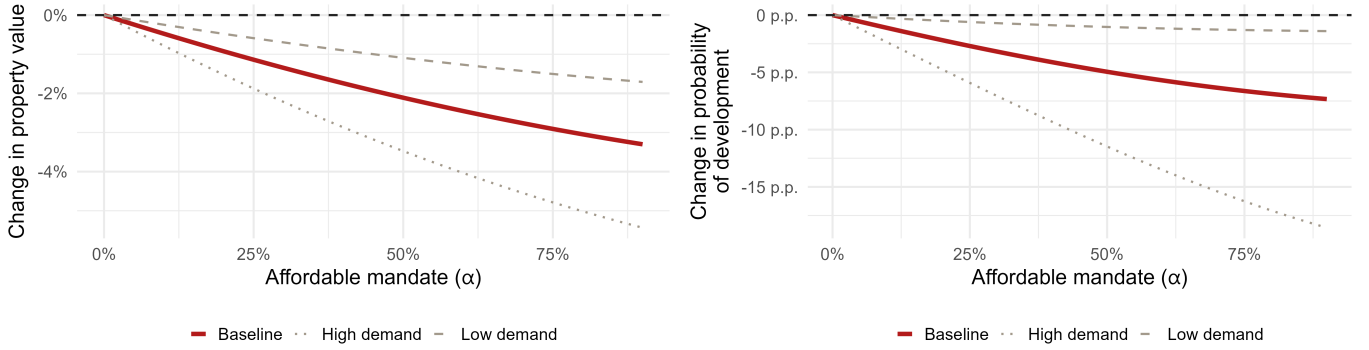
We study a representative parcel near its redevelopment threshold and use the parametrization from Section 3. The calibration allows density to raise both revenues and costs. It also allows the affordability mandate α to reduce rents and construction costs. Figure I shows how each policy instrument changes property values (left panels) and the ten-year redevelopment probability (right panels). The figure also shows how these effects vary with initial demand, by moving potential rent R above or below the baseline.

Panel (a) of Figure I shows that a higher affordable share requirement (α) lowers both property values and redevelopment probabilities. The mandate lowers property values by shrinking the redevelopment option, as the value of the existing structure does not change. Parcels with higher demand lose more value and face larger reductions in redevelopment probability, so developers delay replacement more often on those parcels. The effect is also stronger for parcels closer to the redevelopment threshold, where the mandate binds most tightly.

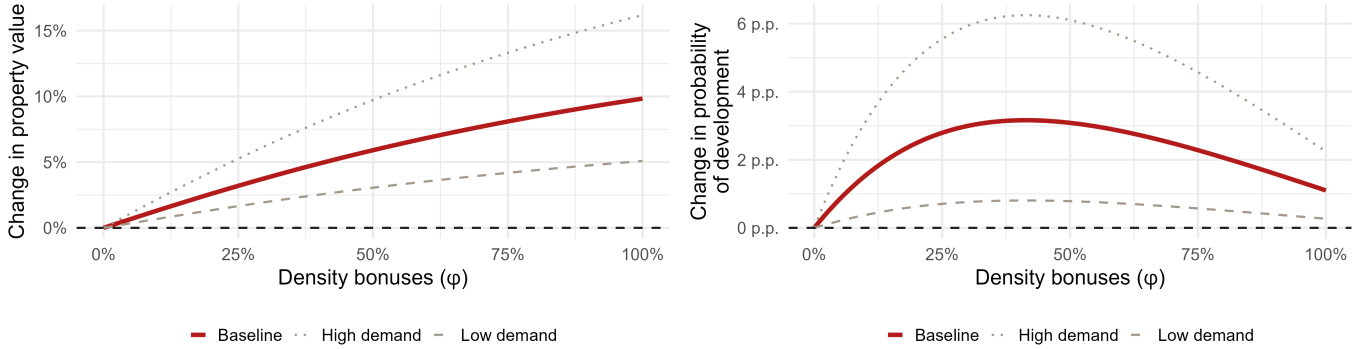
Panel (b) of Figure I shows that a higher density bonus (ϕ) raises property values by making redevelopment more valuable.¹³ However, the effect on redevelopment probability is ambiguous.

¹³In this simple single-parcel analysis, additional supply does not affect market rents, unlike in the citywide model

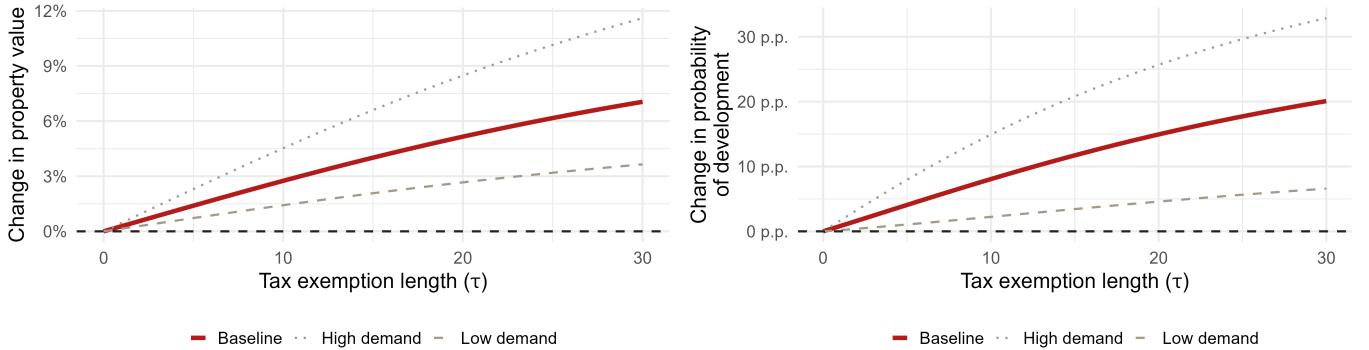
Figure I: Property Value and Development Probability with Respect to Policy Instruments



A. Mandatory share for affordable housing (α)



B. Density bonuses (ϕ)



C. Length of tax exemptions (τ)

Note: These figures show the effects of policy instruments on property value (left panels) and the probability of development within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, aged 150, with a rental growth of 4%, a volatility of 15%, and an annual potential rent \$50 per square foot. All other parameters and elasticities are reported in Table I.

that includes endogenous amenity spillovers and citywide supply effects. In other words, the stochastic potential rent R_t is here assumed to remain unaffected by the policy ϕ .

The probability can rise or fall with ϕ , depending on the parcel characteristics and local housing demand. Because developers retain the option to build at higher density, property value always rises. When ϕ is large, developers often delay construction because they wait until rents are high enough to justify a taller project. The delay is largest for parcels near the redevelopment threshold. On those parcels, extra density raises the value of waiting and pushes replacement further into the future.

Panel (c) of Figure I shows that a tax exemption raises property values. It also accelerates redevelopment by lowering the rent level that triggers replacement. The value gain comes entirely from the redevelopment option because the exemption applies only to new buildings. The effect on development probability is strongest for parcels already close to their redevelopment threshold.

We then study the trade-off between *Carrots* and *Sticks* at the parcel level. We focus on how affordability mandates interact with density bonuses and tax incentives. The left panels of Appendix Figure B2 show that, for any affordability mandate, one can choose a density bonus or a tax exemption that keeps property values unchanged. At those combinations, developers are indifferent between the status-quo and the combination of instruments. The right panels also show that there are combinations that keep redevelopment incentives unchanged. The two indifference curves, however, do not coincide. This gap shows how policies with the same value to developers can still generate different redevelopment choices. That wedge makes it hard to align private incentives with public objectives. Although zoning in New York City is primarily as-of-right (i.e., development allowed without discretionary review), these parcel-level results illustrate how the model extends naturally to discretionary zoning systems in other cities.

Finally, Appendix B.4 also provides numerical analysis of how parcel-level outcomes spill over into space and reshape housing composition. To illustrate these spillovers, we perform an analysis in a stylized city, forcing 1% of parcels to redevelop and study how redevelopment probabilities adjust on all remaining parcels. In a build-out city, we observe that the aggregate supply response to this exogenous shock is small due to the high implicit costs of forgoing income from the income-generating buildings. Yet, the spatial reallocation is heterogeneous. Redevelopment incentives decline more sharply in high-demand areas through the citywide supply channel but rise locally around the shocked parcels through amenity spillovers, causing redevelopment to cluster in space.

3 Model Quantification

We calibrate the model to New York City’s housing market and focus on affordable-housing mandates as the central policy. First, we specify the functional forms $\Gamma(\cdot)$, $\gamma(\cdot)$, and $\kappa(\cdot)$ that govern the instantaneous return and costs to redevelopment. Second, we calibrate the parameter vector Θ using microdata, estimates from the literature, and selected empirical moments. Table I reports the resulting parameter values. Third, we construct the parcel-level dataset and summarize its sources and summary statistics in Table II.

3.1 Parametrization

3.1.1 The rental income scaling function

The underlying potential rent per square foot R is scaled by $\Gamma(\cdot)$, which maps parcel characteristics into annual rental income. We parametrize this scaling function as

$$\Gamma(F, z, \alpha, L) = LF^{(1+\epsilon_F)}z(1 - \alpha\epsilon_\alpha). \quad (12)$$

Here L is parcel size, F is floor-area ratio, $z \in (0, 1)$ is the rental depreciation index, and α is the share of affordable units in the building.¹⁴ The parameter $\epsilon_F > 0$ is the elasticity of rental incomes with respect to building height, while $\epsilon_\alpha > 0$ captures the rental discount associated with subsidized units. We set ϵ_F at 0.07 using estimates from Danton and Himbert (2018), Ahlfeldt and Barr (2022), and Ahlfeldt et al. (2026). The positive value of ϵ_F reflects the capitalization of height-related amenities, such as improved views and reduced street-level noise, into market rents.

The rental discount parameter ϵ_α could, in principle, vary with policy $\mathcal{P}$. To limit the counterfactual space, we vary the share of affordable units α , while holding ϵ_α fixed. We follow Favilukis et al. (2023), who report that rent-stabilized units in New York City rent at an average discount of 26.6% relative to market-rate units, while rent-controlled units rent at an average discount of 77.7%. To span this range, we consider $\epsilon_\alpha \in \{0.25, 0.50, 0.75\}$. Our baseline counterfactuals set $\epsilon_\alpha = 0.25$. Section 5 examines how results vary with the rental discount, with particular attention to developers’ voluntary adoption of policy $\mathcal{P}$.

¹⁴The theoretical model abstracts from the size of the parcel because the optimal decision rule is unaffected by it (i.e., the expansion threshold R^{e*} is homogeneous of degree zero in L). However, it affects aggregate outcomes, such as total housing production.

3.1.2 The demolition cost function

We parametrize demolition costs as

$$\gamma(F_B, L) = k_d L F_B^{\eta_D}, \quad (13)$$

where η_D governs how demolition costs scale with existing built density F_B , and $k_d > 0$ is a cost shifter. We calibrate η_D using information from developer interviews. They reported that fixed cost components (those that do not scale with structure size) account for approximately 50–60% of total demolition costs. To match this feature, we choose η_D so that the average ratio of the cost of demolishing a building with $F_B = 0.1$ to the cost of demolishing the full structure equals 55%. Formally,

$$\frac{1}{n_B} \sum_{i: F_{B,i} > 0} \frac{\gamma(0.10, L_i)}{\gamma(F_{B,i}, L_i)} = 0.55,$$

where $n_B \equiv \#\{i : F_{B,i} > 0\}$. This condition yields $\eta_D = 0.296$, implying substantial returns to scale in demolition. Developers also reported that demolition costs typically account for 13–20% of total redevelopment costs when an existing structure is removed. This range pins down the relationship between the demolition cost shifter k_d and the construction cost shifter k_c . We determine these parameters jointly in Section 3.2.

3.1.3 The construction cost function

We parametrize annual construction costs as

$$\kappa(F, \alpha, d, L) = \frac{1}{d} k_c L F^{(1+\eta_F)} (1 - \alpha \eta_\alpha), \quad (14)$$

where k_c is the baseline construction cost per square foot, $\eta_F > 0$ governs how costs scale with building height, and η_α captures cost differences between market-rate and affordable units. We set $\eta_F = 0.60$ to reflect the convex relationship between height and construction costs. This value is consistent with estimates in [Ahlfeldt and McMillen \(2018\)](#), [Ahlfeldt et al. \(2026\)](#), and [Ahlfeldt and Barr \(2022\)](#), which report elasticities in the range of 0.55–0.61 for tall residential buildings.

The parameter η_α captures systematic differences in per-square-foot construction expenditures between market-rate and subsidized units. We calibrate η_α using data from the National Apartment Association’s Income and Expenses Report, as summarized in the RCLCO report [The Affordable Housing Asset Class](#). This report documents capital expenditures per square foot for market-rate and subsidized units from 2012 to 2021. On average, subsidized units receive 19.9% lower capital expenditure per square foot than market-rate units. We treat this differential as a

Table I: Baseline Model Elasticities and Parameters

Parameters	Description	Value	Methods & main sources
<u>The Rental Scaling function</u>			
ε_F	Density elasticity of rent	0.07	Calibration – Literature
ε_z	Age elasticity of rent	-0.093	Estimation – Table C2
ε_α	Affordable units rental discount	{0.25; 0.50; 0.75}	Calibration – Literature
<u>The demolition cost function</u>			
η_D	Demolition cost elasticity of density	0.296	Target: Fixed cost demolition (55%)
k_d	Baseline demolition cost per density unit	73.91	Target: Share of development cost (15%)
<u>The construction cost function</u>			
η_F	Density elasticity of construction cost	0.60	Calibration – Literature
η_α	Affordable housing construction cost discount	0.20	Estimation - RCLCO report
k_c	Baseline construction cost per density unit	436.37	Target: Land-value ratio (23.68%)
<u>Spatial supply spillovers</u>			
ζ_S	Global Supply elasticity of rent	-0.50	Calibration – Literature
ζ_A	Local Supply elasticity of rent	0.10	Calibration – Literature
	Distance for local supply spillovers	500 feet	Calibration – Literature
<u>Others</u>			
ψ	Real estate tax rate (% of rental income)	20.15	Estimation - DoF & Costar
d	Construction periods (years)	3.08	Estimation – DCP
r	Risk free rate (%)	1.90	Estimation – FED

reasonable proxy for construction cost differences and set $\eta_\alpha = 0.20$.

3.2 Calibration of Other Parameters

Amenity and demand elasticities (ζ_S and ζ_A). As described in Section 1.3, new construction affects rents and redevelopment incentives through two channels. A citywide supply effect lowers rents and is governed by the inverse demand elasticity $\zeta_S < 0$. A local amenity effect raises rents and is governed by $\zeta_A > 0$. No empirical studies separately identify these two components. Instead, they recover their combined effect at local spatial scales. Recent evidence for New York City guides our choice of the net elasticity. Li (2022) shows that an unanticipated increase in nearby housing supply reduces local rents within a 500-foot radius. Rollet (2025) finds a larger negative elasticity at a similar spatial scale. Mense (2025), Asquith et al. (2023), Anenberg and Kung (2020), and Pennington (2025) provide related evidence of negative, though limited, rent elasticity of supply. Taken together, these studies imply that the net effect of new construction on nearby rents is negative on average. Local amenity gains partially offset supply effects but do not overturn them.

Guided by this evidence, we impose $\zeta_S + \zeta_A = -0.40$. We define the neighbors of parcel i

as all parcels located within 500 feet of its centroid. We incorporate endogenous amenity effects (Almagro and Domínguez-Iino, 2025; Diamond, 2016; Gechter and Tsivanidis, 2023), which have been shown to be associated with new supply in New York City (Liao, 2025), setting a baseline value of $\zeta_A = 0.10$. This specification allows local redevelopment to increase nearby rents and thereby strengthen redevelopment incentives, as in Pennington (2025). Given the lack of separate empirical identification of ζ_S and ζ_A , we treat their decomposition as a modeling choice. We explore alternative decompositions in Appendix D.2.

Construction length (d) We calibrate the construction period using the [DCP Housing Database Project-Level Files](#). We define construction duration as the interval between permit issuance and building completion. During this period, developers incur construction costs but earn no rental income. We restrict the sample to multi-family buildings developed by private for-profit firms. We exclude projects completed in less than three months or more than ten years. We set $d = 3.08$ years, which matches the average time between permit issuance and completion.

Risk-free rate (r) We calibrate the risk-free rate r using the average one-year yield on U.S. Treasury securities reported by the Board of Governors of the Federal Reserve System ([FRED](#)). The average yield over the period from January 2000 to December 2023 equals 1.90%. We therefore set $r = 0.019$. This rate serves as the discount rate in the risk-neutral valuation.

Property tax rate (ψ) We construct an effective property tax rate on residential properties by combining multiple datasets. We first link assessment data for multi-family properties from PLUTO with statutory tax rates ([NYC Department of Finance](#)) to compute parcel-level tax liabilities since 2000. We then merge these tax liabilities with 36,000 multi-family transactions from CoStar and compute tax rates by dividing tax bills by corresponding sale prices. Since 2000, the median effective tax rate equals 1.1%, with an interquartile range of 0.57–1.83%. The model requires a tax rate on rental flows. We therefore convert the asset-based tax rate into an income-based rate by dividing by the sample’s average capitalization rate of 5.34%. This mapping translates observed property tax liabilities into the effective tax wedge on rental income that enters developers’ decisions.

Baseline construction and demolition costs (k_c and k_d) Three parameters govern the monetary level of the model: the market rent of a non-depreciated square foot, R , the demolition cost, k_d , and the construction cost, k_c . All three quantities are expressed per square foot. Property values (Proposition 1) are homogeneous of degree one in R , k_d , and k_c , and development probabilities (Proposition 3) are homogeneous of degree zero. Only their relative magnitudes therefore matter

for counterfactual analyses. We take the distribution of R_i across parcels as given and calibrate k_d and k_c to match empirical moments. We treat these parameters as constant across neighborhoods, since labor, material, and regulatory costs are broadly similar within the city (Glaeser and Gyourko, 2018).

To pin down k_c , we use PLUTO data on the ratio of land to property values. For each parcel, we compute the model-implied land-value ratio $LandRatio_i = \pi_i(F_{B,i} = 0; \Theta) / \pi_i(F_{B,i} = F_{B,i}; \Theta)$, which compares the value of the parcel with no structure to its value at the observed density $F_{B,i}$. We interpret this ratio as the redevelopment option value when the existing structure is removed. Lower construction cost k_c increases redevelopment incentives and raises land values relative to current property values. We therefore choose k_c so that the average model-implied $LandRatio_i$ matches the observed mean of 23.7%.

To identify k_d , we use evidence that demolition costs account for 13–20% of total development costs when an existing structure is removed. We target the midpoint of this range by imposing

$$\frac{1}{n_B} \sum_{i: F_{B,i} > 0} \frac{\gamma(F_{B,i}, L_i)}{d \cdot \kappa_i(F_i^*, \alpha_{B,i}, d, L_i) + \gamma(F_{B,i}, L_i)} = 0.15$$

in the model. This condition matches the share of the total development costs attributable to demolition. Higher demolition costs k_d increase the share of total redevelopment costs attributable to demolition. Matching the land-value ratio and demolition share targets jointly yields calibrated values of $k_c = 436.37$ and $k_d = 73.91$. The mean value of R_i in the calibrated model is \$35.46. The cost of constructing one unit of density is therefore approximately twelve times annual rental income.

3.3 Parcel-level Data

Each parcel i is characterized by its current density $F_{B,i}$, maximum allowable density $\bar{F}_i$, rental depreciation index $z_{B,i}$, share of current affordable units $\alpha_{B,i}$, and location. These variables determine the value of the existing structure and the constraints on redevelopment. Location determines the potential rent per square foot R_i , and the expected return and volatility of rental income. Together, these variables pin down the evolution of rents and the payoffs to redevelopment. We restrict the sample to parcels zoned or partially zoned for residential use. We exclude parcels designated exclusively for public or other non-residential purposes. The final sample contains 765,830 parcels. Appendix C describes the data construction and cleaning procedure, and includes additional summary statistics of key parcel-level variables.

Table II: Summary Statistics of Lot-level Data

Statistic	Source	Mean	St. Dev.	Min	Median	Max
Current FAR – $F_{B,i}$	PLUTO	1.05	1.18	0.00	0.73	88.55
Maximum FAR – $\bar{F}_i$	PLUTO	1.44	1.43	0.50	0.90	88.55
Lot size (th. sqft) – L_i	PLUTO	3.75	6.50	1.25	2.50	250.00
Age (years)	PLUTO	72.92	36.63	0	84	302
Rental depreciation index (%) – $z_{B,i}$	Estimated	68.98	7.08	58.69	66.08	100.00
Share of aff. units (%) – $\alpha_{B,i}$	DHCR & HPD	0.25	2.98	0	0	50
Annual Rent per sqft (\$)	ACS + PLUTO	24.06	7.99	2.48	23.11	99.24
Annual Potential Rent per sqft (\$) – R_i	Estimated	35.46	11.08	3.41	34.62	127.05
Expected growth (%) – g_i	Costar	2.25	0.45	1.34	2.32	4.10
Caprate (%) – $r - \delta_i$	Costar	5.34	0.34	4.22	5.21	6.17
Volatility (%) – σ_i	Costar	8.21	0.39	7.46	8.26	11.08

Notes: This table presents summary statistics for the 765,830 parcels included in the baseline analysis. The majority of parcel-level data is sourced from the Primary Land Use Tax Lot Output (PLUTO) as of 2023. Age measures the number of years since the last recorded renovation or construction. Rental depreciation index is the rental discount associated with property aging, with 1 representing a newly constructed building. Annual rent per square foot is derived from the 2023 ACS 5-Year data at the census block-group or tract level. Potential rent reflects market-rate rents for one non-depreciated square foot at FAR=1, obtained by adjusting observed rent for the typical building within each geography. Cap rates, rental returns, and volatility are sourced from CoStar, following their definitions for New York City residential markets, as detailed in Appendix C.

Built and maximum allowed density ($F_{B,i}$ and $\bar{F}_i$). We measure current density $F_{B,i}$ and maximum allowable density $\bar{F}_i$ using the built Floor-Area-Ratio (FAR) and zoning FAR from the Primary Land Use Tax Lot Output (PLUTO) dataset. We also record lot size L_i in square feet so that total built space on parcel i equals $s_i = F_{B,i}L_i$. In the baseline calibration, the average built density $F_{B,i}$ equals 1.05. The average building size is approximately 4,750 square feet. The average maximum allowable density $\bar{F}_i$ equals 1.44. A total of 36.8% of parcels are built to their maximum allowable density. As Figure C1 shows, unused capacity is spatially heterogeneous, with greater scope for densification in areas such as the Southern Bronx.

Rental depreciation index ($z_{B,i}$). We model depreciation as a reduction in rental income as the structure ages. Specifically, we set $z_i = \text{Age}_i^{\epsilon_z}$, $\epsilon_z < 0$ where Age_i is the number of years since construction or last recorded renovation. Older buildings, therefore, generate lower effective rental income. This functional form implies faster depreciation for younger buildings. Lopez and Yoshida (2022) provides evidence consistent with this pattern. To estimate ϵ_z , we follow Rosenthal (2014) and regress transaction prices on building age in a hedonic pricing framework, as described in Appendix C.2. The estimated elasticity ranges from -0.096 to -0.089 . These estimates imply an average annual decline of approximately 2% over the first 10 years, and slower depreciation thereafter. We set ϵ_z to -0.093 in the baseline calibration. This value yields a mean $z_{B,i}$ of 0.69 and

a standard deviation of 0.07. Lower values of $z_{B,i}$, concentrated mostly in Queens and Brooklyn (Panel B of Figure C1), reduce the value of the existing structure and increase the likelihood of redevelopment by lowering the expansion threshold R_i^{e*} .

Current share of affordable units ($\alpha_{B,i}$). We construct $\alpha_{B,i}$ from multiple data sources, because no single dataset reports building-level affordability shares. We first use publications from the New York State Division of Housing and Community Renewal (DHCR) to identify buildings that receive a 421-a or a J-51 tax exemption. These programs provide property tax benefits in exchange for affordable units. We identify 4,611 parcels with a 421-a exemption and 1,486 parcels with a J-51 exemption. We assign $\alpha_{B,i} = 30\%$ for 421-a properties and $\alpha_{B,i} = 50\%$ for J-51 properties, to reflect the stronger affordability requirement under J-51. We then incorporate buildings that the Department of Housing Preservation & Development reports as participating in Mandatory Inclusionary Housing (MIH) or Voluntary Inclusionary Housing (VIH) programs. These programs also require the provision of affordable units. We identify 104 MIH/VIH buildings and assign $\alpha_{B,i} = 30\%$ for those parcels. Because VIH allows FAR bonuses to be transferred to parcels other than those hosting affordable units, we assign a positive $\alpha_{B,i}$ only to the generating sites.

Potential rent per square foot (R_i). We estimate R_i using block-group-level median rents from the 2023 American Community Survey (ACS) 5-year data. Appendix C.3 provides details. We first convert reported rents to a per-square-foot measure by normalizing ACS rents using the median residential unit size in the block group of i (UnitSize_i), computed from parcel-level data in PLUTO. The average annual rent per square foot in the sample equals \$24.06. The variable R_i however represents the potential rent of *market-rate*, non-depreciated space at $\text{FAR} = 1$. We, therefore, adjust observed rent per square foot in geography j to remove the effects of depreciation, built density, and existing affordability requirements. For instance, low observed rent per square foot in a block-group may reflect low potential rent R_i or a high prevalence of obsolete structures. Specifically, we define

$$R_{i(j)} = \frac{\text{Rent}_{i(j)}}{\Gamma\left(\tilde{F}_{B,i(j)}, \tilde{z}_{B,i(j)}, \tilde{\alpha}_{B,i(j)}\right)},$$

where $\text{Rent}_{i(j)}$ denotes the average observed rent per square foot in geography j . The terms $\tilde{F}_{B,i(j)}$, $\tilde{z}_{B,i(j)}$, and $\tilde{\alpha}_{B,i(j)}$ denote the weighted median built density, rental depreciation index, and affordable unit share across parcels in geography j , respectively. We weight each parcel by built square footage $F_{B,i}L_i$ so that representative characteristics reflect the stock of space occupied by renters. After this adjustment, the average annual potential rent in the sample equals \$35.46 per square foot.

Rental return and volatility (δ_i , g_i and σ_i). We calibrate the baseline pricing drift δ'_i using observed capitalization rates $caprate_i$ defined as the rent-to-price ratio. We set $\delta'_i = r - \text{CapRate}_i$ where r is the risk-free rate. This specification treats δ'_i as a reduced-form, risk-neutral measure of expected rental growth that incorporates both fundamentals and the required risk premia. We obtain capitalization rates from CoStar market data and assign them to each parcel based on its location in one of the 37 CoStar-defined New York submarkets. Appendix C.4 describes the assignment procedure and reports summary statistics. We calibrate the baseline physical rent growth rate g'_i using Costar data with $g'_i = \text{TotalReturn}_i - \text{CapRate}_i$. The average implied rent growth rate across submarkets equals 2.25%. This measure captures developers' expectations about future rental growth under the status-quo policy. We estimate volatility σ_i using asset and rental returns for multifamily properties over the past 20 years in each submarket. We compute standard deviations of these return series and denote them by σ_i . The average annual volatility across submarkets equals 8.21%.

4 Baseline Model Outcomes and Validation

We compute baseline outcomes under policy $\mathcal{P}_0 = \{\alpha = 0, \phi = 0, \tau = 0\}$, which serves as the reference point for all counterfactuals. At the parcel-level, we recover three objects: $\mathbf{P}_{ih}^*(\mathcal{P}_0)$, $F_i^*(\mathcal{P}_0)$, $\pi_i(\mathcal{P}_0)$. The redevelopment probability $\mathbf{P}_{ih}^*(\mathcal{P}_0)$ describes the optimal probability that parcel i is replaced within horizon h . The optimal density $F_i^*(\mathcal{P}_0)$ determines how intensively the parcel is rebuilt once redevelopment occurs. The parcel value $\pi_i(\mathcal{P}_0)$ combines the continuation value of the existing structure and the option value of redevelopment: $\pi_i(\mathcal{P}_0) = \pi_i^{\text{existing}}(\mathcal{P}_0) + \pi_i^{\text{option}}(\mathcal{P}_0)$. Together, these objects characterize redevelopment as the margin through which policy affects supply and rents.

For each parcel i , we compute from these objects the expected housing supply $s_{ih}(\mathcal{P}_0)$ and the expected rental income $\text{RentInc}_{ih}(\mathcal{P}_0)$ following Equations (9) and (10), respectively. To ease the interpretation of our quantity results, we report outcomes in housing units rather than square footage by scaling the expected supply $s_{ih}(\mathcal{P}_0)$ by block-group median unit size UnitSize_i . This transformation aids interpretation but leaves the redevelopment decisions unchanged. Rents are reported in dollars per month per unit unless otherwise specified. Appendix C.5 describes the aggregation in detail.

Table III: Baseline Calibration Outcomes and Robustness to Parameter Changes

	UnitCons (# of Unit)	UnitNet (# of Unit)	Rent (\$ per month)	$PV = \sum_i \pi_i$ (\$ Bn.)	Option (% of PV)
	(1)	(2)	(3)	(4)	(5)
<i>Baseline</i>	168,395	125,685	2,682	1,574.13	0.81
<u>Varying key underlying model's elasticities</u>					
<i>Shorter duration time ($d = d/2$)</i>	275,227	181,601	2,727	1,578.91	1.11
<i>Higher cost to height ($\eta_F = 2\eta_F$)</i>	192,387	135,036	2,673	1,581.01	1.24
<i>Higher return to height ($\epsilon_F = 2\epsilon_F$)</i>	196,595	143,401	2,903	1,703.79	0.79
<i>Higher depreciation ($\epsilon_z = 2\epsilon_z$)</i>	321,393	177,542	2,010	1,127.91	1.55
<u>Other model benchmarks</u>					
<i>No endogenous density choice ($F_i^* = \bar{F}_i$)</i>	171,581	130,594	2,689	1,569.88	0.54
<i>Adding an option to renovate</i>	172,469	125,595	2,684	1,574.14	0.81
<i>No development ($k_c = \infty$)[†]</i>	0	0	2,644	1,602.23	0
<i>No maximum allowable density ($\bar{F}_i = \infty$)[†]</i>	419,066	323,510	2,798	1,528.31	1.76

Notes: This table reports the aggregate outcomes of our baseline calibration with a horizon of 10 years and additional sensitivity analyses performed by varying underlying model parameters. Column (1) shows the expected number of housing units completed at the 10-year horizon, Column (2) shows expected net unit addition, and Column (3) shows expected monthly rent. Column (4) shows the sum of the parcel present values, and Column (5) shows the average value of the redevelopment option. The first row reports results under the baseline calibration described in Table I, and subsequent rows report the same outcomes under alternative parameter assumptions. [†]Except for the last two model specifications, the rent drift is held fixed at its baseline value in these exercises and does not adjust with excess supply.

4.1 Baseline Results

Housing production and expected rents. Table III, row 1, reports the baseline outcomes under $\mathcal{P}_0$. We obtain 168,395 additional housing units over ten years, in line with New York City's historical production of 10,000–28,000 units per year since 2010 ([NYC housing production](#)). After accounting for demolitions, we obtain a net supply addition of 125,685 units. We find that construction concentrates in locations with high potential rents R_i and on parcels with obsolete and underutilized structures. Parcels in these locations reach the redevelopment threshold R_i^{e*} more often and therefore exhibit higher redevelopment probabilities $P_{ih}(\mathcal{P}_0)$. Panel A of Figure C4 shows this spatial pattern.

Average rent rises from \$2,107 to \$2,682 per housing unit per month. This increase follows directly from the expected rent of Equation (10). The first term captures non-redeveloped properties. Depreciation lowers $\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}, L_i)$ over time and reduces rents on the old structure. The second term captures redeveloped properties. Replacement raises rents because $\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}, L_i) < \Gamma(F_i^*, 1, \alpha_{B,i}, L_i)$. The growth term g'_i increases $\mathbb{E}_{\mathbb{P}}\{R_{ih}\}$ for both states.¹⁵

¹⁵In the baseline scenario, δ'_i and g'_i do not vary with future expected supply, $\Delta S(\mathcal{P}_0, t)$, because the calibrated cap

Asset valuation, and option value. Our calibration implies a total residential asset value of \$1.58 trillion, close to the Department of Finance’s reported valuation of \$1.83 trillion ([Tax Lot Assessment Portal](#)). At the parcel level, values satisfy $\pi_i(\mathcal{P}_0) = \pi_i^{\text{existing}}(\mathcal{P}_0) + \pi_i^{\text{option}}(\mathcal{P}_0)$. The redevelopment option accounts for 0.81% of total residential asset value. Thus, most of the parcel’s value comes from the existing structure rather than from growth options. This option to delay limits redevelopment and density growth, and also constrains the supply response to policy.

Parcel-level heterogeneity. Redevelopment varies across parcels through the optimal redevelopment probability $\mathbf{P}_i^*$ and density F_i^* . Older buildings redevelop more often. Depreciation lowers the existing payoff $\Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i)$, which reduces the continuation value π_i^{existing} . This lowers the expansion threshold R_i^{e*} and increases the redevelopment probability. Parcels with unused capacity $(\bar{F}_i - F_{B,i})$ also redevelop more often. Higher feasible density raises the post-redevelopment payoff $\Gamma(F_i^*, 1, \alpha_{B,i}, L_i)$, increases the option value π_i^{option} , and lowers R_i^{e*} . High existing density has the opposite effect. It delays redevelopment. It raises the opportunity cost of demolition through the value of current cash flows, $\pi_i^{\text{existing}} \propto (1-\psi)/(r-\delta_i)\Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i)R_i$. A higher continuation value increases R_i^{e*} and reduces $\mathbf{P}_i^*$. High-rent locations redevelop more often because higher potential rent increases the likelihood that $R_{ih} \geq R_i^{e*}$ and raises $\mathbf{P}_i^*$. Figure C5 and Table C4 show these patterns across built space and potential rent. The model thus generates a rich cross-section. Some parcels sit on valuable but “patient” options, while others sit on deteriorated cash flows that push owners to rebuild.

4.2 Robustness to underlying parameters

The remaining panels of Table III assess how changes in parameters affect redevelopment. In these scenarios, we hold developers’ expectations about future aggregate and local supply fixed and thus do not update $\mathbb{E}_{\mathbb{P}}\{R_{ih}\}$ through equilibrium feedback. This approach isolates the direct effect of construction technology and demand parameters on the redevelopment decisions and, thus, aggregate outcomes. The objective of these exercises is to illustrate how baseline outcomes vary with underlying parameters, not to evaluate a counterfactual in which a given parameter shifts exogenously.

4.2.1 Sensitivity to elasticity assumptions

Shorter construction durations d increase redevelopment. A lower d raises the option value π_i^{option} through discounting which lowers the rent threshold R_i^{e*} , which increases $\mathbf{P}_i^*$. Redevelopment occurs earlier and more often.

rate already incorporates developers’ expectations under the status quo.

Higher curvature of construction costs with respect to height raises marginal costs $\kappa(F, \alpha, L)$. This change reduces the optimal density F_i^* . At the same time, developers respond to this lower optimal density by lowering the rent threshold and expect to redevelop their parcels sooner. The timing margin dominates the scale margin. Total expected supply, therefore, increases despite lower density per project. These results underscore the central role of construction technology in shaping urban outcomes (Ahlfeldt et al., 2026).

Higher returns to height increase $\Gamma(F, z, \alpha, L)$ which raises the payoff to redevelopment if $F_i^* > F_{B,i}$. For most parcels, this shift reduces rent thresholds and increases the redevelopment probability. Faster depreciation works through the existing assets. It lowers $\Gamma(F, z, \alpha, L)$, reduces the continuation value $\pi_i^{existing}$, and again lowers the replacement threshold. Both changes make redevelopment more attractive, either by improving the upside or by eroding the value of the existing asset.

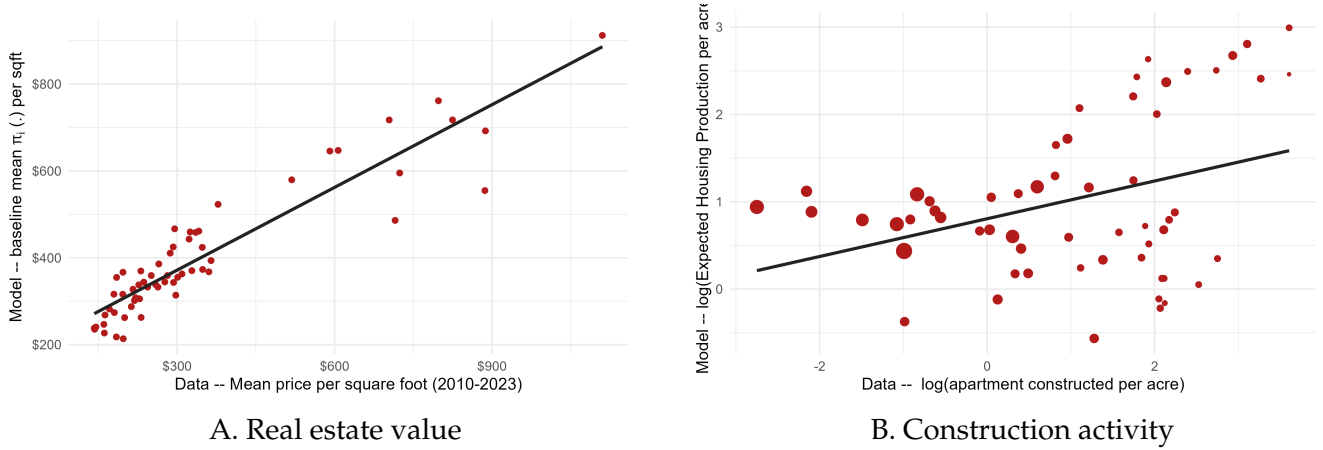
4.2.2 Alternative models

The final panel of Table III reports baseline outcomes under alternative model assumptions. The first row imposes a density constraint $F_i^* = \bar{F}_i$ upon redevelopment (i.e., developers do not choose the optimal density). Aggregate outcomes remain close to the baseline, although the sum of present values declines. This comparison shows that optimizing F_i^* matters mainly for surplus, not for citywide volumes. In New York City, zoning is often binding, so developers already choose densities close to the regulatory limit (Brueckner et al., 2024). Fixing $F_i^* = \bar{F}_i$ therefore trims gains from optimization but does not greatly alter output.

The second row shows the results of a model in which we not only permit redevelopment but also the renovation of the current asset. We detail this model extension in Appendix E and also present our main counterfactual policy results under this extended model. Renovation is the preferred option on many parcels, yet it has limited aggregate implications for housing supply because this option is preferred for parcels unlikely to redevelop. Aggregate housing supply remains essentially unchanged. Total expected production (renovation and reconstruction) equals 172,469 units, and 125,595 units net of demolitions. The larger difference between the two numbers indicates that, under this model, a greater share of the housing stock consists of renovated units. The expected citywide rent is marginally higher with renovation, reflecting quality improvements in the existing stock rather than a contraction of housing supply. Under this model extension, the present value of parcels also increases marginally, reflecting the additional option available to developers.

The third row of Table III shuts down redevelopment by setting the cost of construction

Figure II: Model Validation



Note: Panel A shows the relation between the log mean price per housing unit of Multi-Family buildings, using transaction from CoStar from 2010 to 2023, and the log of the model-implied mean price per unit in each community district. Panel B plots the log number of housing units constructed in each community district between 2016 and 2025, scaled by community district acreage, against the corresponding model-implied construction outcomes based on the 2015 calibration.

prohibitively high ($k_c = \infty$). We use this comparison to isolate the contribution of replacement relative to the supply effect, and thus apply the supply effect feedback ($\zeta_S + \zeta_A < 0$). In this case, the expected rent is 1.4% lower than in the baseline despite the supply being held fixed. Without redevelopment, parcels remain at $F_{B,i}$. With redevelopment, some parcels move to F_i^* . Because $\Gamma(F_i^*, 1, \alpha_{B,i}, L_i) > \Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i)$, new construction raises average rents by replacing older, lower-rent buildings with newer and taller ones. In the absence of redevelopment, rents fall because the stock deteriorates.

The final row of Table III removes the density constraint. Construction increases by a factor of 2.5. This result reflects a higher redevelopment payoff, since unconstrained density raises $\Gamma(F^*, 1, \alpha_{B,i}, L_i)$. We include the negative supply effect ($\zeta_S + \zeta_A < 0$) to avoid unrealistic infinite construction and ensure equilibrium. Even in this case, average rents rise to \$2,798. The supply response remains too small to offset the replacement effect.

Taken together, the results of Table III show that the main driver of outcomes is the replacement margin. Density and redevelopment constraints matter, but they operate largely through shifting the rent threshold R_i^{e*} and the redevelopment probability P_i^* , rather than through aggregate changes in realized supply.

4.3 Model Validation

Does the model predict real estate values? We compare predicted asset values to observed transaction prices to validate the model. The object of interest is the parcel value $\pi_i(\mathcal{P}_0; \Theta)$, which governs the redevelopment decisions. For each community district j , we compute the average model-implied value per housing unit:

$$\widehat{PricePerUnit}_j = \frac{1}{n_j} \sum_{i \in j} \frac{\pi_i(\mathcal{P}_0; \Theta)}{F_{B,i} L_i / \text{UnitSize}_i}.$$

The denominator converts parcel value into a per-unit measure using built density and local unit size UnitSize_i . Panel A of Figure II compares $\widehat{PricePerUnit}_j$ to observed prices per unit from CoStar transactions from 2010 to 2023. The correlation equals 0.93. The model captures cross-neighborhood variation in property values. This fit is central to our contribution because developers' redevelopment decisions rely on maximizing π_i . By matching cross-sectional variations in values, we show that the model ranks parcels correctly by their incentives to redevelop across policy environments.

Does the model predict real estate construction activity? We compare model-implied construction to observed construction to validate the model's predictions for redevelopment activity. The key object is the expected number of units constructed that depends on both the development probability $\mathbf{P}_i(\mathcal{P}_0)$ and the optimal density $F_i^*(\mathcal{P}_0)$. We measure observed construction using data on net housing units built by the private sector in each community district between 2016 and 2025 (NYC Planning). We recalibrate parcel characteristics to their 2015 values using the procedure described in Section 3. For each district j , the model-implied number of housing units constructed is

$$\widehat{\text{UnitNet}}_j(\mathcal{P}_0, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}_0) \frac{L_i(F_i^*(\mathcal{P}_0) - F_{B,i})}{\text{UnitSize}_i},$$

expressed in housing units. Citywide, our model predicts net construction of 210,609 housing units. There were 199,992 housing units net of demolition during this period. Panel B of Figure II compares $\widehat{\text{UnitNet}}_j(\mathcal{P}, h)$ to observed construction, with both measures scaled by district acreage. The model matches the cross-sectional distribution of construction across neighborhoods. The model assigns redevelopment to the correct locations by capturing variations in $\mathbf{P}_{ih}(\mathcal{P}_0) F_i^*(\mathcal{P}_0)$ across parcels. We provide this model fit by borough in Figure C6 of the Appendix. It shows an even greater fit for all boroughs except for the Bronx, where a large share of housing production during the 2016–2025 period was associated with programs such as the Harlem River Waterfront District.

5 Policy Counterfactuals

We now examine how alternative policy regimes reshape the housing stock through parcel-level redevelopment decisions. At $t = 0$, the city moves from the baseline environment $\mathcal{P}_0$ to a counterfactual regime $\mathcal{P} = [\alpha \geq 0, \phi \geq 0, \tau \geq 0]$, where α denotes the affordable housing requirement, ϕ the density bonus, and τ the duration of the post-redevelopment property tax exemption. A policy change affects outcomes only by changing the developer's optimization problem. It shifts the value of redevelopment relative to the continuation value of the existing structure, alters the optimal redevelopment threshold R^{e*} , and also changes the optimal redevelopment density F^* . Each developer's expected choice, in turn, changes expectations about future rental growth on nearby and citywide parcels.

We present all counterfactual results relative to the baseline. For example, the ten-year change in expected net housing production under $\mathcal{P}$ is

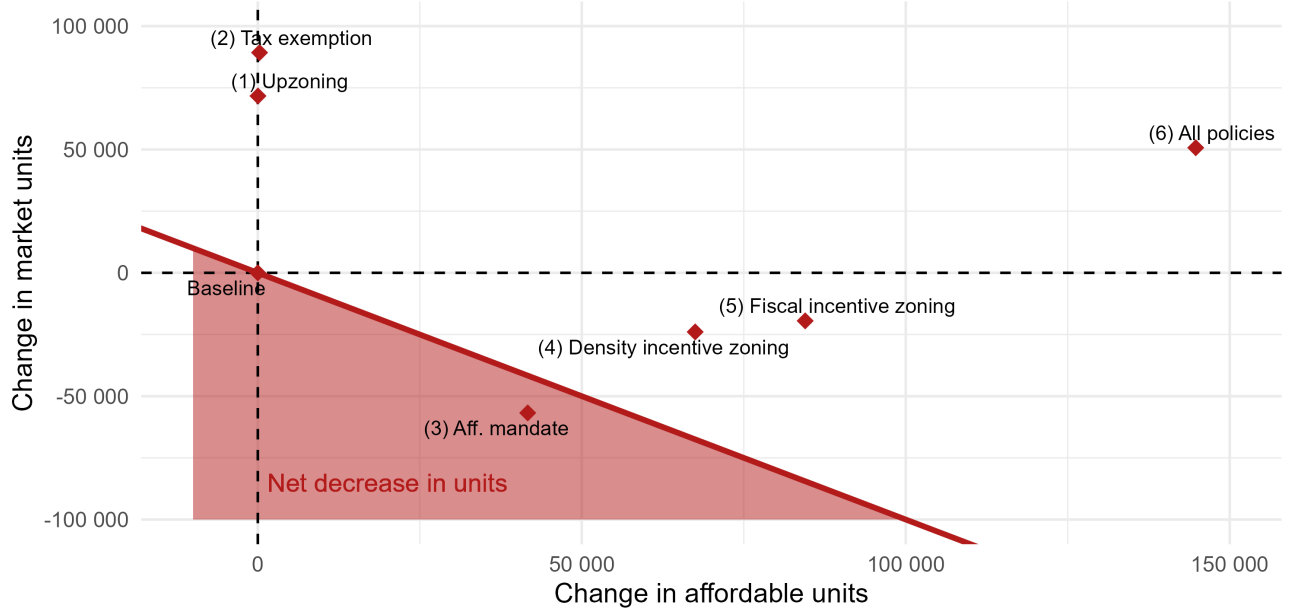
$$\Delta \text{UnitNet}(\mathcal{P}|t = 10) = \text{UnitNet}(\mathcal{P}|t = 10) - \text{UnitNet}(\mathcal{P}_0|t = 10)$$

We report monetary outcomes as percentage deviations from the baseline to facilitate comparisons across policy bundles. Table D3 in the Appendix reports corresponding numerical outcomes for each policy.

Although the model allows any combination of policy instruments, we focus on six policy environments that span the main approaches used in practice. Policy (1) is upzoning, which raises allowable density by 50%. Policy (2) is a 20-year property tax exemption. Policy (3) is an affordability mandate that requires 30% of units in new construction to be affordable. Policy (4) is a Density Incentive Zoning, which combines the affordability mandate with the density bonus. Policy (5) is Fiscal Incentive Zoning, which combines the affordability mandate with the tax exemption. Policy (6) is the combined incentive policy that stacks all three instruments.

We organize the results in three blocks. We first study housing production and distinguish between market-rate and affordable units. We then study expected rents and decompose rent changes into the model's four channels: the affordable share effect, the aggregate supply effect, the local amenity effect, and the replacement effect. We then turn to policy tradeoffs and show that policies that hold housing production fixed generally do not hold rents fixed. This non-coincidence sits at the core of the paper's economics. In a built-out city, policy does not operate on an abstract aggregate supply curve. It operates through the transformation of the housing stock by shifting which parcels are redeveloped, when they are redeveloped, and what replaces the existing stock.

Figure III: Policy Counterfactuals — Housing Production



Note: This scatter plot shows the expected change in affordable housing units (x-axis) and market-rate housing units (y-axis) relative to the baseline over a 10-year horizon. Policy (1) Upzoning corresponds to a 50% increase in allowable density; Policy (2) Tax exemption corresponds to a 20-year property tax exemption; and Policy (3) corresponds to a 30% affordability mandate. Policies (4), (5), and (6) combine these instruments.

5.1 Housing Production

5.1.1 Market-rate vs affordable units: aggregate effects

Figure III summarizes how policy affects aggregate housing production through the replacement margin by plotting changes in affordable units against changes in market-rate units. Each point corresponds to a policy regime. The negative 45-degree line provides a benchmark for interpreting how gains in affordable units map into changes in market-rate housing units. Movements along this line correspond to one-for-one reallocation between affordable and market-rate units. Along the line, aggregate supply remains unchanged. Consequently, points above the line indicate net expansions of the housing stock, while points below the line indicate net contractions relative to the baseline.

The figure shows that policy affects aggregate production primarily by shifting redevelopment thresholds $R_i^{e*}(\mathcal{P})$, which determine the set of parcels that redevelop. Policies that reduce $R_i^{e*}(\mathcal{P})$ activate redevelopment on marginal parcels and expand the housing stock. Policies that increase $R_i^{e*}(\mathcal{P})$ deactivate these parcels and contract supply. Unilateral upzoning ($\phi > 0$) and tax exemptions ($\tau > 0$) both increase market-rate housing, but they operate through distinct margins

of the redevelopment decision. Tax exemptions lower the redevelopment threshold $R_i^{e*}(\mathcal{P})$ by increasing post-construction returns. This effect raises redevelopment probabilities $P_i(\mathcal{P})$ and shifts mass along the extensive margin. Upzoning increases the value of redevelopment by allowing higher density $F_i^*(\mathcal{P})$. At the same time, it raises the option value of waiting. This effect increases $R_i^{e*}(\mathcal{P})$ for some parcels and delays redevelopment. As a result, both policies increase the supply of market-rate housing, but only tax exemptions systematically accelerate redevelopment across all parcels.

A standalone affordability mandate ($\alpha > 0$) shifts production toward affordable units but lies below the negative 45-degree line. The mandate reduces the net return to redevelopment by affecting $\Gamma(F, z, \alpha, L)$, thereby lowering post-development rental income. This increases the redevelopment threshold and reduces the probability of redevelopment. The contraction in total housing supply arises from fewer redevelopment events rather than smaller projects. Affordable housing production increases only because it is bundled with redevelopment that still occurs on high-value parcels.

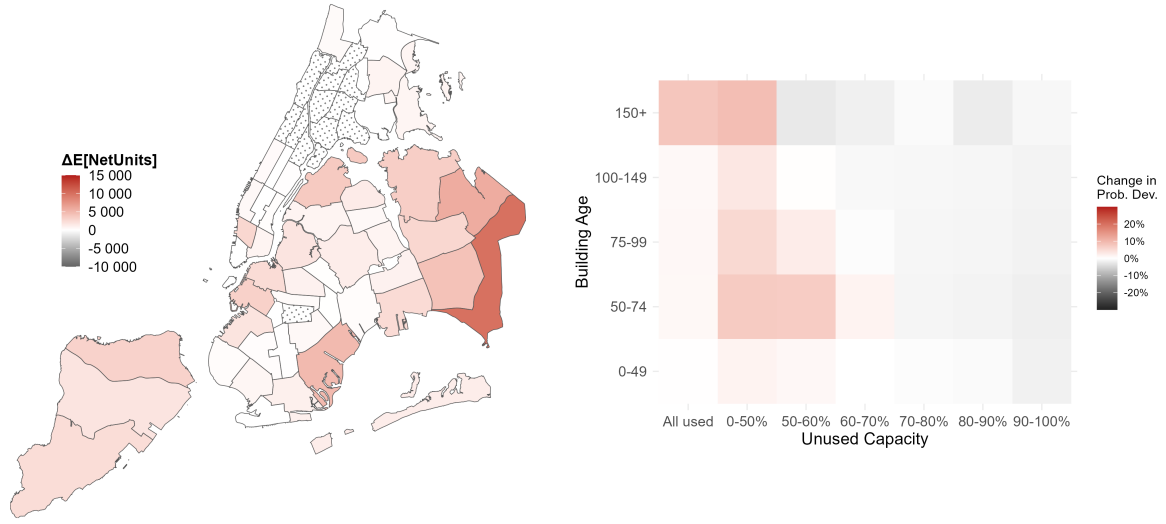
Policies that combine mandates with incentives occupy intermediate positions. Density Incentive Zoning (α, ϕ) and Fiscal Incentive Zoning (α, τ) both lie above the negative 45-degree line, indicating net expansions of the housing stock. In these regimes, incentives partially offset the increase in $R_i^{e*}(\mathcal{P})$ induced by the mandate. However, affordable housing gains arise primarily through reallocation. Increases in affordable units are accompanied by reductions in market-rate units within redeveloped projects, reflecting the constraint imposed by α . Only the Combined Incentive Policy (α, ϕ, τ) lies in the upper-right quadrant. This policy both lowers $R_i^{e*}(\mathcal{P})$ and increases $F_i^*(\mathcal{P})$ for most parcels. It activates redevelopment on a broad set of parcels while increasing density conditional on redevelopment. As a result, it expands both affordable and market-rate housing simultaneously.

5.1.2 Spatial heterogeneity is driven by parcel heterogeneity

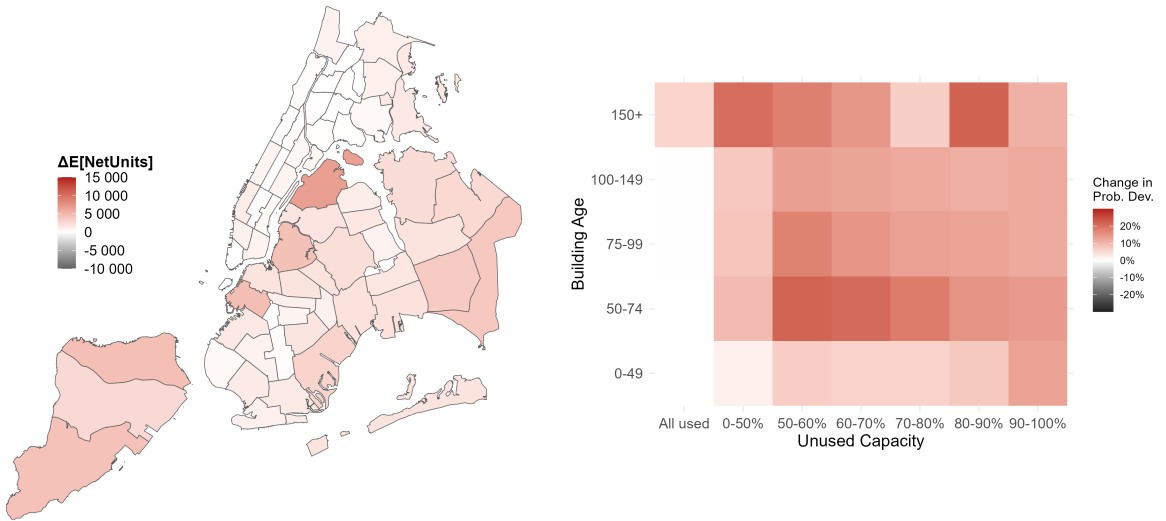
Figure IV shows that aggregate housing production masks substantial spatial heterogeneity. This heterogeneity arises endogenously from parcel-level differences in redevelopment incentives. Even when policy is applied uniformly across the city, redevelopment responses vary because parcels differ in their redevelopment threshold, in their existing density, and in their level of depreciation.

The left-hand panels of Figure IV map housing production changes across two benchmark policies: upzoning ($\phi > 0$) and tax exemptions ($\tau > 0$). Although both policies increase aggregate housing production, their spatial incidence differs sharply. Tax exemptions generate concentrated

Figure IV: Policy Counterfactuals — Housing Production Heterogeneity



A. Upzoning ($\phi = .50$)



B. Tax exemptions ($\tau = 20$)

Note: The maps report the expected net change in the number of housing units under each counterfactual policy, aggregated at the community-district level. The right-hand panels report changes in redevelopment probabilities, grouping parcels by deciles of unused density ($\bar{F}_i - F_{B,i}$) and building age. Panel A reports results for an upzoning policy allowing a 50% increase in maximum density, and Panel B reports results for a 20-year exemption from property taxes on newly developed buildings.

responses in areas with older buildings and substantial unused density, such as Western Brooklyn and Queens. Upzoning reduces expected housing production in the Bronx and Harlem while

increasing it mostly in Queens.

The right-hand panels of Figure IV make this mechanism explicit by grouping parcels into deciles of unused density and building age. Tax exemptions raise redevelopment probabilities for parcels with older structures and substantial unused density. Upzoning, by contrast, reduces redevelopment probabilities for many parcels in the same deciles. This divergence reflects that tax incentives generate additional housing production primarily on the extensive margin, while density bonuses operate on the intensive margin.

Thus, spatial heterogeneity arises because aggregate housing production results from individual redevelopment decisions. Policies do not shift all parcels uniformly. They move some parcels across their redevelopment threshold while leaving others unaffected. Tax incentives concentrate redevelopment in areas where parcels are already close to redevelopment. Density bonuses increase the value of redevelopment but often delay its timing. As a result, two policies that generate similar aggregate supply expansions can produce very different spatial patterns of development.

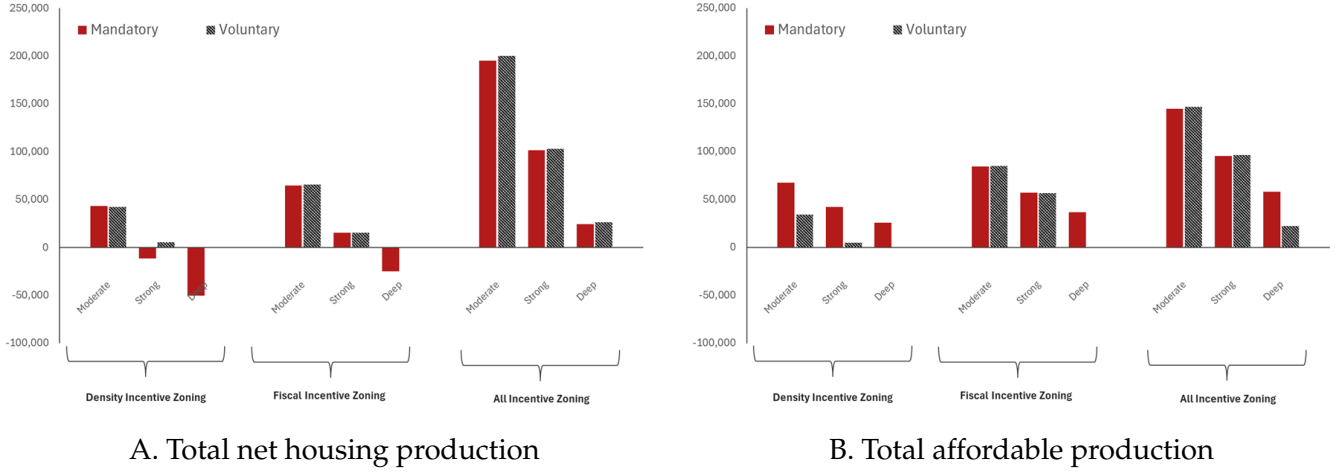
Targeted Counterfactuals. To further highlight the importance of aligning policy instruments with parcel characteristics, we re-conduct our main unilateral policy counterfactuals on three targeted subsets: the 20% of parcels with the highest underlying demand (top quintile of R_i), the 20% most obsolete parcels (bottom quintile of $z_{B,i}$), and the 20% least utilized buildings (bottom quintile of $F_{B,i}/\bar{F}_i$). Appendix D.4 presents the full results and shows that instrument effectiveness depends critically on parcel characteristics. Density bonuses generate the largest housing production when it targets high-value parcels, where additional floor area is most profitable. Tax exemptions are strongest when they target underbuilt parcels, where they lower the barrier to entry on the development margin.

5.1.3 Voluntary *Carrot & Stick* policies and affordability levels

We next examine how voluntary adoption and the depth of affordability requirements affect housing production under the *Carrot & Stick* policies. We model voluntary regimes by allowing developers to adopt a policy bundle $\mathcal{P}$ only if it increases property value relative to the baseline, that is, if $\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$. Parcels for which regulatory requirements dominate incentives remain under the baseline regulatory environment.¹⁶ This adoption rule implies that policy affects out-

¹⁶The voluntary-adoption counterfactual introduces a binary participation decision that can create discontinuities in the fixed-point iteration described in Appendix A.6. To ensure numerical stability, we smooth adoption beliefs during the iteration using a logit approximation and recover the binary adoption rule once the algorithm has converged. Appendix A.8 provides details.

Figure V: Affordability Levels and *Voluntary* Policy Interventions



Notes: The figures report changes in expected housing production relative to the baseline calibration $\mathcal{P}_0$ over a 10-year horizon. Panel A reports the change in total housing units net of demolitions, while Panel B reports the change in affordable housing units. Bars are grouped by policy regime, Density Incentive Zoning, Fiscal Incentive Zoning, and the Combined Incentive Policy, and by the depth of the affordability requirement. Affordability depth is parameterized by the rent discount ϵ_α , which scales the rental income of affordable units relative to market-rate units: $\epsilon_\alpha = 0.25$ (moderate), $\epsilon_\alpha = 0.50$ (strong), and $\epsilon_\alpha = 0.75$ (deep). Red bars correspond to mandatory policy regimes, in which all new construction must comply with the specified requirements. Stripped black bars correspond to voluntary policy regimes, in which developers choose whether to adopt the policy bundle or remain under the baseline regulatory environment. In voluntary counterfactuals, adoption decisions are made assuming all other parcels adopt the policy.

comes not only by shifting redevelopment incentives, but also by determining which parcels are exposed to those incentives. We provide additional results on the types of parcels that adopt different policy bundles in Appendix D.3. We also vary the depth of affordability requirements through the rent discount ϵ_α , holding the affordable-share requirement fixed. Figure V summarizes the results. Panel A reports changes in total net housing production, and Panel B reports changes in net affordable housing production. Within each policy environment, outcomes are shown for moderate, strong, and deep affordability requirements.

Under mandatory regimes, increasing ϵ_α reduces the value of redevelopment by lowering effective rental income. This lower post-development income reduces redevelopment incentives for all parcels. As a result, both total and affordable housing production decline. In several cases, total housing production becomes negative relative to the baseline even as affordable production remains positive, indicating that the contraction is driven by a reduction in market-rate units rather than through a collapse of affordable construction.

Under voluntary regimes, the same forces operate in addition to the participation condition. At moderate affordability levels, voluntary adoption yields housing production outcomes

similar to mandatory regimes because most parcels that redevelop also satisfy the participation condition. As affordability requirements deepen, participation declines sharply as an increasing fraction of parcels no longer satisfies $\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$. The policy becomes ineffective at the margin because it fails to induce redevelopment on parcels near their threshold. As a result, both total and affordable housing production converge back toward the baseline.

The Combined Incentive Policy provides an exception. By increasing the value of redevelopment through both density bonus ϕ and tax exemptions τ , it offsets the decline in $\pi_i(\mathcal{P})$ induced by high ϵ_α . This keeps the participation condition satisfied for a broader set of parcels. As a result, redevelopment persists even under deep affordability requirements ($\epsilon_\alpha = 0.75$). Relative to the other regimes, stacking incentives preserves redevelopment across a broader set of parcels, allowing affordable production to remain positive despite stringent requirements.

5.2 Expected Rent

5.2.1 Average expected housing rent and its decomposition

Figure VI reports the change in average citywide expected rent relative to the baseline $\Delta\text{Rent}(h; \mathcal{P}) = \text{Rent}(h; \mathcal{P})/\text{Rent}(h; \mathcal{P}_0) - 1$. Across counterfactuals, policies generate different rent responses even when they have similar effects on aggregate housing production. The difference arises because policy reshapes the housing stock through the replacement margin.

To formalize this mechanism, recall that expected rental income for parcel i at horizon h is given by:

$$\text{RentInc}_i(h, \mathcal{P}) = \mathbf{E}_{\mathbb{P}}\{R_{ih}\} \left[(1 - \mathbf{P}_i^*(h; \mathcal{P})) \Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}, L_i) + \mathbf{P}_i^*(h; \mathcal{P}) \Gamma(F_i^*, 1, \alpha, L_i) \right]$$

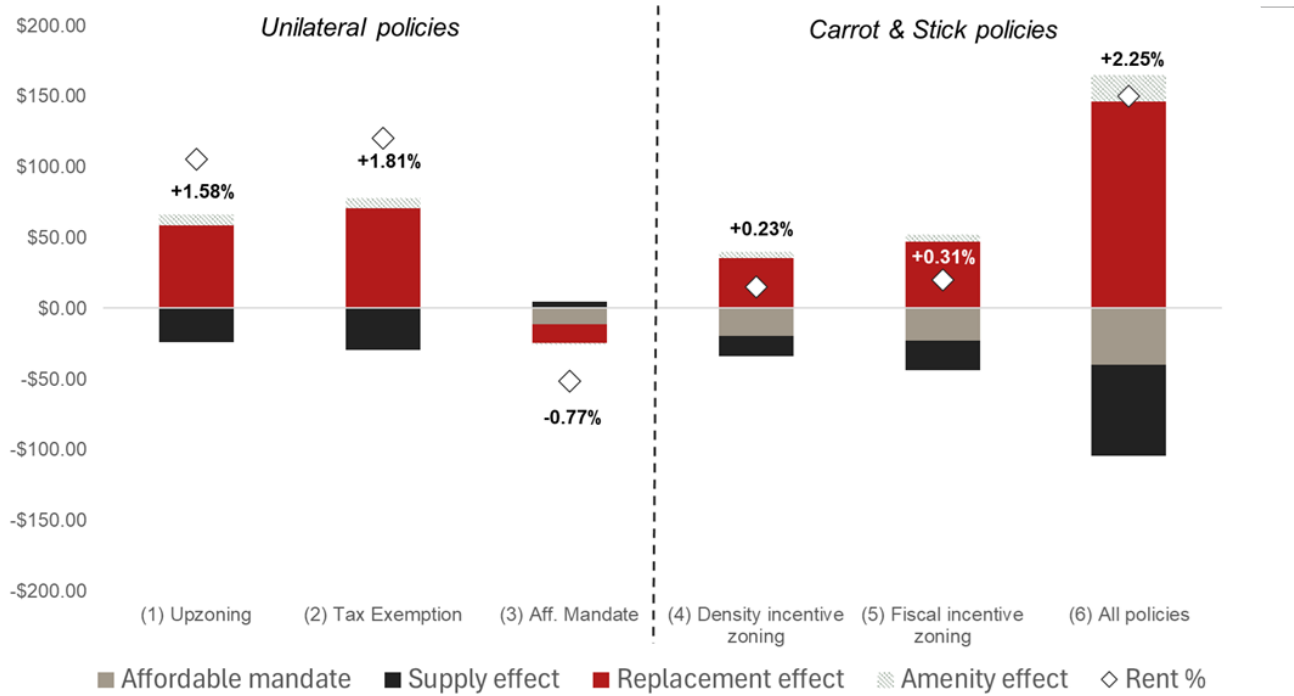
where $\mathbf{P}_i^*(h; \mathcal{P})$ is the probability of redevelopment and $F_i^*(\mathcal{P})$ is the optimal redevelopment density. This expression shows that policy affects rents through three channels. First, changes in supply around i and in the city impact expected potential rent $\mathbf{E}_{\mathbb{P}}\{R_{ih}\}$. Second, policy changes the probability that a parcel redevelops. Third, conditional on redevelopment, it replaces depreciated housing with new construction that generates higher rental income. We decompose changes in average expected rent into four components:

$$\Delta\text{Rent}(h; \mathcal{P}) = \Delta\text{Rent}^{\text{aff.}} + \Delta\text{Rent}^{\text{supply}} + \Delta\text{Rent}^{\text{amenity}} + \Delta\text{Rent}^{\text{replace}}$$

The affordable-share effect reflects the inclusion of below-market units in new construction, which lowers average rents. The aggregate supply effect captures downward pressure on rents

from higher citywide housing supply governed by the inverse-demand elasticity $\zeta_S < 0$. The amenity effect captures local rent increases from nearby development and is governed by $\zeta_A > 0$. The replacement effect captures the rent increase that occurs when redevelopment replaces depreciated units $z_{B,i} < 1$ with new units ($z = 1$).

Figure VI: Counterfactual Results — Expected Rents



Note: The points report the average expected rent change under each counterfactual policy relative to the baseline at a 10-year horizon. The stacked bars decompose this change into four components. The Affordable Mandate component isolates the direct effect of the affordable share α on rental income, holding expected construction activity fixed. The Supply effect captures the change in expected rents attributable to additional citywide housing supply. The Amenity effect is computed analogously and reflects the impact of local supply increases on neighborhood rents. The Replacement effect is defined as the residual component and captures rent changes associated with replacing older, depreciated structures with new construction, including changes in redevelopment probabilities and endogenous updates to expected rental growth. Policies (1)–(3) correspond to unilateral interventions—upzoning, tax exemption, and affordability mandate—while policies (4)–(6) combine these instruments.

Figure VI shows that the replacement effect is large enough to dominate the supply effect in many counterfactuals. Under policies that strongly increase redevelopment activity—such as upzoning (Policy [1]), tax exemptions (Policy [2]), and the combined policy (Policy [6])—average expected rents increase by 1.58%, 1.81%, and 2.25%, respectively.¹⁷ In these cases, rents rise because redevelopment replaces a larger share of the housing stock with new, higher-rent units.

By contrast, policies that combine affordability mandates with incentives generate modest

¹⁷Figure D2 provides spatial context by mapping changes in average expected rent across community districts. Neighborhoods experiencing larger increases in redevelopment activity—such as parts of Brooklyn, Queens, and Staten Island—exhibit larger rent responses, while districts with limited redevelopment display more muted changes.

rent increases. Under Density Incentive Zoning and Fiscal Incentive Zoning, average rents increase by 0.23% and 0.31%, respectively. These policies dampen the replacement effect by weakening redevelopment incentives, and they lower average rents directly by introducing below-market units.

Within our calibrated model, the decomposition shows that more housing supply does not necessarily lower average rents. For example, under upzoning, the replacement effect raises rents by approximately \$60 per unit per month, while the supply effect lowers rents by approximately \$24. The net effect is positive because redevelopment shifts the composition of the housing stock toward higher-rent units. Introducing an affordability mandate (Policy [4]) dampens this increase by reallocating a share of new construction to below-market units, which, together with the supply effect, offsets the replacement effect.

Unit normalization does not explain this result. Our model operates in per-square-foot terms, but we present results in housing units by dividing expected floor area by a fixed parcel-specific unit size. One might be concerned that the replacement effect is an artifact of this unit size normalization. We respond to this concern in two ways. First, we replicate the rent decomposition in square foot terms, which yields the same quantitative pattern, with the replacement effect dominating the supply effect across all policies (Figure D8). Second, we introduce a unit-size adjustment parameter β , defined as the ratio of the new unit size to the existing unit size, and reconduct our counterfactual analysis for different values of β . The results detailed in Appendix D.5 show that average expected rents always increase with additional supply induced by the policies. This holds even when new units are 25% smaller than the current stock.

5.2.2 Rent distribution

We next study how policy affects the rent distribution, not just its average. Table IV reports the share of expected housing supply in each rent quintile under policy $\mathcal{P}$, where quintile cutoffs are defined using the baseline distribution of rent per unit under $\mathcal{P}_0$. This normalization lets us interpret distributional changes as shifts in the composition of the housing stock:

$$\Delta \text{shr}_q(\mathcal{P}) = \text{shr}_q(\mathcal{P}) - \text{shr}_q(\mathcal{P}_0),$$

where $\text{shr}_q(\mathcal{P})$ denotes the share of housing units in rent quintile q .

Policies without affordability mandates push the housing stock toward both tails of the distribution. Under upzoning (Policy [1]) and tax exemptions (Policy [2]), the share of units in the top quintile increases. This reflects the expansion of new market-rate construction, which enters

Table IV: Counterfactual Results – Change in Rent Distribution

	Q1	Q2	Q3	Q4	Q5
Range	\$0–\$1,840	\$1,840–\$2,260	\$2,260–\$2,700	\$2,700–\$3,550	\$3,550+
Baseline	20%	20%	20%	20%	20%
	Unilateral Policy				
(1) <i>Upzoning</i> ($\phi = 50\%$)	+0.28	–0.11	–0.86	–0.85	+1.53
(2) <i>Tax exemption</i> ($\tau = 20$)	+0.23	–0.39	–1.23	–0.64	+2.02
(3) <i>Affordability mandate</i> ($\alpha = 30\%$)	+0.01	+0.08	+0.35	+0.33	–0.77
	Mandatory Carrot & Stick Zoning				
(4) <i>Density Incentive Zoning</i>	+0.24	+0.01	–0.27	–0.30	+0.32
(5) <i>Fiscal Incentive Zoning</i>	+0.23	–0.13	–0.53	–0.08	+0.52
(6) <i>All Policies</i>	+0.69	–0.50	–1.79	–1.02	+2.62

Notes: This table reports the 10-year horizon change in the share of expected housing supply falling within each rent quintile relative to the baseline calibration P_0 , expressed in percentage points. Quintile thresholds are defined from the baseline rent distribution. A positive value indicates a relatively larger share of units in that rent segment compared to baseline. By construction, each row sums to zero.

the housing stock at higher rent levels due to the absence of depreciation. At the same time, the share of units in the bottom quintile also rises modestly. That increase reflects the aggregate supply effect, which lowers rents for existing units. As a result, the distribution stretches outward, with gains at both the lower and upper ends and losses concentrated in the middle quintiles.

Policies that include affordability mandates (Policies [4] - [6]) reshape the distribution in a similar way, but they add a further compositional effect. Because of the requirement to have affordable units in new construction, the top quintile increases to a lesser extent. Yet the bottom quintile is roughly unaffected by the affordable mandate addition. To further examine this pattern, Figure D1 plots changes in the expected number of housing units across \$500 rent bins where positive (negative) values indicate increases (decreases) in the mass of units in each rent interval relative to the baseline. The figure decomposes these changes into market-rate units and units subject to affordability mandates.

One key result is that newly created affordable units do not concentrate in the lower tail of the rent distribution. Under both Density Incentive Zoning and Fiscal Incentive Zoning, most additional affordable units are located above the mean rent. This pattern follows directly from the redevelopment decision rule. Redevelopment occurs only on parcels where it remains profitable, typically in high-demand locations. In these locations, even discounted rents remain well above the lower tail of the distribution. In our model, where redevelopment requires private profitability, the existing housing stock continues to supply most units in the lower end of the rent distribution.

bution. Newly built affordable units occupy intermediate or upper portions of the distribution because they are tied to redevelopment on high-value parcels. As a result, increases in affordable housing supply do not necessarily improve affordability at the lower end of the rent distribution. New affordable units arise through redevelopment on high-value parcels, not through production targeted at the bottom of the market.

5.2.3 Robustness to demand elasticity assumptions

In the model, expected rents depend on both the housing stock and the reduced-form demand responses that map supply into rent changes. Developers internalize these effects when making redevelopment decisions. Two parameters govern these responses: the citywide inverse demand elasticity $\zeta_S < 0$ and the local amenity elasticity $\zeta_A > 0$, which captures how nearby development affects local rents. This subsection examines how alternative values of these elasticities affect rent and housing production outcomes.

When does the supply effect dominate rent outcomes? We first examine how alternative values of ζ_S affect rent outcomes and housing production. Figure D3 reports the change in average expected rent $\Delta\text{Rent}(h; \mathcal{P})$, and total housing production $\Delta\text{UnitNet}(h; \mathcal{P})$, as functions of ζ_S , focusing on upzoning (left panel) and tax exemptions (right panel). Under the baseline calibration, both policies raise average expected rents even though they expand housing supply. This result reflects the replacement effect. As a result, the composition effect dominates the aggregate supply effect. Absent any supply effects, this effect would even be stronger, as there would be no disincentive for developers when supply rises.

As $|\zeta_S|$ increases, the downward pressure from aggregate supply strengthens. For sufficiently large values of $|\zeta_S|$, the supply effect can offset the replacement effect and push average rents down. However, this occurs only when $|\zeta_S|$ approaches 2, an unrealistic parametrization in light of the current literature. At these values, redevelopment incentives weaken. Lower expected future rents increase the redevelopment threshold and reduces redevelopment probabilities and housing production.

As a result, parameter values that generate large rent declines also generate weak housing production responses. The constraint comes from endogenous rent feedback. Policies that put strong downward pressure on rents also reduce the expected return to redevelopment. Large rent declines therefore coincide with weaker redevelopment responses.

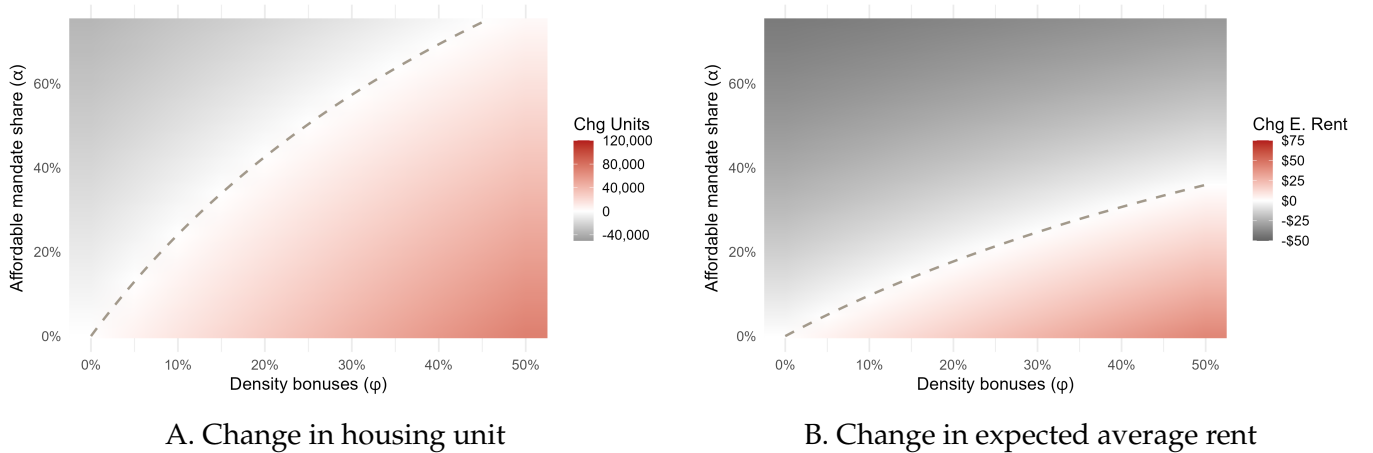
Amenity spillovers and spatial clustering. We next turn to local amenity effects. To quantify spatial clustering in rents, we compute Moran’s I statistic using inverse-distance weights with a

250-meter radius. Under the baseline policy, Moran's I equals 0.53, which indicates strong spatial correlation in rents across housing units. Figure D4 reports how this spatial correlation changes under upzoning and tax exemptions for different values of ζ_A . The two policies have opposite effects. Upzoning increases spatial correlation, while tax exemptions reduce it. This divergence reflects the different margins on which the two policies operate. Upzoning concentrates redevelopment in high-value areas, where new construction clusters near existing high-rent units. Tax exemptions target older, lower-density parcels, which are more prevalent in lower-rent neighborhoods. As ζ_A increases, each pattern becomes more pronounced. When ζ_A approaches the magnitude of $|\zeta_S|$, spatial correlation rises by approximately six percentage points under upzoning, corresponding to a 10 percent increase relative to the baseline.

The extent to which supply reduces rents depends on how rent growth responds to changes in the housing stock. Strong supply elasticities reduce rents but also discourage redevelopment by lowering expected returns. As a result, large rent reductions occur only when redevelopment activity is weak. At the same time, local spillovers shape where redevelopment occurs. Policies that concentrate development in high-rent areas increase spatial inequality in rents. Policies that target lower-value parcels reduce it.

5.3 Rent and Housing Production Trade-offs

Figure VII: Continuous Policies Outcomes - Density Incentive Zoning



Note: These heatmaps report changes in model outcomes under a Density Incentive Zoning program across continuous values of the density bonus ϕ (horizontal axis) and the affordable housing requirement α (vertical axis). All changes are measured relative to the status-quo policy $\mathcal{P}_0$ at the 10-year horizon. Panel A shows the net change in the expected number of housing units $\text{UnitNet}(P)$. Panel B reports the change in average expected rents per housing unit. Dotted contour lines indicate combinations of (ϕ, α) that leave the corresponding outcome equal to its baseline level under the status-quo policy $\mathcal{P}_0$.

The preceding analysis evaluates a discrete set of policy bundles. We now characterize the

trade-offs that arise when policy instruments vary continuously. Let policy be defined by $\mathcal{P} = (\alpha, \phi, \tau)$. The model maps each policy into a pair of outcomes for housing production and rents: $\{\Delta\text{UnitNet}(\mathcal{P}); \Delta\text{Rent}(\mathcal{P})\}$. Figure VII plots these outcomes for Density Incentive Zoning as a function of the density bonus ϕ (horizontal axis) and the affordability requirement α (vertical axis). Figure D11 reports analogous results for policies that combine tax exemptions τ with affordability mandates.

Changes in ϕ and α shift redevelopment incentives through their effect on the redevelopment threshold $R_i^{e*}(\mathcal{P})$ and optimal density $F_i^*(\mathcal{P})$. Two forces determine policy outcomes. The first operates through the extensive margin: policies that reduce $R_i^{e*}(\mathcal{P})$ increase redevelopment probabilities and expand the set of active parcels. The second operates through the intensive margin: policies that increase allowable density raise $F_i^*(\mathcal{P})$ conditional on redevelopment. These forces generally move in opposite directions across instruments.

Figure VII shows that these opposing forces generate a trade-off between housing production and rent outcomes. Policies that strongly increase redevelopment also increase the replacement of existing housing stock, which raises average rents through the replacement effect. Policies that limit redevelopment weaken the replacement effect but also reduce total housing production. This trade-off implies that policy cannot simultaneously maximize housing production and minimize average rents. Increasing supply through redevelopment raises the share of new, higher-rent units in the housing stock. Reducing rents instead requires limiting redevelopment or imposing affordability requirements that weaken the replacement effect. These objectives conflict because they operate through the same underlying decision: whether and when parcels redevelop.

Figure VII also illustrates that policy effects are not locally linear. Increasing ϕ and α proportionally does not produce proportional changes in outcomes. Nonlinearity arises because redevelopment responds to threshold crossings rather than marginal adjustments. Once a parcel crosses the redevelopment threshold, further increases in incentives have limited effects on its decision, while small changes near the threshold can have large effects.

Policy design involves a trade-off between activating redevelopment and controlling the composition of the housing stock. Policies that stimulate redevelopment increase supply but also speed up the replacement of low-rent units with higher-rent new construction. Policies that constrain redevelopment preserve lower-rent units but limit supply expansion. Because both outcomes are governed by the same redevelopment decision, policy cannot fully target them independently. This trade-off is a direct consequence of the replacement margin: the same redevelopment decisions determine both housing production and rent outcomes. We next quantify how these trade-offs map into the incidence of policy across landowners and the public sector.

6 Policy Incidence and the Cost of Mandated Affordable Units

Having established how policy affects redevelopment and rents through the replacement margin, we now examine how the same mechanisms determine the cost and incidence of affordability mandates. The analysis remains outside a welfare framework. Instead, the focus is on how policy-induced changes in redevelopment decisions translate into changes in private land values and in the fiscal position of the city. Because policies alter the timing and intensity of replacement, they determine how the costs of affordability mandates are distributed across agents.

6.1 Public and Private Costs of Policy Interventions

We evaluate how each policy shifts the distribution of costs between the public sector and private landowners. The empirical objects are the present value of property tax revenues and parcel values, which together capture how redevelopment responses are reflected in fiscal outcomes and in land value capitalization.

$$\Delta PV^{\text{Tax}}(\mathcal{P}) = \frac{\sum_i PV_i^{\text{Tax}}(\mathcal{P})}{\sum_i PV_i^{\text{Tax}}(\mathcal{P}_0)} - 1 \quad ; \quad \Delta PV(\mathcal{P}) = \frac{\sum_i \pi_i(\mathcal{P})}{\sum_i \pi_i(\mathcal{P}_0)} - 1 \quad (15)$$

Because policies shift redevelopment thresholds, they determine whether these costs are absorbed through fiscal channels or are capitalized into land values.

Table V reports both dimensions for each counterfactual policy. Upzoning (Policy [1]) modestly raises tax revenues by 0.05%, but reduces aggregate residential asset values by -1.20%. This decrease reflects the downward pressure of additional supply on existing properties, consistent with land-value dilution concerns following land-use deregulation (Ahlfeldt et al., 2026; Duranton and Puga, 2023). Tax exemptions (Policy [2]) generate the largest fiscal cost among unilateral interventions (-3.71%), reflecting foregone tax revenues during the exemption window. The affordability mandate (Policy [3]) has limited fiscal effects (-0.28%) but generates the smallest share of losers (14%), as reduced redevelopment activity limits supply expansion and supports expected rents. The Combined Incentive Zoning (Policy [6]) yields the largest decline in both parcel values (-2.74%) and tax revenues (-5.35%), alongside the widest dispersion of outcomes, with large gains among a small set of redeveloping parcels and substantial losses among the majority.

Heterogeneous Impact on Landowners. Aggregate statistics in Table V conceal substantial heterogeneity in how policies affect individual parcels. Across policies, between 84% and 88% of parcels experience declines in present value, while a small minority capture gains. Two forces account for this pattern. The supply effect operates broadly, reducing expected rental income and

Table V: Counterfactual Results – Costs of Interventions

		Private Gain/Loss				Public
		ΔPV	Shr. PV-losers	Avg. Gain	Avg. Loss	ΔPV^{Tax}
		(%)	(%)	(000's \$ per parcel)		(%)
Unilateral Policy						
(1)	Upzoning ($\phi = 50\%$)	-1.20	87	47.04	-35.73	0.05
(2)	Tax exemption ($\tau = 20$)	-0.93	85.9	110.31	-40.27	-3.71
(3)	Affordability mandate ($\alpha = 30\%$)	0.17	14.0	7.13	-18.57	-0.28
Mandatory Carrot & Stick Zoning						
(4)	Density Incentive Zoning	-0.85	87.5	30.45	-24.31	-0.40
(5)	Fiscal Incentive Zoning	-0.75	87.3	78.23	-29.15	-3.36
(6)	Combined Incentive Zoning	-2.74	83.6	117.64	-90.58	-5.35

Notes: This table reports the private and public cost outcomes under each counterfactual policy relative to the baseline calibration P_0 with a horizon of 10 years. Each row corresponds to a counterfactual regulatory environment $\mathcal{P}$. ΔPV reports the percentage change in the aggregate present value of residential parcels. The share of PV-losers is the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(P_0)$). Average gain and average loss report the average change in present value (in thousands of dollars per parcel) among PV-winners and PV-losers, respectively. ΔPV^{Tax} reports the percentage change in the present value of property tax revenues collected by the city.

generating diffuse losses. In contrast, the redevelopment option generates concentrated gains for parcels near the redevelopment threshold, where incentives shift the optimal timing decision.

Appendix Figure D12 illustrates this asymmetry under the Combined Incentive Policy. The distribution of present value changes is highly skewed. Losses cluster near zero, while gains exhibit a long right tail, reflecting large option value increases for parcels that switch redevelopment status. The average loss among losing parcels is \$750.58k per acre, compared to \$1,292.96k per acre among winners, implying a ratio of approximately 1.7. Gains are therefore concentrated among a limited set of parcels that become marginal under the policy. This uneven distribution aligns with political economy constraints emphasized in Diamond and McQuade (2019) and Fischel (2002).

6.2 Cost of Mandated Affordable Units

The previous results are not directly comparable across policies, as interventions differ in their effects on total housing supply. To isolate the cost of mandated affordable units, we focus on policy combinations that hold total expected net housing supply fixed, $\Delta \text{NetUnit}(h; \mathcal{P}) = 0$. Under this restriction, policies reallocate units from market-rate to affordable without changing aggregate supply. This isolates the incidence of one additional affordable unit from broader density effects. Abstracting from supply growth allows us to specifically set aside changes in con-

sumer surplus and in agglomeration externalities from densification (Duranton and Puga, 2020). Under this controlled comparison, a welfare analysis reduces to incidence on landowners and on the fiscal position of the public sector.

We construct the cost measure as follows. For each affordability mandate $\alpha \in \{0.01, \dots, 0.50\}$, we search over the incentive ϕ (holding $\tau = 0$) or τ (holding $\phi = 0$) that restores total housing supply to its baseline level. The existence of such a compensating incentive follows from the monotonicity of redevelopment incentives established in Appendix A.9. For a fixed mandate α , increasing either ϕ or τ weakly raises parcel values and expands the set of parcels that redevelop. As a result, aggregate housing supply is weakly increasing in incentives and continuous in aggregate. Panel A of Figure VIII shows the level of incentives required to maintain supply neutrality as the affordability mandate increases. For instance, replacing roughly 40,000 market-rate units with affordable ones (equivalent to an α of 25%) requires a density bonus of 10% or a tax exemption of 2.40 years. Both density bonuses and tax exemptions must increase with the mandate. The convexity of both schedules indicates that maintaining supply neutrality requires progressively larger incentives as affordability requirements increase.

With each policy bundle, we compute $\Delta\text{Aff.Net}(\mathcal{P})$, the change in expected affordable units, together with the private cost $\Delta PV(\mathcal{P})$ and the fiscal cost $\Delta PV^{\text{Tax}}(\mathcal{P})$. The cost per mandated affordable unit is then summarized as

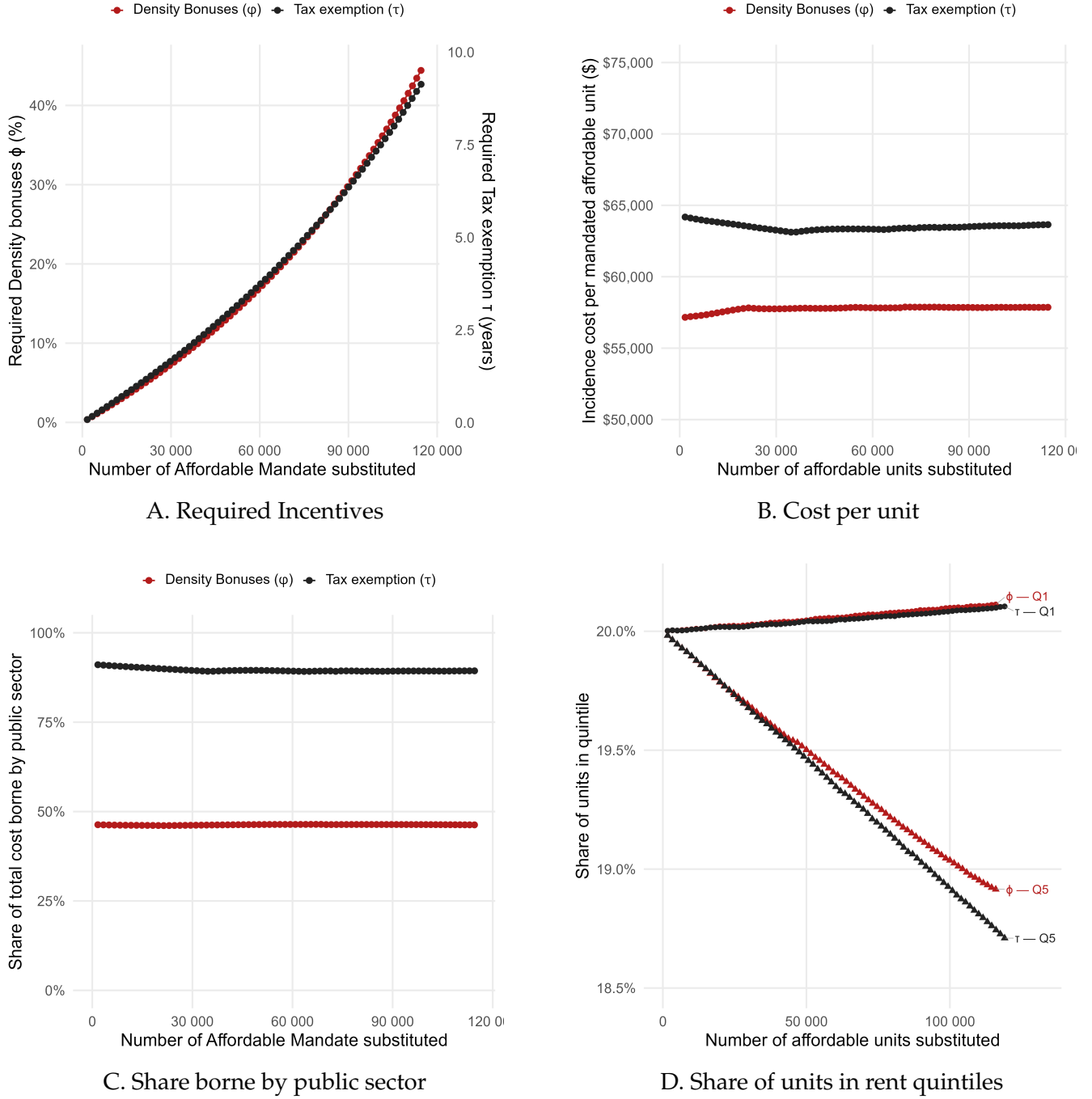
$$\omega(\mathcal{P}) = \frac{\Delta PV(\mathcal{P}) + \Delta PV^{\text{Tax}}(\mathcal{P})}{\Delta\text{Aff.Net}(\mathcal{P})}, \quad \Delta\text{Aff.Net}(\mathcal{P}) > 0. \quad (16)$$

The wedge ω measures the total cost of generating one additional affordable unit while holding aggregate supply fixed. It therefore isolates the cost of the mandate from broader rent or density effects, and provides a metric for comparing policies that differ in their design (ϕ versus τ) but are equated on their supply implications. We further decompose ω into private and public components. The private component captures the reduction in parcel values borne by landowners, and the public component captures the fiscal cost of the incentive:

$$\omega^{\text{private}}(\mathcal{P}) = \frac{\Delta PV(\mathcal{P})}{\Delta\text{Aff.Net}(\mathcal{P})}, \quad \omega^{\text{public}}(\mathcal{P}) = \frac{\Delta PV^{\text{Tax}}(\mathcal{P})}{\Delta\text{Aff.Net}(\mathcal{P})}.$$

Panel B reports the cost per mandated affordable unit, $\omega(\mathcal{P})$. The per-unit cost remains relatively stable across mandate levels and policy instruments, averaging approximately \$57,700 under density-based policies and \$63,500 under fiscal policies. Panel C shows how this cost is distributed between the public and private sectors. Under density-based policies, the burden is

Figure VIII: Cost of Mandated Affordable Housing



Note: These figures show outcomes against the number of affordable units substituted for market-rate units, holding total net housing supply fixed ($\Delta \text{NetUnit}(h; \mathcal{P}) = 0$). Panel A shows the level of the incentives required to achieve supply neutrality under various affordability mandates α . Panel B reports the welfare cost per mandated affordable unit, $\omega(\mathcal{P})$. Panel C shows the share of this cost borne by the public sector. Panel D reports the share of units in the bottom and top quintiles of the baseline rent distribution.

roughly evenly split. Landowners lose about \$30,000 per unit through negative capitalization, while the city experiences a comparable reduction in the present value of tax revenues. In contrast, under fiscal policies, most of the cost is borne by the public sector where up to 90% of the cost arises from reduced tax revenues. As the mandate increases, a larger share of the burden is absorbed by private landowners through declining parcel values, but this share remains limited, rarely exceeds 15%, and reaches only about \$6,800 per unit. This asymmetry reflects differences in how policy instruments allocate costs across agents, a distinction that has been linked to political constraints on land-use reform (Fischel, 2002). Fiscal instruments benefit landowners but are fiscally costly, while regulatory instruments preserve public revenues at the expense of private asset values.

Finally, Panel D shows the share of housing units in the bottom and top quintiles of the baseline rent distribution. Despite the increase in the number of affordable units, the share of units in the bottom quintile does not increase substantially. At most, this share rises to 20.08%, which highlights that increases in affordable housing supply do not translate into large changes at the lower end of the rent distribution. On the contrary, the share of units in the highest quintile decreases systematically with the strength of the affordability mandate. At $\alpha = 50\%$, nearly 35,000 units that would have fallen in the top quintile of the baseline were “transferred” to lower segments. This result reflects that new affordable units are generated through redevelopment on higher-value parcels, and therefore do not substantially increase the mass of units in the lowest segment of the rent distribution but deter the development of higher-end units.

6.3 Synthesis and Policy Implications

The results reflect a common mechanism: policies operate by shifting redevelopment thresholds rather than by mechanically adding units. Redevelopment is endogenous, forward-looking, and selective, targeting parcels with higher unused density and more depreciated structures. New construction therefore predominantly takes the form of replacement. As a consequence, supply expansions do not translate into lower observed rents. Rent outcomes reflect the interaction of supply effects, local spillovers, and the replacement effect as new units enter at higher quality and higher rents than the structure they replace.

Affordability outcomes depend on the composition of the housing stock. Low-rent housing is largely provided by aging structures, while affordability mandates reallocate access to new construction at discounted rents. This reallocation has limited impact on the lower tail of the rent distribution and entails non-trivial costs, borne either through land value reductions, fiscal expenditures, or both.

Policy design is therefore inherently multidimensional. Density bonuses expand supply but affect composition. Fiscal incentives accelerate redevelopment at a fiscal cost. Mandates generate targeted affordability but weaken redevelopment incentives and redistribute costs. Because redevelopment operates at the parcel level, policy mixes determine not only aggregate outcomes but also the distribution of costs across agents. These patterns arise directly from the replacement margin and link policy design to redevelopment timing, rent dynamics, and incidence.

7 Conclusion

We develop a forward-looking spatial framework in which housing policy operates through endogenous redevelopment decisions. The model embeds parcel-level investment choices in a spatial environment with aggregate and local spillovers. It evaluates how regulatory and fiscal instruments affect the timing, scale, and location of development. In this setting, housing supply operates through the transformation of the existing housing stock that arises from parcel-level redevelopment decisions. Developers choose whether and when to redevelop and how intensively to build. We refer to these choices as the replacement margin. Policy shifts this margin by changing the conditions under which parcels redevelop and what replaces the existing stock. This mechanism determines housing composition, rent dynamics, and policy incidence.

These implications follow directly from stock transformation through the replacement margin. Policies that expand construction do not mechanically lower average rents once changes in composition are taken into account, because redevelopment replaces lower-rent units with higher-rent housing. Affordability mandates primarily reallocate access to new construction rather than expanding the lowest segment of the rent distribution. These policies also generate non-trivial costs. The policy instrument determines whether those costs fall on landowners, the public sector, or both.

These findings underscore that housing policy design is inherently multidimensional. Instruments that appear similar in terms of aggregate supply can generate different redevelopment dynamics, rent dynamics, and incidence outcomes because they operate on different margins of the replacement decision. For that reason, policy evaluation should track not only how much housing is built, but also how the housing stock is transformed, which units are replaced, who gains access to new construction, and who bears the cost. Although we calibrate the model to New York City, the framework we provide can be applied to any housing market in which investment is irreversible, forward-looking, and spatially interconnected.

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Online Appendix

A Proofs

A.1 Foundations for Risk-Neutral Pricing of Rents.

To obtain analytical valuations of development options, we represent rent R_t as a geometric Brownian motion under the risk-neutral measure $\mathbb{Q}$. Rents are not traded claims, but their risk can be priced consistently through a common kernel linked to traded real-estate instruments. Residential property values are therefore the present value of net rents:

$$V_t = \mathbf{E}_{\mathbb{Q}} \left[\int_0^{\infty} e^{-ru} R_{t+u} du \right] = \frac{R_t}{r - \delta}, \quad \delta < r.$$

This parallels rent-capitalization approaches used to relate the rental and asset markets through a capitalization rate – or rent-to-price ratio (Büchler et al., 2021; DiPasquale and Wheaton, 1992). Thus, with V_t inheriting rent dynamics and applying Itô's Lemma gives

$$dV_t = \delta V_t dt + \sigma V_t dW_t.$$

Four assumptions underlie this representation: (1) asset value derives solely from net rental income; (2) rents equal net operating income, so there is no wedge with value; (3) rents are discounted at a constant r ; and (4) rent risk is spanned by traded real-estate instruments such as REITs or CMBS. The measure $\mathbb{Q}$ is thus viewed as a pricing device calibrated to observed valuations. Although real estate markets are incomplete, *good-deal bounds* narrow admissible measures, permitting valuation under $\mathbb{Q}$ without perfect replication (Cochrane and Saa-Requejo, 2000). It will follow from the empirical work that we calibrate δ from observed capitalization rates c_{obs} via $\delta = r - c_{\text{obs}}$, treating δ as a reduced-form risk premium.

A.2 Proof of Proposition 1: Parcel Value

Let $T_{R^e} \equiv \inf\{t \geq 0 : R_t = R^e\}$ with $R^e > R$, and $H_{R^e} \equiv T_{R^e} + d$, with $d > 0$.

Under $\mathbb{Q}$, $dR_t = \delta R_t dt + \sigma R_t dW_t^{\mathbb{Q}}$ with $r > \delta$, so

$$R_t = R \exp\left(\left(\delta - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{Q}}\right) = R e^{\sigma Z_t}, \quad Z_t \equiv bt + W_t^{\mathbb{Q}}, \quad b \equiv \frac{\delta - \frac{1}{2}\sigma^2}{\sigma}.$$

Define

$$a \equiv \frac{1}{\sigma} \ln\left(\frac{R^e}{R}\right) > 0, \quad T_{R^e} = \inf\{t \geq 0 : Z_t = a\} \equiv T_a.$$

Parcel value is

$$\pi(R, F_B, z_B, \alpha_B, F; \mathcal{P}, \Theta) = \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_0^{T_{R^e}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right\}}_{\text{Value of existing property}}$$

$$\begin{aligned}
& - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ e^{-rT_{Re}} \gamma(F_B) \right\}}_{\text{Demolition costs}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{T_{Re}}^{H_{Re}} e^{-ru} \kappa(F, \alpha) du \right\}}_{\text{Construction costs}} \\
& + \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{Re}}^{\infty} e^{-ru} \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Value of new property gross of tax}} - \underbrace{\mathbf{E}_{\mathbb{Q}} \left\{ \int_{H_{Re}+\tau}^{\infty} e^{-ru} \psi \Gamma(F, 1, \alpha) R_u du \right\}}_{\text{Property taxes after } \tau \text{ years of exemption}}. \quad (\text{A.1})
\end{aligned}$$

Drift removal. Define an auxiliary measure $\tilde{\mathbb{P}}$ by

$$\left. \frac{d\mathbb{Q}}{d\tilde{\mathbb{P}}} \right|_{\mathcal{F}_t} = \exp \left(bZ_t - \frac{1}{2}b^2t \right), \quad t \geq 0, \quad (\text{A.2})$$

so Z_t is standard Brownian motion under $\tilde{\mathbb{P}}$. Let

$$\lambda \equiv \sqrt{2r + b^2}, \quad \xi \equiv \frac{b - \lambda}{\sigma}.$$

1. Existing cash flows.

$$\mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_{Re}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right] = (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R \mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_a} e^{-ru} e^{\sigma Z_u} du \right].$$

Using (A.2),

$$\mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_a} e^{-ru} e^{\sigma Z_u} du \right] = \mathbf{E}_{\tilde{\mathbb{P}}} \left[\int_0^{T_a} e^{-\frac{\lambda^2}{2}u} e^{(\sigma+b)Z_u} du \right] = \mathbf{E}_{\tilde{\mathbb{P}}} \left[\int_0^{\infty} \dots du \right] - \mathbf{E}_{\tilde{\mathbb{P}}} \left[\int_{T_a}^{\infty} \dots du \right].$$

The resolvent identity yields

$$\mathbf{E}_{\tilde{\mathbb{P}}} \left[\int_0^{\infty} e^{-\frac{\lambda^2}{2}u} e^{(\sigma+b)Z_u} du \right] = \frac{1}{\lambda} \int_{-\infty}^{\infty} e^{-|y|\lambda} e^{(\sigma+b)y} dy = \frac{1}{r - \delta}. \quad (\text{A.3})$$

By the strong Markov property at T_a and $Z_{T_a} = a$,

$$\mathbf{E}_{\tilde{\mathbb{P}}} \left[\int_{T_a}^{\infty} e^{-\frac{\lambda^2}{2}u} e^{(\sigma+b)Z_u} du \right] = e^{(\sigma+b)a} \mathbf{E}_{\tilde{\mathbb{P}}} \left[e^{-\frac{\lambda^2}{2}T_a} \right] \cdot \frac{1}{r - \delta},$$

and the Laplace transform of T_a gives $\mathbf{E}_{\tilde{\mathbb{P}}} [e^{-(\lambda^2/2)T_a}] = e^{-\lambda a}$. Hence

$$\mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_a} e^{-ru} e^{\sigma Z_u} du \right] = \frac{1}{r - \delta} [1 - e^{(\sigma+b-\lambda)a}] = \frac{1}{r - \delta} \left[1 - \left(\frac{R^e}{R} \right)^{1+\xi} \right].$$

Therefore,

$$\mathbf{E}_{\mathbb{Q}} \left[\int_0^{T_{Re}} e^{-ru} (1 - \psi) \Gamma(F_B, z_B, \alpha_B) R_u du \right] = \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta} \left[R - R^e \left(\frac{R^e}{R} \right)^{\xi} \right]. \quad (\text{A.4})$$

2. Demolition.

$$\mathbf{E}_{\mathbb{Q}}[e^{-rT_{Re}}] = \mathbf{E}_{\tilde{\mathbb{P}}}[e^{-\frac{\lambda^2}{2}T_a}] = e^{-\lambda a} = \left(\frac{R^e}{R}\right)^{\xi},$$

so

$$\mathbf{E}_{\mathbb{Q}}[e^{-rT_{Re}}\gamma(F_B)] = \gamma(F_B)\left(\frac{R^e}{R}\right)^{\xi}. \quad (\text{A.5})$$

3. Construction costs on $[T_{Re}, T_{Re} + d]$.

$$\mathbf{E}_{\mathbb{Q}}\left[\int_{T_{Re}}^{T_{Re}+d} e^{-ru}\kappa(F, \alpha) du\right] = \kappa(F, \alpha)\mathbf{E}_{\mathbb{Q}}[e^{-rT_{Re}}] \int_0^d e^{-rs} ds = \frac{\kappa(F, \alpha)}{r}(1 - e^{-rd})\left(\frac{R^e}{R}\right)^{\xi}.$$

4. New value after completion. Conditioning on $\mathcal{F}_{T_{Re}}$ and using $R_{T_{Re}} = R^e$,

$$\mathbf{E}_{\mathbb{Q}}\left[\int_{T_{Re}+d}^{\infty} e^{-ru}R_u du\right] = \mathbf{E}_{\mathbb{Q}}[e^{-rT_{Re}}] \int_d^{\infty} e^{-rs}\mathbf{E}_{\mathbb{Q}}[R_{T_{Re}+s} | R_{T_{Re}} = R^e] ds = \left(\frac{R^e}{R}\right)^{\xi} \frac{R^e}{r-\delta} e^{-(r-\delta)d},$$

hence

$$\mathbf{E}_{\mathbb{Q}}\left[\int_{H_{Re}}^{\infty} e^{-ru}\Gamma(F, 1, \alpha)R_u du\right] = \Gamma(F, 1, \alpha)\frac{R^e}{r-\delta}e^{-(r-\delta)d}\left(\frac{R^e}{R}\right)^{\xi}. \quad (\text{A.6})$$

5. Taxes after τ . By the same argument with lower limit $d + \tau$,

$$\mathbf{E}_{\mathbb{Q}}\left[\int_{H_{Re}+\tau}^{\infty} e^{-ru}\psi\Gamma(F, 1, \alpha)R_u du\right] = \psi\Gamma(F, 1, \alpha)\frac{R^e}{r-\delta}e^{-(r-\delta)(d+\tau)}\left(\frac{R^e}{R}\right)^{\xi}. \quad (\text{A.7})$$

Closed form. Combining (A.4), (A.5), Step 3, (A.6), and (A.7) gives

$$\begin{aligned} \pi(R, R^e, F, F_B, z_B, \alpha_B; \mathcal{P}, \Theta) &= \Gamma(F_B, z_B, \alpha_B)\frac{1-\psi}{r-\delta}\left[R - R^e\left(\frac{R^e}{R}\right)^{\xi}\right] - \gamma(F_B)\left(\frac{R^e}{R}\right)^{\xi} \\ &\quad - \frac{\kappa(F, \alpha)}{r}(1 - e^{-rd})\left(\frac{R^e}{R}\right)^{\xi} + \Gamma(F, 1, \alpha)\frac{R^e}{r-\delta}e^{-(r-\delta)d}\left(\frac{R^e}{R}\right)^{\xi} \\ &\quad - \psi\Gamma(F, 1, \alpha)\frac{R^e}{r-\delta}e^{-(r-\delta)(d+\tau)}\left(\frac{R^e}{R}\right)^{\xi}. \end{aligned} \quad (\text{A.8})$$

A.3 Proof of Proposition 2: Optimal Expansion Barrier

Fix $F \leq \bar{F}$ and regime $\mathcal{P}$. Under deterministic time-to-build d , parcel value is

$$\begin{aligned} \pi &= \Gamma(F_B, z_B, \alpha_B)\frac{1-\psi}{r-\delta}\left[R - R^e\left(\frac{R^e}{R}\right)^{\xi}\right] - \gamma(F_B)\left(\frac{R^e}{R}\right)^{\xi} \\ &\quad - \frac{\kappa(F, \alpha)}{r}(1 - e^{-rd})\left(\frac{R^e}{R}\right)^{\xi} + \Gamma(F, 1, \alpha)\frac{R^e}{r-\delta}e^{-(r-\delta)d}\left(\frac{R^e}{R}\right)^{\xi} \end{aligned}$$

$$- \psi \Gamma(F, 1, \alpha) \frac{R^e}{r - \delta} e^{-(r-\delta)(d+\tau)} \left(\frac{R^e}{R} \right)^\xi, \quad (\text{A.9})$$

with $\xi = \frac{b-\lambda}{\sigma}$, $b = \frac{\delta - \frac{1}{2}\sigma^2}{\sigma}$, $\lambda = \sqrt{2r + b^2}$.

Define

$$K(F) = \gamma(F_B) + \frac{\kappa(F, \alpha)}{r} (1 - e^{-rd}),$$

$$\vartheta = \Gamma(F, 1, \alpha) \frac{e^{-(r-\delta)d} - \psi e^{-(r-\delta)(d+\tau)}}{r - \delta} - \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta}.$$

Then

$$\pi = \Gamma(F_B, z_B, \alpha_B) \frac{1 - \psi}{r - \delta} R + \left[\vartheta R^e - K(F) \right] \left(\frac{R^e}{R} \right)^\xi.$$

Maximizing in R^e is equivalent to maximizing

$$g(R^e) = \left[\vartheta R^e - K(F) \right] \left(\frac{R^e}{R} \right)^\xi.$$

FOC.

$$g'(R^e) = \vartheta \left(\frac{R^e}{R} \right)^\xi + \left[\vartheta R^e - K(F) \right] \frac{\xi}{R} \left(\frac{R^e}{R} \right)^{\xi-1}.$$

Setting $g'(R^e) = 0$ yields

$$\vartheta(1 + \xi)R^e = \xi K(F),$$

so, for $\vartheta > 0$,

$$R^{e*}(F) = \frac{\xi}{1 + \xi} \frac{K(F)}{\vartheta}.$$

SOC. Since

$$g(R^e) = R^{-\xi} \left(\vartheta (R^e)^{1+\xi} - K(F) (R^e)^\xi \right),$$

$$g''(R^e) = R^{-\xi} \left[\vartheta(1 + \xi) \xi (R^e)^{\xi-1} - K(F) \xi (\xi - 1) (R^e)^{\xi-2} \right].$$

Using the FOC,

$$g''(R^{e*}) = R^{-\xi} \vartheta (1 + \xi) (R^{e*})^{\xi-1}.$$

Under $r > \delta$, $\xi < -1$, hence $(1 + \xi) < 0$ and $g''(R^{e*}) < 0$. Thus R^{e*} is a strict local maximizer.

A.4 Proof of Proposition 3: Probability of Development

Under the physical measure $\mathbb{P}$, rent follows

$$dR_t = g(h; \mathcal{P}) R_t dt + \sigma R_t dW_t^\mathbb{P}, \quad R_t = R \exp \left((g(h; \mathcal{P}) - \frac{1}{2} \sigma^2) t + \sigma W_t^\mathbb{P} \right).$$

Define

$$X_t = \frac{1}{\sigma} \ln \left(\frac{R_t}{R} \right) = m(h; \mathcal{P}) t + W_t^\mathbb{P}, \quad m(h; \mathcal{P}) = \frac{g(h; \mathcal{P}) - \frac{1}{2} \sigma^2}{\sigma}.$$

Let $R^{e*} > R$ and

$$a = \frac{1}{\sigma} \ln\left(\frac{R^{e*}}{R}\right) > 0.$$

The initiation time is

$$T_{R^{e*}} = \inf\{t \geq 0 : R_t = R^{e*}\} = \inf\{t \geq 0 : X_t = a\} \equiv T_a.$$

For Brownian motion with drift, the first-passage distribution satisfies

$$\mathbf{P}_{\mathbb{P}}(T_a \leq t) = \mathcal{N}\left(\frac{m(h; \mathcal{P})t - a}{\sqrt{t}}\right) + \exp(2m(h; \mathcal{P})a) \mathcal{N}\left(-\frac{m(h; \mathcal{P})t + a}{\sqrt{t}}\right), \quad t > 0.$$

Applying this result to the stopping time $T_{R^{e*}}$ gives the redevelopment probability $\mathbf{P}_{\mathbb{P}}(T_{R^{e*}} \leq t)$.

With deterministic construction length d , completion occurs at $H = T + d$. Therefore, for horizon h ,

$$\mathbf{P}_{\mathbb{P}}(H \leq h) = \begin{cases} \mathbf{P}_{\mathbb{P}}(T \leq h - d), & h > d, \\ 0, & h \leq d. \end{cases}$$

Substituting $t = h - d$ yields

$$\mathbf{P}_{\mathbb{P}}(H \leq h) = \mathcal{N}\left(\frac{m(h; \mathcal{P})(h - d) - a}{\sqrt{h - d}}\right) + \exp(2m(h; \mathcal{P})a) \mathcal{N}\left(-\frac{m(h; \mathcal{P})(h - d) + a}{\sqrt{h - d}}\right),$$

which establishes the result.

A.5 Optimal Density Conditional on a Fixed-expansion Threshold

Fix R^e and deterministic time-to-build d (so $H_{R^e} = T_{R^e} + d$). From (A.8), the terms that depend on F can be written as

$$\pi(F; R^e) = \text{const} - D(d) \kappa(F, \alpha) + M(d, \tau) \Gamma(F, 1, \alpha),$$

where const is independent of F and

$$D(d) \equiv \frac{1 - e^{-rd}}{r} \left(\frac{R^e}{R}\right)^{\xi}, \quad M(d, \tau) \equiv \frac{e^{-(r-\delta)d}(1 - \psi e^{-(r-\delta)\tau})}{r - \delta} R^e \left(\frac{R^e}{R}\right)^{\xi},$$

with $\xi = (b - \lambda)/\sigma$ as in the main text. Using the calibration functional forms from subsection 3.1

$$\kappa(F, \alpha) = \frac{k_c L}{d} (1 - \alpha \eta_{\alpha}) F^{1+\eta_F}, \quad \Gamma(F, 1, \alpha) = L (1 - \alpha \epsilon_{\alpha}) F^{1+\epsilon_F},$$

the objective becomes $\pi(F; R^e) = \text{const} - K_1 F^{1+\eta_F} + K_2 F^{1+\epsilon_F}$ with

$$K_1 = D(d) \frac{k_c L}{d} (1 - \alpha \eta_{\alpha}), \quad K_2 = M(d, \tau) L (1 - \alpha \epsilon_{\alpha}).$$

Assuming an interior optimum with $\eta_F \neq \epsilon_F$, the first-order condition is

$$-(1 + \eta_F) K_1 F^{\eta_F} + (1 + \epsilon_F) K_2 F^{\epsilon_F} = 0,$$

which yields

$$\begin{aligned} F^*(R^e) &= \left[\frac{(1 + \epsilon_F)K_2}{(1 + \eta_F)K_1} \right]^{\frac{1}{\eta_F - \epsilon_F}} \\ &= \left[\frac{rd}{k_c(1 - e^{-rd})} \cdot \frac{e^{-(r-\delta)d}(1 - \psi e^{-(r-\delta)\tau})}{r - \delta} \cdot R^e \cdot \frac{(1 + \epsilon_F)(1 - \alpha\epsilon_\alpha)}{(1 + \eta_F)(1 - \alpha\eta_\alpha)} \right]^{\frac{1}{\eta_F - \epsilon_F}} \end{aligned}$$

As in the main text, once R^{e*} is substituted back as a function of F , the joint choice of (R^e, F) generally does not admit a closed-form solution. The optimal density must therefore be obtained numerically from the reduced-form objective $\pi(R, R^{e*}(F), F, \cdot)$.

A.6 Iterative Algorithm for the Spatial Fixed Point

The equilibrium drift vector is defined as the fixed point of a mapping from expected housing supply to parcel-level redevelopment decisions and back to aggregate supply. Policy-induced construction alters expected supply, which modifies the rent dynamics entering both the valuation problem and the redevelopment probability. These changes affect optimal redevelopment timing and density and therefore expected supply. Equilibrium requires consistency between supply adjustments and developers' responses. This appendix describes the fixed-point algorithm under policy $\mathcal{P}$ at horizon h .

Measures. Under the pricing measure $\mathbb{Q}$, rent evolves as $dR_t = \delta_i R_t dt + \sigma_i R_t dW_t^{\mathbb{Q}}$, where the risk-neutral drift δ_i governs valuation and determines optimal redevelopment thresholds and densities.

Under the physical measure $\mathbb{P}$, rent evolves as $dR_t = g_i R_t dt + \sigma_i R_t dW_t^{\mathbb{P}}$, where the physical growth rate g_i governs expected rent growth and redevelopment probabilities.

Initialization. Let δ'_i denote the baseline pricing drift implied by the cap rate and let g'_i denote the baseline physical rent growth rate. Initialize

$$\delta_i^{(0)} = \delta'_i, \quad g_i^{(0)} = g'_i, \quad \Delta S^{(0)} = 0, \quad \Delta N_i^{(0)} = 0.$$

Iteration $k \geq 1$. Given the previous supply adjustments, compute the common supply-induced drift adjustment

$$\Delta \delta_i^{(k)} = \frac{1}{h} \zeta_S \Delta S^{(k-1)} + \frac{1}{h} \zeta_A \Delta N_i^{(k-1)}.$$

Update both the pricing drift and the physical growth rate:

$$\delta_i^{(k)} = \delta'_i + \Delta \delta_i^{(k)}, \quad g_i^{(k)} = g'_i + \Delta \delta_i^{(k)}.$$

Using $\delta_i^{(k)}$, solve the parcel valuation problem under $\mathbb{Q}$ to obtain the optimal redevelopment trigger and density, $R_i^{e*,(k)}$ and $F_i^{*,(k)}$. Using the physical growth rate $g_i^{(k)}$, compute the probability of

redevelopment

$$\mathbf{P}_i^{(k)}(H \leq h) = \mathbf{P}_{\mathbb{P}}\left(T_i^{(k)} \leq h - d\right).$$

Expected supply on parcel i is therefore

$$s_i^{(k)} = \mathbf{P}_i^{(k)} F_i^{*,(k)} L_i + (1 - \mathbf{P}_i^{(k)}) F_{B,i} L_i.$$

Aggregate and neighborhood supply changes are updated as

$$\Delta S^{(k)} = \ln\left(\sum_i s_i^{(k)}\right) - \ln\left(\sum_i s_i^{(0)}\right), \quad \Delta N_i^{(k)} = \ln\left(\sum_{j \in \mathcal{L}_i} s_j^{(k)}\right) - \ln\left(\sum_{j \in \mathcal{L}_i} s_j^{(0)}\right).$$

where $\mathcal{L}_i = \{j : \|x_j - x_i\| \leq 500 \text{ ft}\}$ denotes the set of parcels constituting the neighborhood of parcel i , where $\|x_j - x_i\|$ is the euclidean distance between centroids.

Convergence. The algorithm iterates until $|\Delta S^{(k)} - \Delta S^{(k-1)}| < \text{tol}$. At convergence the equilibrium parameters satisfy

$$\delta_i(h; \mathcal{P}) = \delta'_i + \Delta \delta_i(h; \mathcal{P}), \quad g_i(h; \mathcal{P}) = g'_i + \Delta \delta_i(h; \mathcal{P}),$$

which aligns supply adjustments with developers' optimal redevelopment responses.

A.7 Proof of Corollary 1: Expected Rental Incomes

Under the physical measure $\mathbb{P}$, rent evolves as

$$dR_t = g_i(h; \mathcal{P}) R_t dt + \sigma R_t dW_t^{\mathbb{P}}, \quad R_t = R_i \exp\left(\left(g_i(h; \mathcal{P}) - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{P}}\right).$$

Since $W_t^{\mathbb{P}} \sim \mathcal{N}(0, t)$ and $\mathbf{E}_{\mathbb{P}}\{e^{\sigma W_t^{\mathbb{P}}}\} = e^{\frac{1}{2}\sigma^2 t}$, the expected rent level at horizon h is $\mathbf{E}_{\mathbb{P}}\{R_{ih}\} = R_i \exp(g_i(h; \mathcal{P})h)$.

Let $\mathbf{P}_i^*(H_{Re^*} \leq h; \mathcal{P})$ denote the completion probability defined in Proposition 3. With deterministic construction time d , completion occurs at $H_{Re^*} = T_{Re^*} + d$, so that $\mathbf{P}_i^*(H_{Re^*} \leq h) = \mathbf{P}_i^*(T_{Re^*} \leq h - d)$ for $h > d$ and 0 otherwise.

Conditional on the redevelopment outcome, if redevelopment has not occurred, rental income at horizon h equals $\Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}) R_{ih}$, and $\Gamma(F_i^*, 1, \alpha) R_{ih}$ if redevelopment is completed. Taking expectations under $\mathbb{P}$ yields

$$\begin{aligned} \mathbf{E}_{\mathbb{P}}\{RentInc_{ih}\} &= \left[1 - \mathbf{P}_i^*(H_{Re^*} \leq h; \mathcal{P})\right] \times \Gamma(F_{B,i}, z_{B,i}(h), \alpha_{B,i}) \times \mathbf{E}_{\mathbb{P}}\{R_{ih}\} \\ &\quad + \mathbf{P}_i^*(H_{Re^*} \leq h; \mathcal{P}) \times \Gamma(F_i^*, 1, \alpha) \times \mathbf{E}_{\mathbb{P}}\{R_{ih}\} \end{aligned}$$

which corresponds to Equation (10).

A.8 Fixed-Point Computation under Voluntary Policy Adoption

In the voluntary-adoption counterfactual, developers choose whether to adopt the policy bundle $\mathcal{P}$ or remain under the baseline regime $\mathcal{P}_0$. Parcel i adopts the policy if and only if it increases project value:

$$\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0).$$

Because equilibrium is computed as a fixed point, adoption decisions depend on expectations about other parcels' participation. The resulting binary rule introduces a discontinuity in the mapping from supply adjustments to aggregate supply, which can generate oscillations in the iteration.

To stabilize the algorithm, we smooth adoption beliefs during the fixed-point iteration using a logit approximation. The perceived probability that parcel i adopts the policy is

$$q_i(\mathcal{P}) = \frac{1}{1 + \exp\left(-\frac{\pi_i(\mathcal{P}) - \pi_i(\mathcal{P}_0)}{\nu}\right)}. \quad (\text{A.10})$$

The probabilities $q_i^{(k)}(\mathcal{P})$ enter the supply update step of the fixed-point iteration described in Appendix A.6. Expected supply on parcel i during iteration k is

$$s_i^{(k)} = q_i^{(k)}(\mathcal{P}) s_i^{(k)}(\mathcal{P}) + (1 - q_i^{(k)}(\mathcal{P}_0)) s_i^{(k)}(\mathcal{P}_0).$$

This smoothing replaces the discontinuous adoption rule during the iteration and yields a continuous mapping from drift adjustments to aggregate supply. Once the fixed point has converged, the smoothing device is removed and the final adoption decision reverts to the deterministic rule

$$\text{parcel } i \text{ adopts } \mathcal{P} \iff \pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0).$$

The parameter ν governs the degree of smoothing. As $\nu \rightarrow 0$, equation (A.10) converges to the deterministic adoption rule.

A.9 Feasibility and Compensating Incentives

This appendix characterizes the properties of parcel value that underlie the feasibility and participation results discussed in Section 6.

Proposition 1 (Continuity, Monotonicity, and Feasibility Thresholds) *Fix a mandate intensity $\alpha > 0$. Let parcel value be defined as in Proposition 1, where for each (α, ϕ, τ) the developer chooses optimal redevelopment timing and density:*

$$\pi_i(\alpha, \phi, \tau) = \max_{0 \leq F \leq \bar{F}(\phi)} \pi(R, R^{e*}(F; \alpha, \phi, \tau), F, F_B, z_B, \alpha_B).$$

Then: (i) $\pi_i(\alpha, \phi, \tau)$ is continuous in (ϕ, τ) on the admissible policy domain; (ii) $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in the fiscal incentive τ ; (iii) $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in the density bonus ϕ through expansion of the feasible set $F \leq \bar{F}(\phi)$; (iv) Define $\varsigma_i(\phi, \tau) = \pi_i(\alpha, \phi, \tau) - \pi_i(\mathcal{P}_0)$. Then the compensating set $\mathcal{C}_i(\alpha) = \{(\phi, \tau) : \varsigma_i(\phi, \tau) \geq 0\}$ is monotone in incentives: if $(\phi, \tau) \in \mathcal{C}_i(\alpha)$ and $(\phi', \tau') \geq (\phi, \tau)$ componentwise, then $(\phi', \tau') \in \mathcal{C}_i(\alpha)$; (v) Feasibility is characterized by parcel-specific thresholds: the

boundary $\{(\phi, \tau) : \varsigma_i(\phi, \tau) = 0\}$ separates infeasible and feasible policy combinations.

The proof proceeds in steps:

Step 1: Continuity. For any fixed construction density F , parcel value $\pi(\cdot)$ is given in Proposition 1. This expression is continuous in (ϕ, τ) on the admissible domain. The optimized value $\pi_i(\alpha, \phi, \tau)$ is obtained by maximizing this continuous function over the compact set $F \in [0, \bar{F}(\phi)]$. Standard arguments imply that the resulting value function is continuous in (ϕ, τ) .

Step 2: Monotonicity in τ . From (4), a longer tax exemption reduces the present value of property taxes paid after redevelopment. Holding F fixed, this increases project value. Since the developer optimizes over F , the optimized value $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in τ .

Step 3: Monotonicity in ϕ . A higher density bonus increases the upper bound $\bar{F}(\phi)$ on feasible construction density. This expands the feasible set of the developer's optimization problem. Since the objective is maximized over this set, enlarging the feasible set cannot reduce the optimized value. Hence $\pi_i(\alpha, \phi, \tau)$ is weakly increasing in ϕ . Note that although a higher density bonus raises optimized parcel value, it may also increase the optimal expansion threshold $R^{e*}(F)$ and delay redevelopment. This affects development probabilities but does not overturn the monotonicity of the value function.

Step 4: Compensating set. Define $\varsigma_i(\phi, \tau) = \pi_i(\alpha, \phi, \tau) - \pi_i(\mathcal{P}_0)$. By Steps 1–3, ς_i is continuous and weakly increasing in each argument. The compensating set $\{(\phi, \tau) : \varsigma_i(\phi, \tau) \geq 0\}$ is therefore a superlevel set of a continuous, monotone function. It follows that if $(\phi, \tau) \in \mathcal{C}_i(\alpha)$, any $(\phi', \tau') \geq (\phi, \tau)$ also belongs to $\mathcal{C}_i(\alpha)$.

Step 5: Threshold structure. Continuity of ς_i implies that the boundary $\{(\phi, \tau) : \varsigma_i(\phi, \tau) = 0\}$ separates the feasible and infeasible regions. This boundary defines parcel-specific thresholds for incentives required to restore feasibility.

B Additional Comparative Statics of the Model

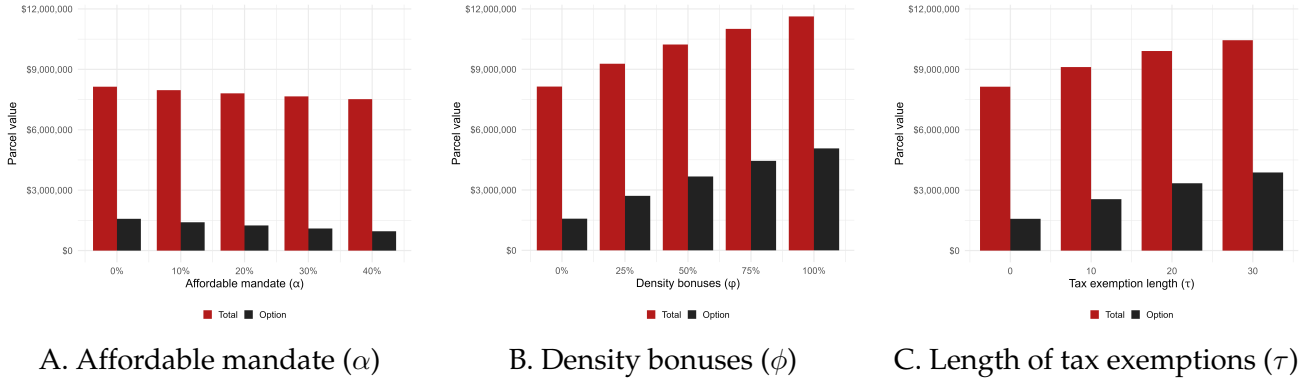
In this appendix, we present additional results and figures illustrating how our model responds to changes in parameters. We first provide more detail on how policy instruments affect model outcomes on a single parcel, report the complete set of comparative statics, and discuss the importance of parcel heterogeneity. We also provide an extended discussion of the role of non-instantaneous construction duration in an investment model under uncertainty. We finally describe how spatial spillovers operate in a stylized city where unexpected supply randomly arrives.

Unless specified otherwise, the comparative statics for the single-parcel are obtained for a parcel close to its redevelopment threshold built at 66% of its capacity, with $F_B = 2$ and $\bar{F} = 3$, an age of 150 years, an expected rental growth rate of 4% (total return of 8% and a caprate of 4%), an annual potential rent per sqft of 50, and a lot size of 2,500 sqft. All other elasticities correspond to the values shown in Table I.

B.1 Policy Instruments Effects

In Table B1, we report changes in parcel value and in the probability of redevelopment associated with the enactment of a single policy. The table shows how these results vary with key elasticity assumptions. Figure B1 highlights the effects of each policy instrument on the value of the redevelopment option, and Figure B2 shows the outcomes when policies are combined.

Figure B1: Policy Instruments Influence the (Re)development Option Value



Note: These figures show the effects the policy instruments on the present value of the current building and the present values of the redevelopment option. This option value is calculated as the difference between the total parcel value and its counterpart, where no development is allowed. These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, a potential rent of 100, and a lot size of 2,500 sqft. All other parameters and elasticities are shown in Table I.

Affordable mandate (α). Mandating affordable units on a given parcel unambiguously reduces both parcel value and the probability of (re)development, because of the decline in the redevelopment option value (Panel A of Figure B1). For instance, Table B1 shows that a 20% mandate lowers parcel value by 0.93% and development probability by 2.19 percentage points. The impact is stronger in high-demand areas where the same 20% mandate leads to a 1.52% drop in value and a 4.75 percentage-point reduction in development probability. Effects are also stronger for parcels

closer to the redevelopment threshold (i.e., with more unused density and older buildings). If the rental discount on affordable units doubles (ϵ_α), a 20% mandate reduces property values by 1.99% and development probability by 5.02 percentage points, because deeper rent discounts further reduce the incentive to develop. In contrast, if developers can reduce construction costs for affordable units (higher η_α), the 20% mandate only modestly lowers property values, suggesting that cost savings can partially offset the mandate's negative effects.

Table B1: Policy's Effects on Single Parcel Outcomes and Sensitivity Analyses

	Change in Property Value (%)	Change in Prob. Dev. (p.p.)
<u>Mandating 20% of Affordable Housing ($\alpha = 0.20$)</u>		
<i>Baseline</i>	-0.93	-2.19
<i>Higher potential rent ($R = 1.25R$)</i>	-1.52	-4.75
<i>Parcel closer to redevelopment ($F_0 = F_0/2$ and $age = 2age$)</i>	-3.08	-3.07
<i>Higher affordable rental discount ($\epsilon_\alpha = 2\epsilon_\alpha$)</i>	-1.99	-5.02
<i>Higher affordable construction discount ($\eta_\alpha = 2\eta_\alpha$)</i>	-0.57	-0.69
<u>Increase Maximum Allowed Density by 50% ($\phi = 0.50$)</u>		
<i>Baseline</i>	5.91	3.08
<i>Higher potential rent ($R = 1.25R$)</i>	9.72	6.10
<i>Parcel closer to redevelopment ($F_0 = F_0/2$ and $age = 2age$)</i>	7.95	-12.82
<i>Higher return to height ($\epsilon_F = 1.5\epsilon_F$)</i>	7.36	4.64
<i>Higher cost from height ($\eta_F = 1.5\eta_F$)</i>	1.57	-0.43
<u>20-years Property Tax Exemption ($\tau = 20$)</u>		
<i>Baseline</i>	5.15	14.95
<i>Higher potential rent ($R = 1.25R$)</i>	8.48	25.68
<i>Parcel closer to redevelopment ($F_0 = F_0/2$ and $age = 2age$)</i>	20.10	21.43
<i>Higher property tax rate ($\psi = 1.5\psi$)</i>	7.92	19.48

Notes: This table reports the change in property value and the change in the probability of development within 10 years with respect to the policy instruments. These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and an annual potential rent of \$50 per square foot. All other parameters and elasticities are shown in Table I.

Density bonuses (ϕ). Although density bonuses unambiguously increase parcel value through a higher redevelopment option value (Figure B1), they may also delay redevelopment as developers may require stronger market potential to justify the higher cost of construction. As an example, the second panel of Table B1 shows that a 50% increase in allowable density ($\phi = 0.5$) raises parcel value by 5.91% and development probability by 3.08 percentage points. In high-demand areas, property values and development probability increase more strongly with ϕ . For parcels closer to the redevelopment threshold, the parcel value still increases, but the probability of redevelopment declines sharply as developers would rationally wait to construct a taller building rather than build a lower one earlier. If the marginal benefit of height is higher (ϵ_F), a ϕ of 50% raises property values by 7.36% and development probability by 4.64 percentage points.

In contrast, if we set a higher height elasticity of construction cost, the parcel value increases modestly with the density restriction relaxation, and redevelopment probabilities remain essentially unchanged. These results highlight that the impact of density policy depends critically on construction technology (Ahlfeldt et al., 2026).

Tax exemptions (τ). Offering tax exemptions consistently increases parcel values and accelerates (re)development. The increase in value arises entirely from the redevelopment option, since the exemption applies only to new construction (Figure B1). As exemplified by the third panel of Table B1, a 20-year exemption increases parcel value by 5.15% and development probability by 14.95 percentage points. These effects are amplified for parcels closer to redevelopment, where probabilities rise by 21.43 percentage points, and in high-demand areas, where values increase by 8.48% and redevelopment probabilities by 25.68 percentage points. As expected, the impact of exemptions grows with higher property tax rates.

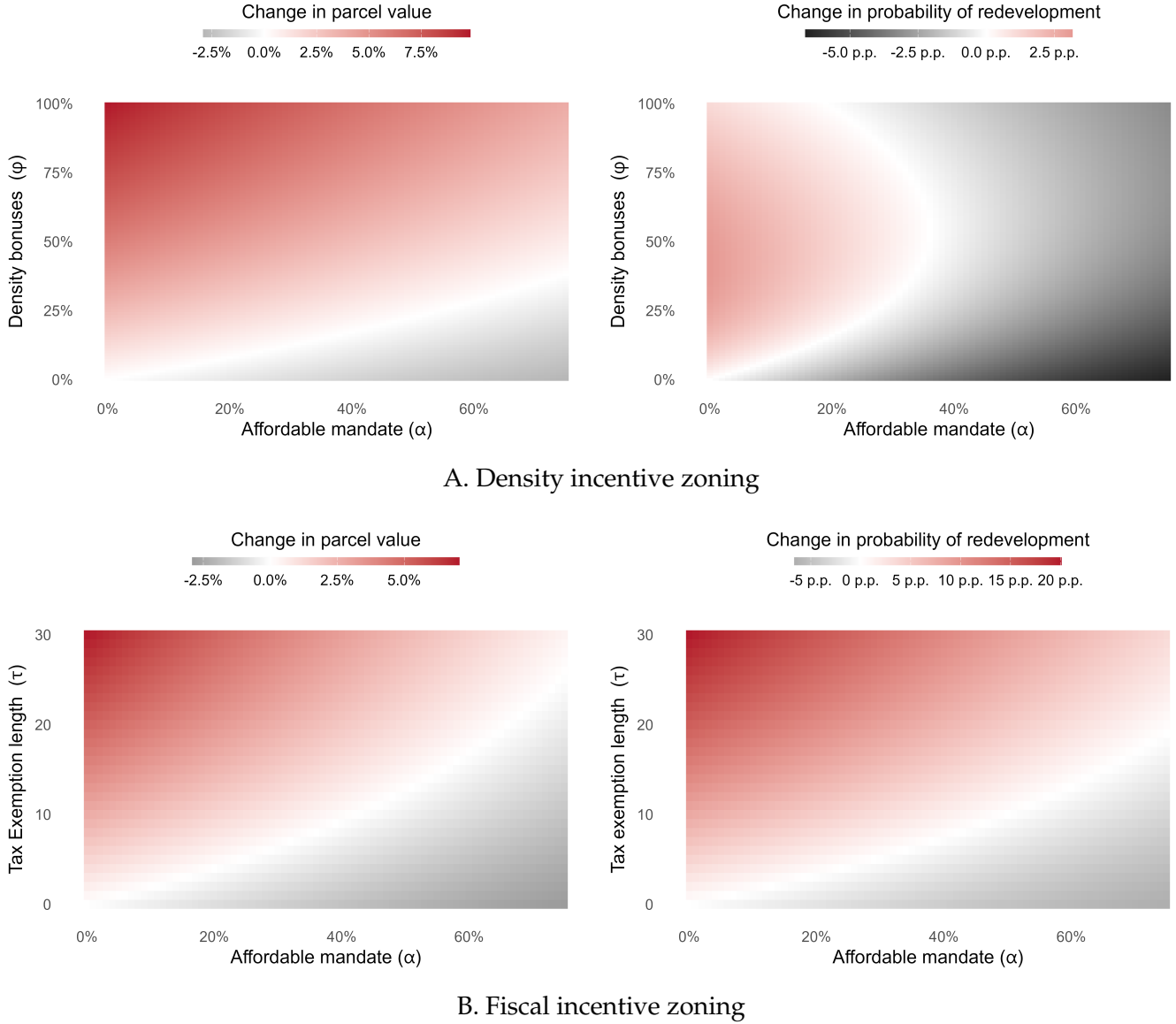
B.2 Parcel Heterogeneity and the Probability of Development

In this subsection, we show how the model predicts parcel value and redevelopment probability as functions of the parcel's characteristics: built form (age and current FAR) and location, which in our framework operates primarily through potential rent (R) and expected rent growth (g). These results highlight that heterogeneity in age, density, and local demand fundamentals is central to explaining spatial variation in redevelopment activity and, therefore, essential for understanding both the distributional and aggregate effects of policy interventions. Figure B4 further illustrates marginal effects with respect to the remaining parameters of the parcel-level model.

Panel A of Figure B3 shows that parcels with low initial built density (F_B) have lower property values than more intensively developed sites but exhibit the highest redevelopment probabilities, reflecting the larger embedded option value of underbuilt parcels. As current density approaches its maximum, the potential for value creation through rebuilding declines sharply, even though the standing property retains a high value. Panel B shows that redevelopment incentives rise with building age. Older structures face greater physical and economic obsolescence, which increases the value of the redevelopment option, while property values decline with age as rental income from existing buildings deteriorates. Together, these panels show that redevelopment becomes more likely when the expected gains from rebuilding outweigh the opportunity cost of preserving the existing structure—an essential mechanism for understanding redevelopment activity across the city.

Spatial heterogeneity further amplifies these dynamics. In our framework, spatial variation operates through potential rent (R) and caprate (c), which together summarize local demand fundamentals. Panels C and D show that higher rent levels (R) or lower caprate (i.e., higher expected rental growth) both increase parcel value and sharply raise redevelopment probabilities, with effects that intensify at higher R and δ . Small differences in neighborhood demand can therefore translate into disproportionately large differences in redevelopment activity, illustrating how micro-level heterogeneity aggregates into uneven spatial patterns of urban growth.

Figure B2: Trade-offs Between Carrots and Sticks in Simple Parcel Analysis

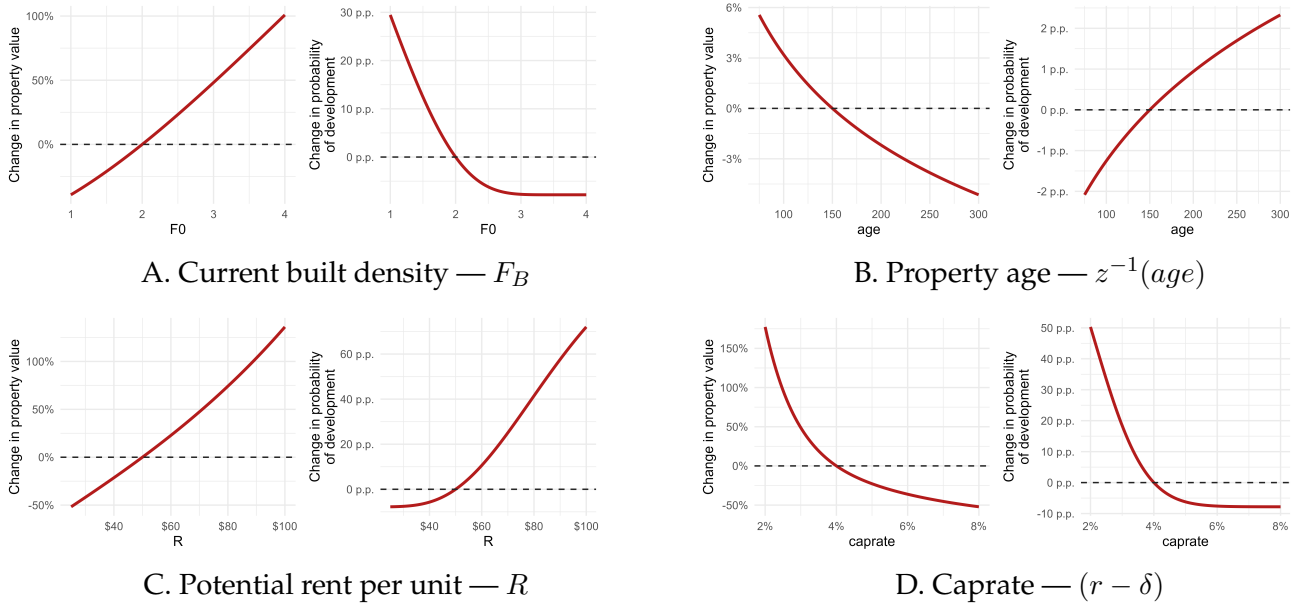


Note: These figures show the effects of combining an affordable mandate with either density bonuses (Panel A) or with a property tax exemption (Panel B) on the change in the property value (left panels) and the change in the probability of development completion within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and an annual potential rent of 50. All other parameters and elasticities are shown in Table I.

B.3 Relevance of Non-instantaneous Construction Duration

The impact of uncertainty on investment behavior changes fundamentally when construction is not instantaneous. Panels A and B of Figure B5 show the probability of development for various construction duration d and cost shifter k_c , respectively. The development probability increases with uncertainty for large d but decreases with uncertainty with instantaneous construction. For long construction periods d , developers hesitate at low uncertainty, waiting for clearer signals. Once uncertainty is high enough, the probability of investment rises sharply.

Figure B3: Effects of key Parcel Heterogeneity on Value and Probability of Development



Note: These figures show the effects of changing the one input parameter on the change in the property value (left panels) and the change in the probability of development starting within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and a potential rent of 50. All other parameters and elasticities are shown in Table I.

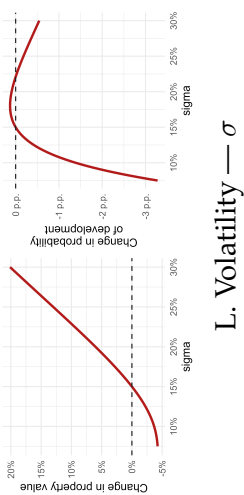
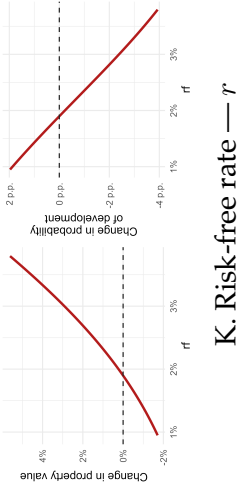
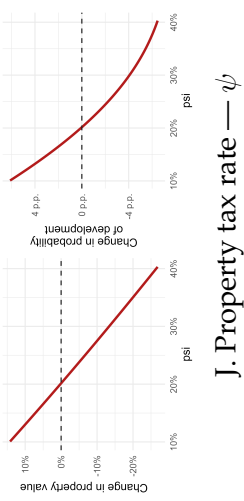
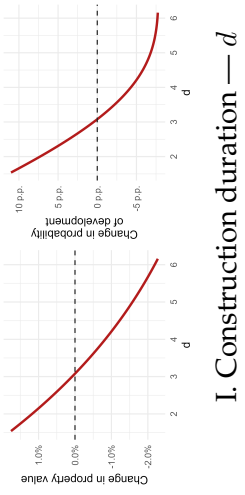
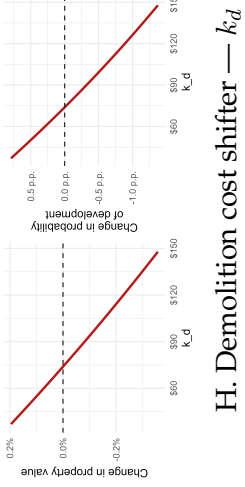
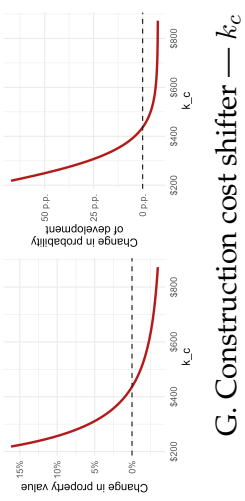
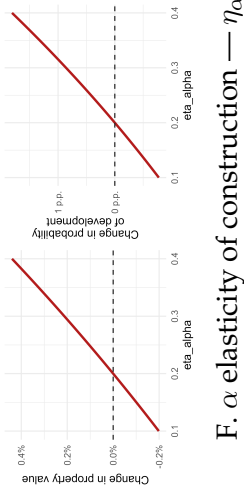
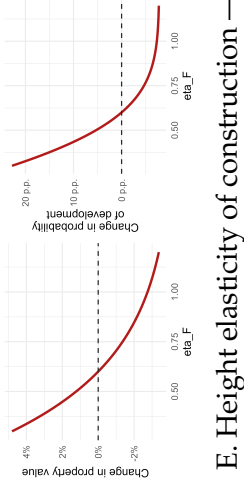
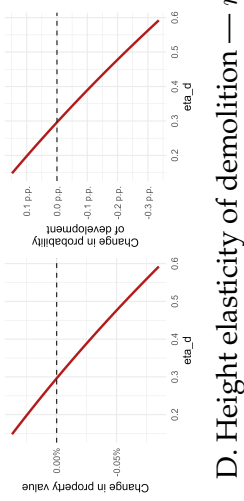
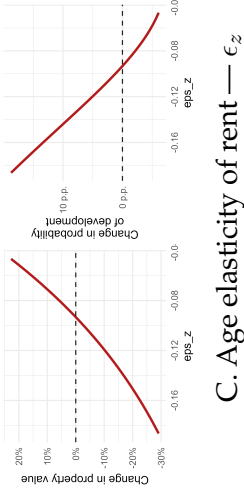
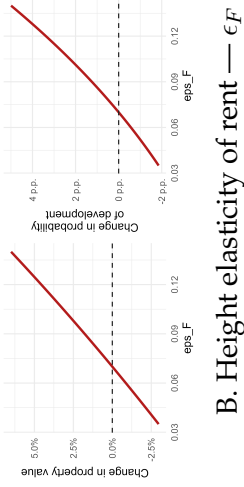
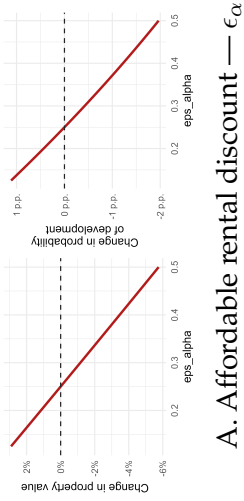
This pattern differs from the *incremental* investment framework in Bar-Ilan and Strange (1999), where uncertainty lowers investment intensity at every stage by discouraging additional capital, paralleling the results of shorter d in Panel A. Unlike with *lumpy* investment, with *incremental* investment, firms make many small decisions over time, each negatively affected by uncertainty. In our setting of *lumpy and lengthy* investment, uncertainty alters timing, but once investment occurs, intensity is unaffected. Models that assume instantaneous or incremental investment, therefore, tend to overstate development probabilities and inflate housing supply forecasts, that variation in construction costs cannot capture.

B.4 Spatial Responses to an Exogenous Supply Shock

We show how an exogenous increase in supply on a subset of parcels propagates across space through the local spillover and citywide supply channels. This exercise isolates how the spatial feedbacks reallocate redevelopment incentives across locations before we calibrate the model to New York City. We force 1% of parcels to redevelop. This shock raises housing supply by 0.92% of the existing stock. We then study how this shock changes construction probabilities on all other parcels. Developers endogenously update their redevelopment decisions in response to both aggregate and local supply effects. We run this experiment in a stylized city where potential rents R_i rise from South to North, while the building stock becomes older and more obsolete from West to East.

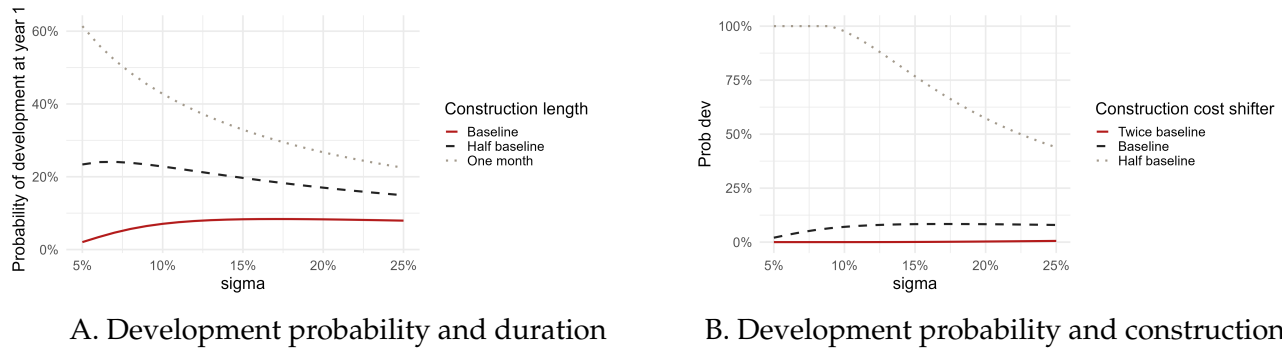
Panel A of Figure B6 shows how construction probabilities change under our preferred elasticity specification ($\zeta_S = -0.50$ and $\zeta_A = 0.10$). Construction probabilities fall on average. The aggregate supply increase lowers expected rents and weakens redevelopment incentives across

Figure B4: Effects of Parameter Changes on Parcel Value and Probability of Development



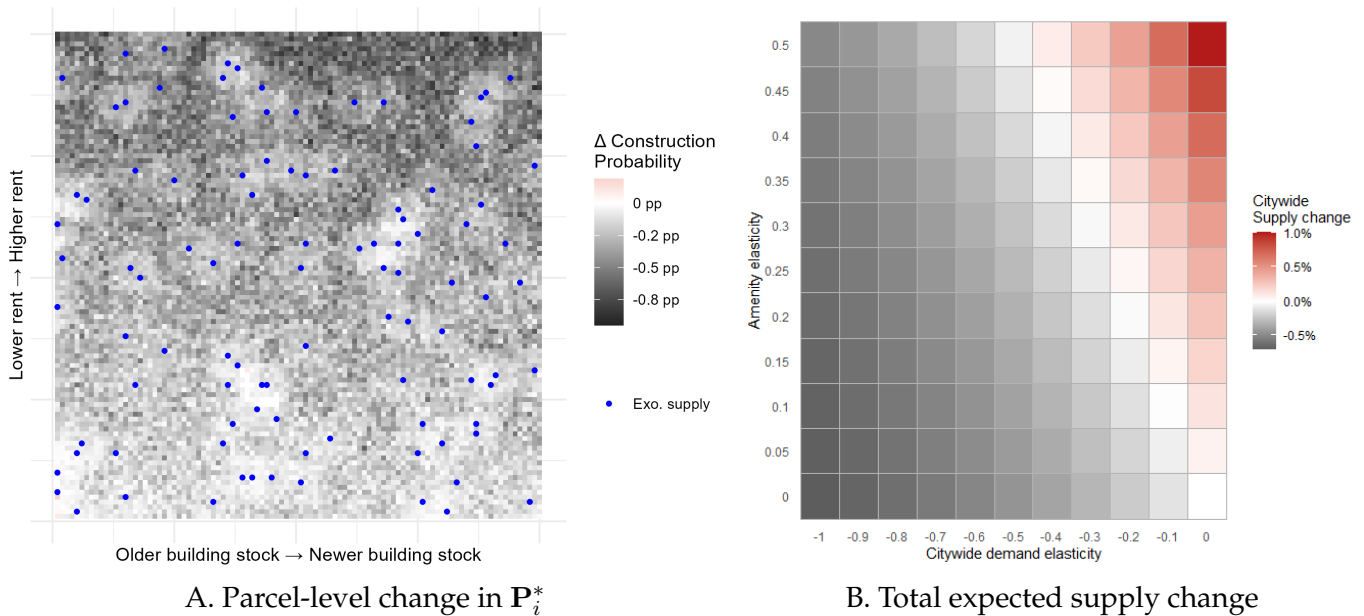
Note: These figures show the effects of changing the one input parameter on the change in the property value (left panels) and the change in the probability of development starting within 10 years (right panels). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and a potential rent of 50. All other parameters and elasticities are shown in Table I.

Figure B5: Uncertainty, Construction Length, Construction Cost, and Probability of Development



Note: This chart shows the effects of varying the rental volatility σ on the the probability of development within 10 years. Each line shows this effect for different assumptions regarding the length of construction d (Panel A) or the baseline cost of construction k_c (Panel B). These figures are obtained for a parcel of land built at 66% of its capacity with $F_B = 2$ and $\bar{F} = 3$, of age 150, with a rental growth of 4%, a volatility of 15%, and a potential rent of 50. All other parameters and elasticities are shown in Table I.

Figure B6: Exogenous Supply Change and Other Parcel Responses



Note: These figures illustrate the spatial propagation of an exogenous housing supply shock in a stylized city. In the policy experiment, 1% of parcels (shown as blue dots in Panel A) are forced to redevelop ($Prob_i = 1$) at a density 100% higher than the current limit, generating an exogenous increase in housing supply equivalent to 0.92% of the current supply. Potential rents R_i are drawn such that they increase from South to North, and building age becomes older (more obsolete) from West to East. All other parameters are similar across parcels ($F_B = 3$, $\bar{F} = 6$, cap rate = 4%, total return = 8%, $\sigma = 15\%$, and $\alpha_B = 0$). All other elasticities and parameters correspond to the baseline values reported in Table I. Panel B shows the corresponding aggregate change in total housing supply (excluding the exogenously redeveloped parcels) as a function of the citywide inverse demand elasticity ζ_S , and the local amenity elasticity ζ_A .

parcels. The decline is largest for parcels with high initial demand (R_i) and for newer buildings. These parcels sit farther from their redevelopment thresholds. At the same time, construction becomes more likely near parcels that are forced to redevelop. Local amenity spillovers drive these

localized increases because they raise the expected rent growth rate, g_i , and increase the value of redevelopment options nearby. As a result, replacement becomes more likely around the initial shock. The effect is strongest when several nearby parcels are forced to redevelop. In those cases, redevelopment clusters in space.

The aggregate response remains, however, small. In a built-out city, redevelopment is expensive. Demolition costs are high. Construction takes time. Panel B of Figure B6 shows how the aggregate supply response varies with the demand elasticity, ζ_S , and the strength of local amenity spillovers, ζ_A . For empirically plausible values of ζ_S , the change in aggregate supply is small. It is typically less than a 1%, even though redevelopment shifts across locations. Stronger amenity spillovers raise local construction incentives but they do not overturn the dominant role of the citywide inverse demand elasticity. Only unrealistically large spillovers overturn this result, and such values are inconsistent with the empirical literature (Asquith et al., 2023; Li, 2022; Mense, 2025).

C Data, Calibration, and Baseline Outcomes

C.1 Parcel-level Data

We begin with the 2023 PLUTO dataset, which contains 857,944 tax lots. We apply a sequence of restrictions to isolate parcels relevant for the counterfactual analysis that we highlight in Table C1. Particularly important:

- **Residential parcels.** We retain lots that are zoned or partially zoned for residential use. To retain parcels eligible for residential use, we keep those with an “R-x” zoning classification in the first through fourth zoning categories. We also retain parcels with observed residential square footage regardless of zoning classification.
- **Complete and consistent information.** We keep parcels that have recorded built year and FAR in addition to parcels without building. We exclude parcels with extreme lot sizes by retaining parcels larger than 1,250 square feet (half of a standard Manhattan lot) and smaller than 250,000 square feet (roughly one hundred times a standard Manhattan lot).
- **Rental and caprate data availability.** We keep parcels with (i) an ACS-based rental estimate and (ii) an observed CoStar caprate measure, leaving 765,830 parcels in the final dataset.

In total, we keep 89.3% of all parcels of New York City. The same procedure is used when constructing the 2015 parcel-level data we use to predict construction activity, which we compare with realized construction as a model fit exercise (89.3% are retained).

Table C1: Parcel-level Cleaning Steps

Category	Parcels	Share of Total	Loss
PLUTO 2023	857,944	100.00%	
with FAR information	825,703	96.24%	-3.76%
residential or partially residential	803,026	93.60%	-2.75%
with acceptable parcel size	778,060	90.69%	-3.11%
with spatial information	777,211	90.59%	-0.11%
with age information	776,872	90.55%	-0.04%
with rental (ACS) data	766,558	89.35%	-1.33%
with return (CoStar) data	765,885	89.27%	-0.09%
with all variables	765,830	89.26%	0.00%

C.2 Rental Obsolescence due to Aging

To estimate rental discount to obsolescence z_i , we follow [Rosenthal \(2014\)](#) and compute the age elasticity of real estate prices using individual transaction data. Although the estimation relies on transaction prices rather than rents, the estimated price discount ϵ_z can be applied directly to rents because the year and location fixed effects account for variations in capitalization rates. We use multifamily transactions from CoStar between 2000 and 2023 and estimate a hedonic pricing regression of the form

$$\log(\text{SalePrice})_{ijt} = \beta_j + \beta_t + \epsilon_z \log(\text{Age}_{it} + 1) + X'_i \beta + \varepsilon_{ijt},$$

where $\log(\text{SalePrice})_{ijt}$ is the log building price in year t and location j , β_j and β_t denote spatial and time fixed effects, Age_{it} is the difference between the sales year and the building's construc-

tion year, and X_i includes covariates such as log building size, log number of floors, a retail-use indicator, and distance to the nearest MTA subway station. The log-log specification allows for faster depreciation at earlier ages, consistent with [Lopez and Yoshida \(2022\)](#).

Table C2: Estimation of the Age Elasticity

	Dependent variable:			
	log(SalePrice)			
	(1)	(2)	(3)	(4)
$\log(\text{Age}_i + 1)$	-0.096*** (0.014)	-0.095*** (0.014)	-0.093*** (0.013)	-0.089*** (0.011)
Controls	X	X	X	X
Community District FE	X	X		
Year FE	X			
Borough x year FE		X		
Community district x year FE			X	
Neighborhoods x year FE				X
Observations	35,383	35,383	35,383	35,383
R ²	0.814	0.819	0.834	0.852
Adjusted R ²	0.813	0.818	0.827	0.836

Notes: This table reports the estimates of $\log(\text{SalePrice})_{ijt} = \beta_j + \beta_t + \epsilon_z \log(\text{Age}_i + 1) + X_i \beta + \varepsilon_{ijt}$, where $\log(\text{SalePrice})$ is the log of the building price transacted in year t in location j , β_j and β_t are spatial and time fixed effects, respectively, Age_{it} is the age of the building at the time of transaction, X_i are covariates including the log size of the building, and the log number of floors, an indicator for whether the property has a retail component, and the distance to the closest MTA metro station. The sample consists of all building transactions from 2000 to 2023 in New York City, collected from Costar. Robust standard errors clustered at the community district and year are presented in parentheses. Estimates followed by ***, **, and * are statistically significant at the 1%, 5%, and 10% levels, respectively.

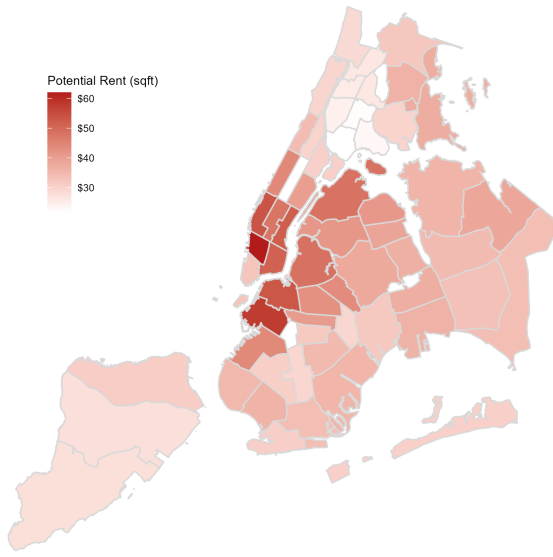
Table C2 reports results across several fixed-effect specifications. Estimates of ϵ_z , which capture the decline in prices due to aging, range from -0.096 to -0.089 , implying an annual rental depreciation pattern that is steeper in the early years (averaging -2% over the first 10 years), consistent with the findings. We set $\epsilon_z = -0.093$ in the baseline calibration and compute $z_i = \text{Age}_i^{\epsilon_z}$ to obtain parcel-level rental depreciation. Spatial variation in z_i is shown in Panel B of Figure C4.

C.3 Rental Data Construction

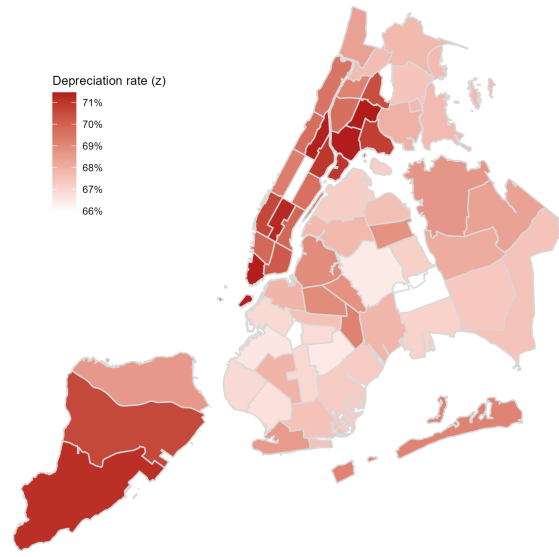
For the model, we need to measure rental income that reflects the potential rent per unit of space, net of the effects of height, obsolescence, and affordable-unit requirements. We use the ACS median gross rent variable (B25064_001) that reports the median contract rent at the block group or census tract level. First, we assign to each parcel either the block group level rent, Rent_{bg} , or the tract level rent, Rent_{tract} , when the former is unavailable. This procedure preserves geographic heterogeneity while maintaining close to full data coverage.

To recover a rent measure consistent with the model, we adjust the ACS data in several steps. First, because ACS rent is reported per dwelling unit, we normalize it by the median

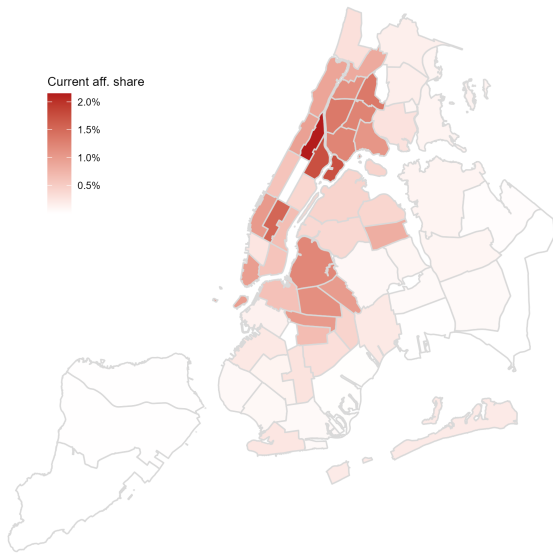
Figure C1: Spatial Variations of Key Variables



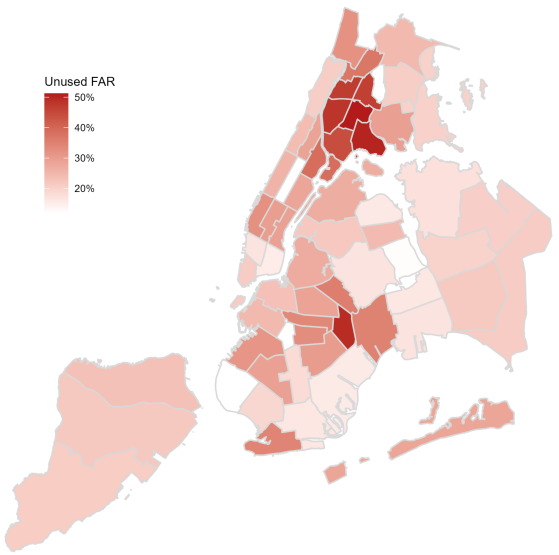
A. Potential rent per unit (R_i)



B. Rental depreciation Rate ($z_{B,i}$)



C. Pre-policy affordable share ($\alpha_{B,i}$)

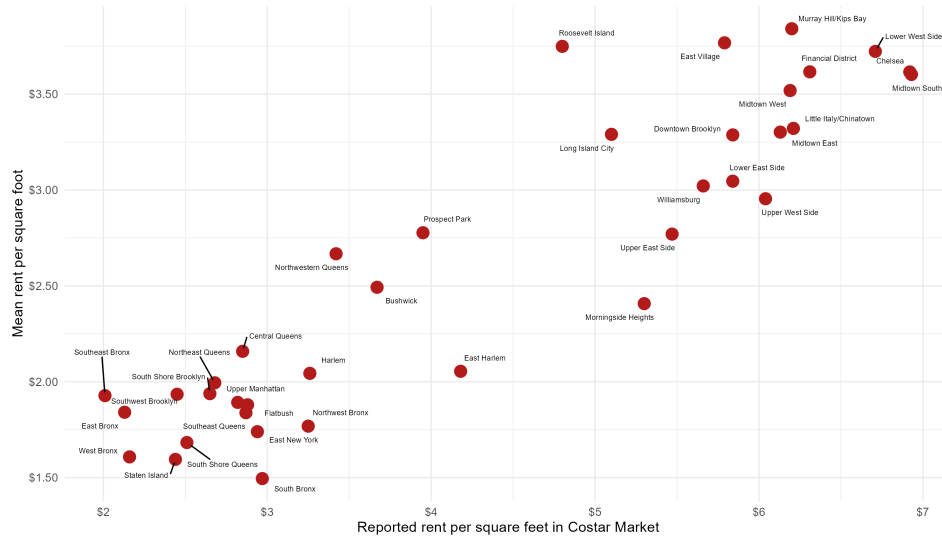


D. Unused density ($1 - (F_{B,i}/\bar{F}_i)$)

Note: These maps show the average across all parcels of each community district of key parameters of the calibration. Panel A shows the potential rent per unit of density R_i that reflects market-rate rents for one non-obsolete unit, obtained by adjusting the annual rent for the average age and size of buildings within each neighborhood. Panel B shows the average discount due to the aging of properties $z_{B,i}$. Panel C shows the average share of buildings with affordable units. Panel D shows the unused density computed as the share of allowed FAR that is currently not built. All statistics are computed with 2023 data.

apartment size in the corresponding geography. We compute the median size of residential units using PLUTO records for buildings with reliable information on total residential square footage and number of units. We then divide each block-group (or tract) median rent by this median size of apartment to obtain the rent per square foot in each geography: $RentPSF_{bg} = MedianRent_{bg}/UnitSize_{bg}$.

Figure C2: Constructed ACS Rents per Square Foot versus CoStar Market Rents



Note: This scatter plot compares our constructed rent per square foot to CoStar market level rents. The vertical axis reports ACS based rents adjusted for median unit size and for average values of density, depreciation, and affordable share. The horizontal axis reports CoStar rents per square foot, aggregated at the market level. A perfect match would place all points on the 45 degree line. Most observations fall below this line, indicating that ACS based rents understate CoStar rents, with the gap strongest in high rent neighborhoods where ACS top coding is most binding. In contrast, lower rent neighborhoods lie close to the benchmark, suggesting that ACS rents provide reliable coverage in these areas. The pattern is consistent with the known censoring of ACS rental data above three thousand five hundred dollars.

We acknowledge that ACS rents are top-coded and therefore underestimate rent levels in the highest value neighborhoods. This bias affects primarily parts of Manhattan and Western Brooklyn. As a result, the calibrated model likely understates property values and redevelopment probabilities in these areas and parcels in these neighborhoods will respond less intensively to policy interventions, as illustrated in Figure I. This downward bias is evident in Figure C2, which compares our constructed ACS rents per square foot to market rents from CoStar, yet the two measures show a positive relation with an R^2 of $= 0.82$. Despite this limitation, the ACS-based construction remains essential because it delivers the finer spatial heterogeneity in rents. Market-based datasets such as CoStar cannot serve this purpose because their spatial units cover much larger geographic areas (37 sub-markets vs. 5,426 different rents with ACS geographies).

We finally convert this "realized" rent per square foot into the potential rent R_i required by the model. The ACS rent is collected through a survey of current residents and therefore reflects rental payments that depend on the existing buildings. Thus, it embeds the effects of current height, depreciation, and the share of affordable units. For instance, if we observe very low rent per square foot in a neighborhood, it might reflect low potential rent R_j or simply the fact that many buildings are obsolete. In our model, R_i is the rental income per square foot of residential space in a building of height 1 ($F_B = 1$), not obsolete ($z_B = 1$), with no affordable units ($\alpha_B = 0$).

To compute this potential rent, we scale the observed rent by the median values of F_B , z_B , and α_B in the corresponding geography. This adjustment removes the influence of existing building characteristics and yields an estimate of the underlying potential rent that would prevail on parcel i . Specifically, we define

$$R_{i(j)} = \frac{\text{Rent}_{i(j)}}{\Gamma(\tilde{F}_{B,i(j)}, \tilde{z}_{B,i(j)}, \tilde{\alpha}_{B,i(j)})},$$

where $R_{i(j)}$ denotes the potential rent assigned to the geography j where parcel i is located, $\tilde{F}_{B,i(j)}$, $\tilde{z}_{B,i(j)}$, and $\tilde{\alpha}_{B,i(j)}$ are the weighted median built density, rental depreciation, and affordable unit share across parcels in geography j , respectively. Weighting by built square footage $F_{B,i} \times L_i$ ensures that the representative building characteristics reflect the stock of space actually inhabited by renters, consistent with the ACS rent measure. Panel A of Figure C1 shows the geographical variation in this constructed variable.

C.4 Returns, Cap Rates, and Rental Growth

We use market-level data from CoStar to construct the return series that underpins the stochastic process for rents. New York City is composed of 37 Costar submarkets that we manually mapped and assigned to individual lots in our database. The dataset covers the last twenty years (2004 to 2023). Returns are computed in three steps.

1. **Asset value per square foot.** For each market m and year t , we construct the implied asset value per square foot using market rents and cap rates:

$$\text{ValuePSF}_{m,t} = \frac{\text{RentPSF}_{m,t}}{\text{Caprate}_{m,t}}.$$

2. **Asset and rental returns.** We compute asset returns and rental returns as follows and report the summary statistics across market in Table C3:

$$\text{AssetReturn}_{m,t} = \frac{\text{ValuePSF}_{m,t}}{\text{ValuePSF}_{m,t-1}} - 1, \quad \text{RentalReturn}_{m,t} = \frac{\text{RentPSF}_{m,t}}{\text{RentPSF}_{m,t-1}} - 1.$$

3. **Total return.** The annual total return in market m is the sum of the two components:

$$\text{TotalReturn}_{m,t} = \text{AssetReturn}_{m,t} + \text{RentalReturn}_{m,t}.$$

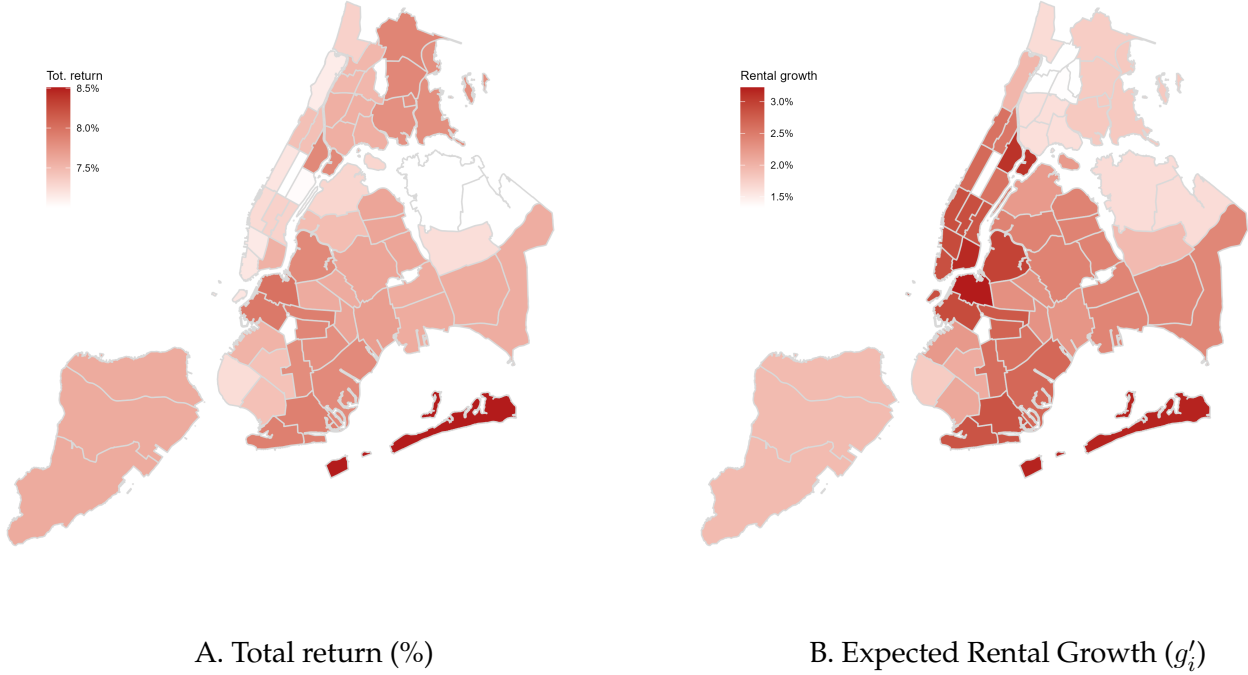
Table C3: Costar Market Level Returns

Statistic	N	Mean	St. Dev.	Min	Median	Max
Asset Return (%)	37	2.39	0.58	1.26	2.46	3.64
Rental Return (%)	37	4.85	0.59	4.05	4.85	5.98
Total Return (%)	37	7.23	0.40	6.60	7.26	8.14

We note that the average total returns across the 37 CoStar markets in New York City is 7.74%. This value falls within the estimates reported by [Lyons et al. \(2026\)](#) and [Amaral et al. \(2025\)](#) for New York City. The advantage of our approach is that it preserves meaningful spatial variation in

returns, rental growth, and volatility. Figure C3 shows that expected total returns decline as one moves away from the city center, a pattern consistent with the argument in [Amaral et al. \(2025\)](#) that dense urban locations tend to deliver lower total returns.

Figure C3: Total Housing Return and Expected Rent Growth



Note: These maps show the average total return (asset and rental returns) in Panel A, and the 2023 estimated rental growth (total return minus 2023 cap rate) in Panel B computed across all parcels of the community districts. These statistics are drawn from Costar market data from 2004 to 2023.

C.5 Aggregating Parcel-Level Outcomes

We aggregate outcomes at the city level and for each neighborhood j . The two main outcomes are the expected number of net additional housing units, $\text{UnitNet}_j(\mathcal{P}, h)$, and the average expected rent per housing unit, $\text{Rent}_j(\mathcal{P}, h)$. We also compute related measures, including: the expected total supply $S_j(\mathcal{P}, h)$ in square foot, the present value of all parcels $PV(\mathcal{P})$, the present value of the property taxes $PV^{\text{Tax}}(\mathcal{P})$, and the average share of option value $\text{Option}(\mathcal{P})$. Aggregated outcomes are defined from the parcel-level outcomes as:

$$S_j(\mathcal{P}, h) = \sum_{i \in j} ((1 - \mathbf{P}_{ih}^*(\mathcal{P}))F_{B,i}L_i + \mathbf{P}_{ih}^*(\mathcal{P})F_i^*(\mathcal{P})L_i) \quad (\text{C.1})$$

$$\text{UnitConst}_j(\mathcal{P}, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i F_i^*(\mathcal{P})}{\text{UnitSize}_i} \quad (\text{C.2})$$

$$\text{UnitNet}_j(\mathcal{P}, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i (F_i^*(\mathcal{P}) - F_{B,i})}{\text{UnitSize}_i} \quad (\text{C.3})$$

$$\text{AffNet}_j(\mathcal{P}, h) = \sum_{i \in j} \mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i(F_i^*(\mathcal{P}) - F_{B,i}) \cdot \max(\alpha_{B,i}, \alpha)}{\text{UnitSize}_i} \quad (\text{C.4})$$

$$\text{Exp.Unit}_j(\mathcal{P}, h) = \sum_{i \in j} \left(\mathbf{P}_{ih}^*(\mathcal{P}) \frac{L_i(F_i^*(\mathcal{P}))}{\text{UnitSize}_i} + (1 - \mathbf{P}_{ih}^*(\mathcal{P})) \frac{L_i F_{B,i}}{\text{UnitSize}_i} \right) \quad (\text{C.5})$$

$$\text{Rent}_j(\mathcal{P}, h) = \frac{\sum_{i \in j} \text{RentInc}_{ih}(\mathcal{P})}{\text{Exp.Unit}_j(\mathcal{P}, h)} \quad (\text{C.6})$$

$$\text{RentPerSqft}_j(\mathcal{P}, h) = \frac{\sum_{i \in j} \text{RentInc}_{ih}(\mathcal{P})}{S_j(\mathcal{P}, h)} \quad (\text{C.7})$$

$$PV_j(\mathcal{P}) = \sum_{i \in j} \pi_i(\mathcal{P}; \Theta) \quad (\text{C.8})$$

$$\text{Option}_j(\mathcal{P}) = \frac{\sum_{i \in j} \pi_i^{\text{option}}(\mathcal{P})}{PV_j(\mathcal{P})} \quad (\text{C.9})$$

$$PV_j^{\text{Tax}}(\mathcal{P}) = \sum_{i \in j} \pi_i^{\text{Tax}}(\mathcal{P}). \quad (\text{C.10})$$

where $\pi_i^{\text{option}}(\mathcal{P})$ is the redevelopment option value computed by the difference between the property value π_i and its counterpart when no building can be rebuilt (i.e., $k_c = \infty$). The present value of the taxes for a parcel is computed as the present value of the tax components on the existing and new buildings, accounting for possible tax exemption:

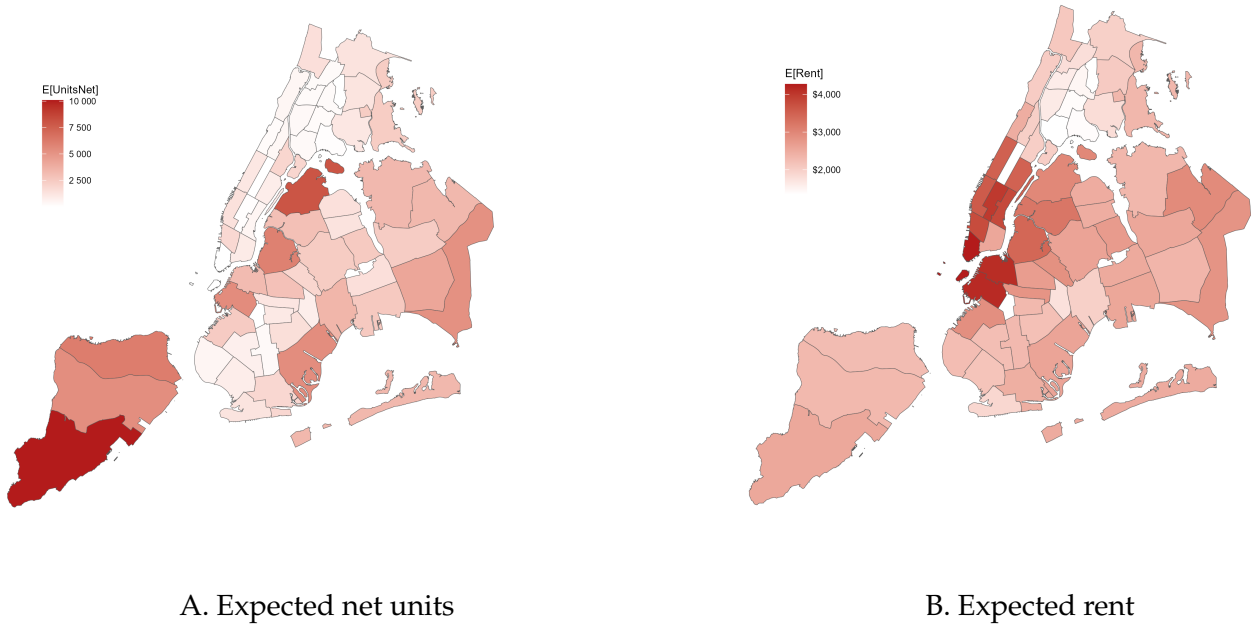
$$\pi_i^{\text{Tax}}(\mathcal{P}) = \underbrace{-\Gamma(F_{B,i}, z_{B,i}, \alpha_{B,i}, L_i) \frac{\psi}{r - \delta_i} \left[R_i - R_i^{e*} \left(\frac{R_i^{e*}}{R_i} \right)^\xi \right]}_{\text{Tax on existing property}} - \underbrace{\psi \Gamma(F_i^*, 1, \alpha, L_i) \frac{R_i^{e*}}{r - \delta_i} e^{-(r - \delta_i)(d + \tau)} \left(\frac{R_i^{e*}}{R_i} \right)^\xi}_{\text{Property taxes after } \tau \text{ years of exemption}}$$

Table C4: Partial Correlations Between Inputs and Outcomes of Baseline Model

	Baseline Outcomes:				
	Prob	F opt	log(PV)	Option	PV/F
	(1)	(2)	(3)	(4)	(5)
Rent potential (R_i)	0.289	−0.015	0.457	0.150	0.792
Rent growth (g_i)	0.056	0.020	0.037	0.028	0.096
Rental depreciation index ($z_{B,i}$)	−0.058	−0.017	0.274	−0.016	0.328
Current density ($F_{B,i}$)	−0.140	0.825	0.652	−0.077	0.087
Non-built parcel ($1_{F_{B,i}=0}$)	0.685	−0.0001	−0.377	0.972	

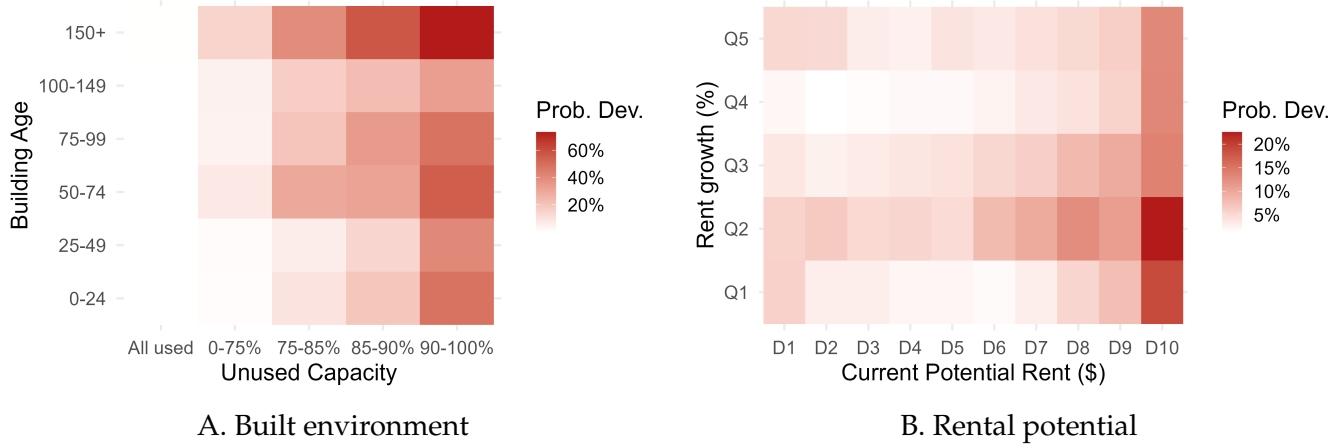
Notes: This table reports the partial correlations between key inputs (rows) and outcomes (columns) of our baseline model results. The partial correlations are conditional on community districts as well as all other inputs listed in the rows. It uses the outcomes of the model with horizon 10-years for the 765,830 parcels under the baseline calibration described in Tables II and I. In Column (1), the outcome is the estimated probability of redevelopment decision $\mathbf{P}_i(\mathcal{P}_0; \Theta)$. In Column (2), the optimal density chosen endogenously to maximize property value ($F^*(\mathcal{P}_0; \Theta)$). In Column (3), the log of the present value of the parcel $\pi_i(\mathcal{P}_0; \Theta)$. In Column (4), the option value share of total value $\tilde{\pi}_i(\mathcal{P}_0; \Theta)/\pi_i(\mathcal{P}_0; \Theta)$. In Column (5), the present value per square foot of built units $\pi_i(\mathcal{P}_0; \Theta)/F_i L_i$.

Figure C4: Baseline - Market-level Aggregated Outcomes



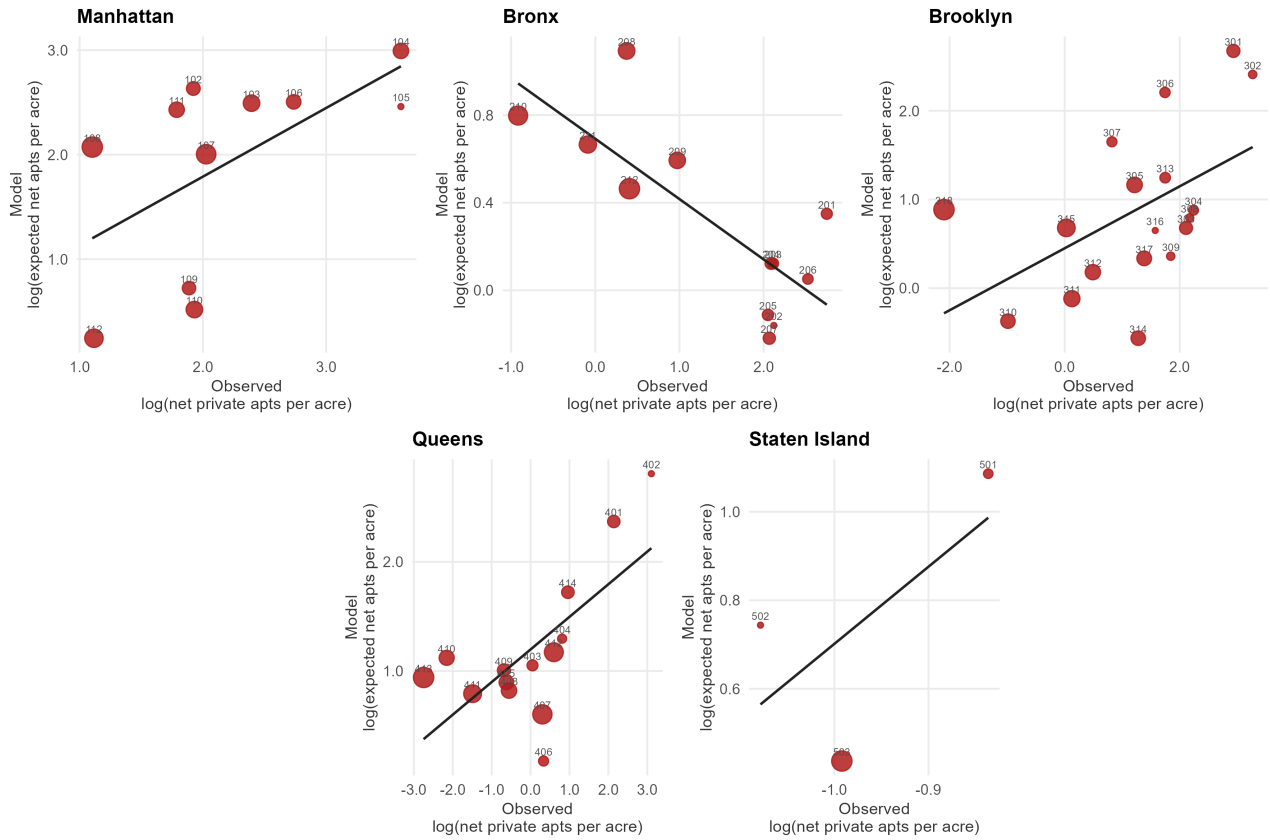
Note: These maps show the market-level aggregated outcomes of our baseline simulation as described in Table I. Panel A shows the expected net units addition and Panel B shows the expected rents per units.

Figure C5: Parcel Heterogeneity and Determinants of Probability of Development



Note: These panels show the unconditional average probability of development computed for each grid cell. Panel A splits cells into deciles based on the current density ($F_{B,i}$) and the current age of the building ($z_{B,i}$). Panel B splits cells into deciles based on the potential rent (R_i) and rental growth (g_i) using the model baseline calibration with parameters described in Table I.

Figure C6: Expected and realized construction by borough

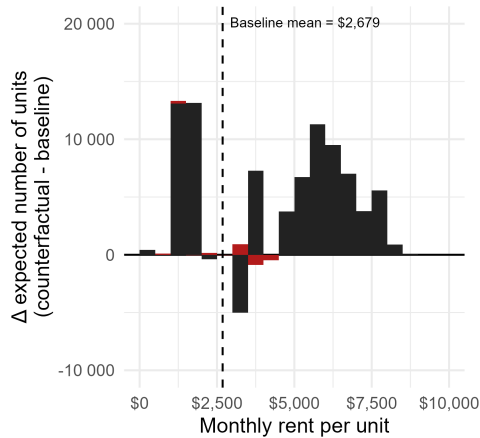


Note: This scatter plot shows, for each borough, the log number of housing units constructed in each community district between 2016 and 2025, scaled by community district residential acreage, against the corresponding model-implied construction outcomes based on the 2015 calibration.

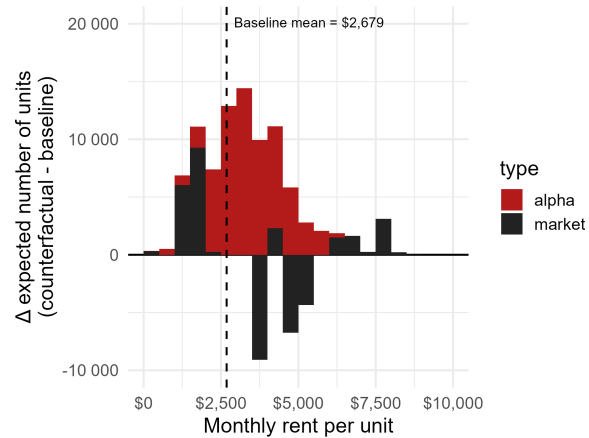
D Counterfactual Results – Additional Tables & Figures

D.1 Distribution of rent

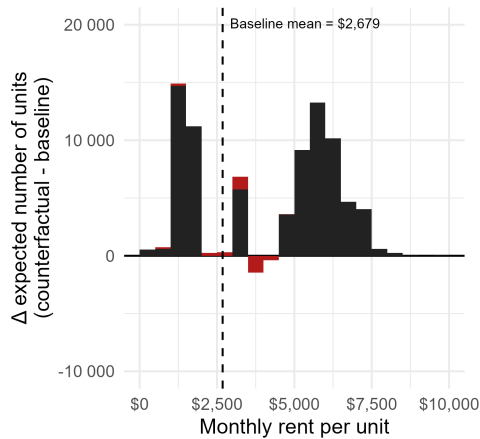
Figure D1: Counterfactuals — Change in Distribution of Rent



A. Upzoning ($\phi = .50$)



B. Density incentive zoning ($\alpha = 0.3$ & $\phi = .50$)



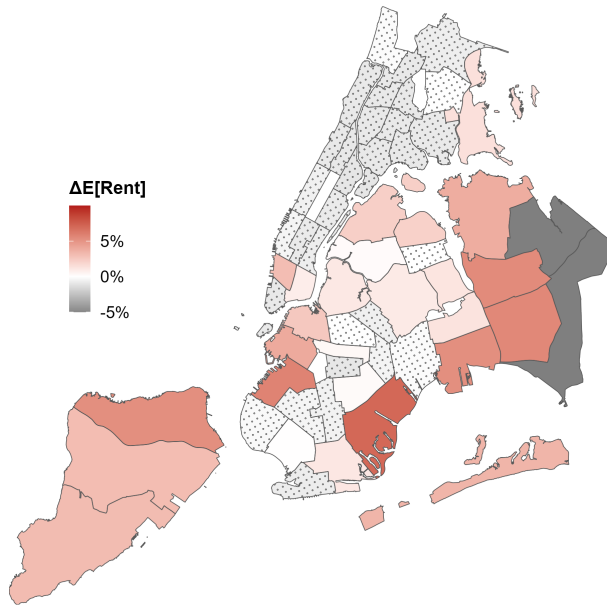
C. Tax exemption ($\tau = 20$)



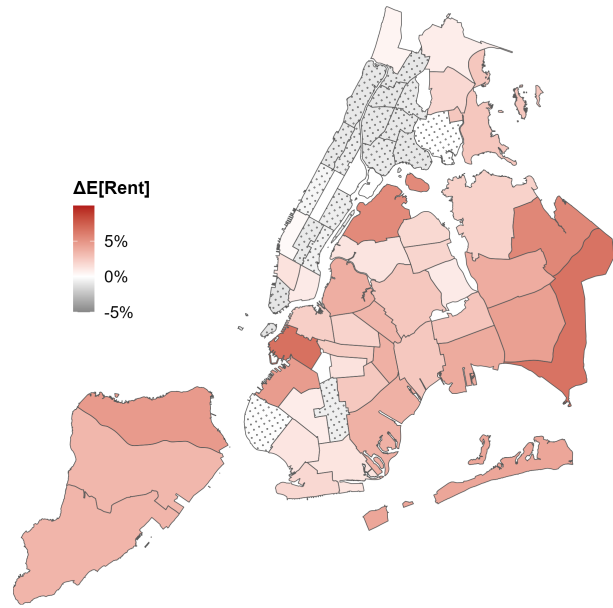
D. Fiscal incentive zoning ($\alpha = 0.30$ & $\tau = 20$)

Note: The histograms report the change in the expected number of apartments in each \$500 rent bin under the counterfactual policy relative to the baseline, evaluated at a 10-year horizon. Bars above (below) zero indicate rent bins in which the expected number of units increases (decreases). Changes are decomposed into market-rate units (black bars) and units subject to affordability mandates (red bars).

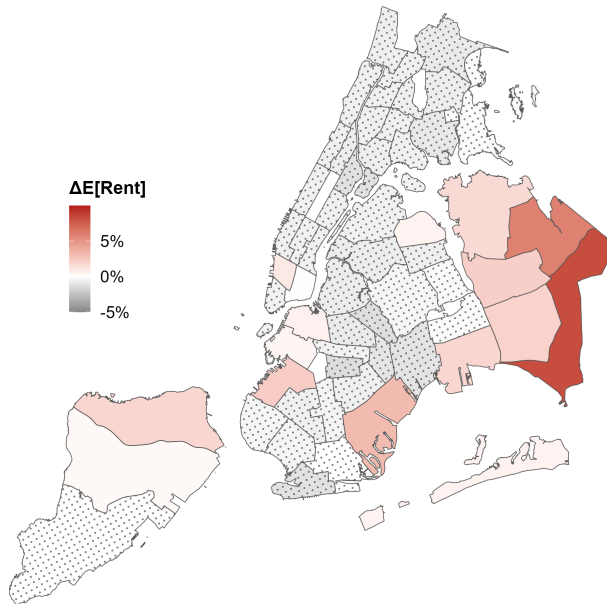
Figure D2: Counterfactuals — Change in Expected Average Rent



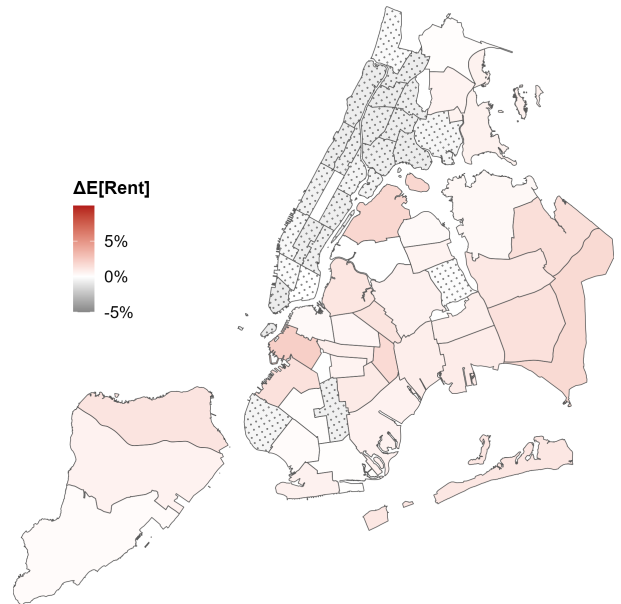
A. Upzoning ($\phi = .50$)



B. Tax exemptions ($\tau = 20$)



C. Density incentive zoning ($\alpha = 0.3$ & $\phi = .50$)

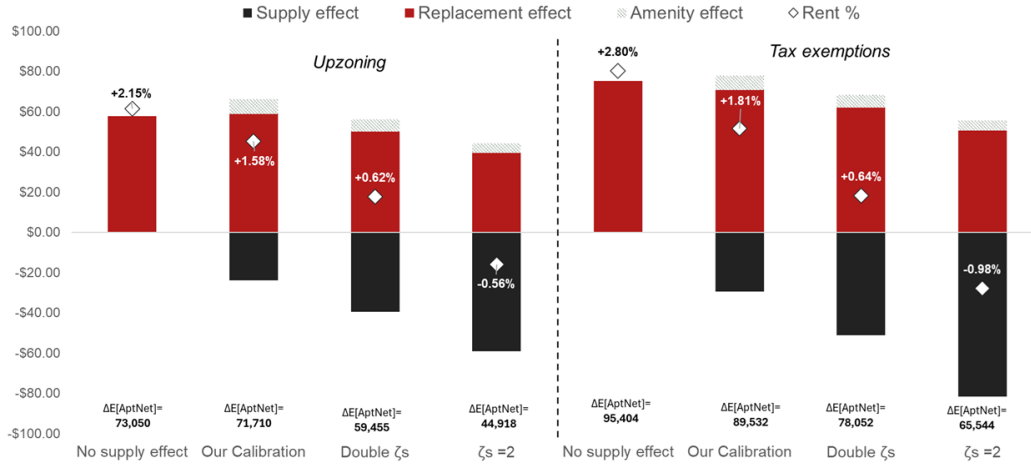


D. Fiscal incentive zoning ($\alpha = 0.3$ & $\tau = 20$)

Note: These maps show the percentage change in average expected rent per unit for different counterfactual policy analyses. It shows for each community district the change compared to the baseline outcomes with horizon 10-year.

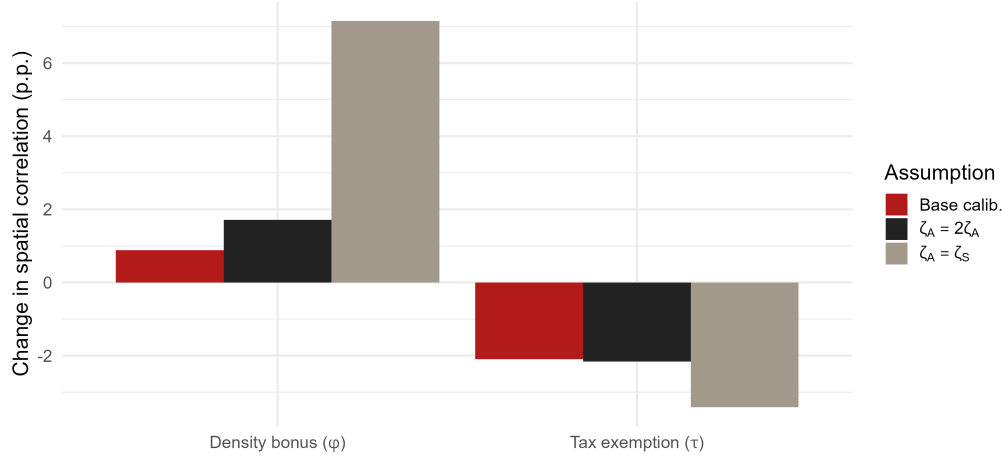
D.2 Robustness to different demand and spillover elasticity parameters

Figure D3: Demand Elasticity, Rents, and Housing Production



Note: The points report the change in average expected rent under each policy relative to the baseline, evaluated at a 10-year horizon, together with its decomposition into the main channels described in the text. For each counterfactual, upzoning in the left panel and tax exemption in the right panel, outcomes are reported for alternative values of the citywide inverse demand elasticity $\zeta_S = \{0, -0.50, -1.00, -2.00\}$ as indicated in the bottom row. Endogenous changes in housing production associated with a policy/elasticity bundle are reported at the bottom.

Figure D4: Local Spillover Strength and the Spatial Correlation of Rents

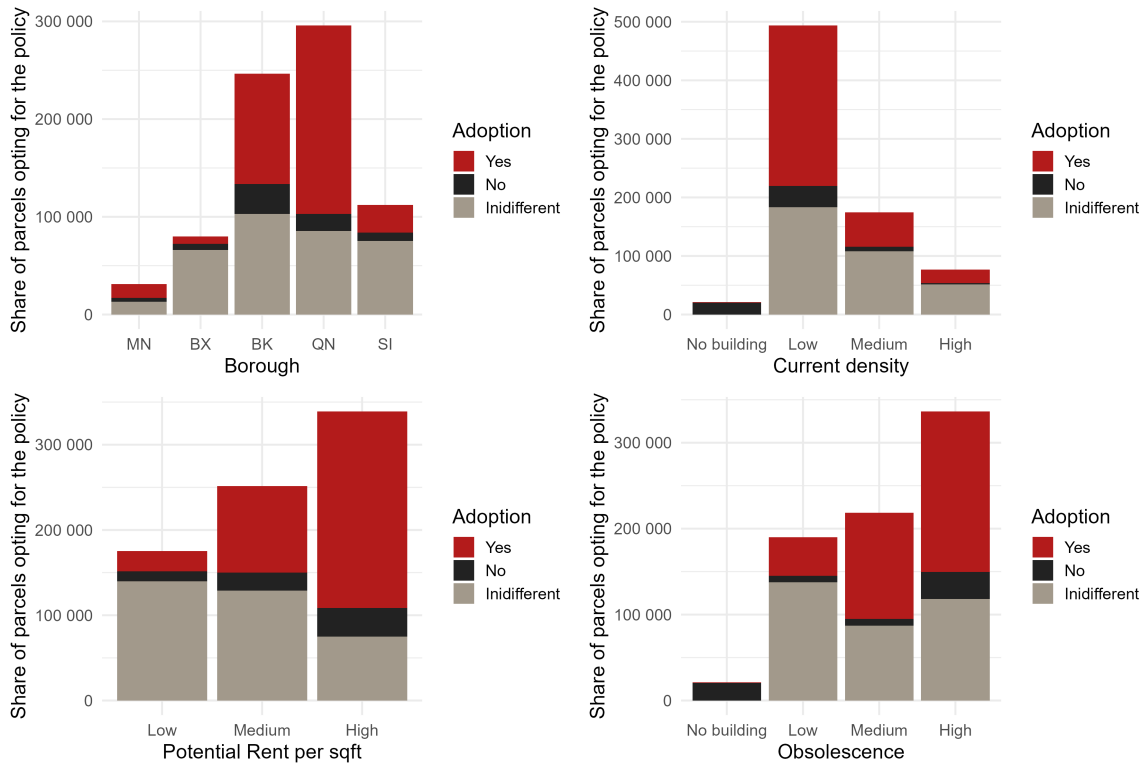


Note: The bars report changes in the spatial correlation of construction probabilities. Spatial correlations are measured using Moran's I , computed with inverse-distance weights over all parcels within 250 meters of each target parcel. Red bars correspond to the baseline calibration, with $\zeta_A = 0.10$ and $\zeta_S = -0.50$. Black bars use $\zeta_A = 0.20$ and $\zeta_S = -0.60$, holding the sum of elasticities constant. Grey bars report results with $\zeta_A = 0.50$ and $\zeta_S = -0.50$. All values are expressed as deviations from the baseline spatial correlation at the 10-year horizon.

D.3 Voluntary Adoption of Policies

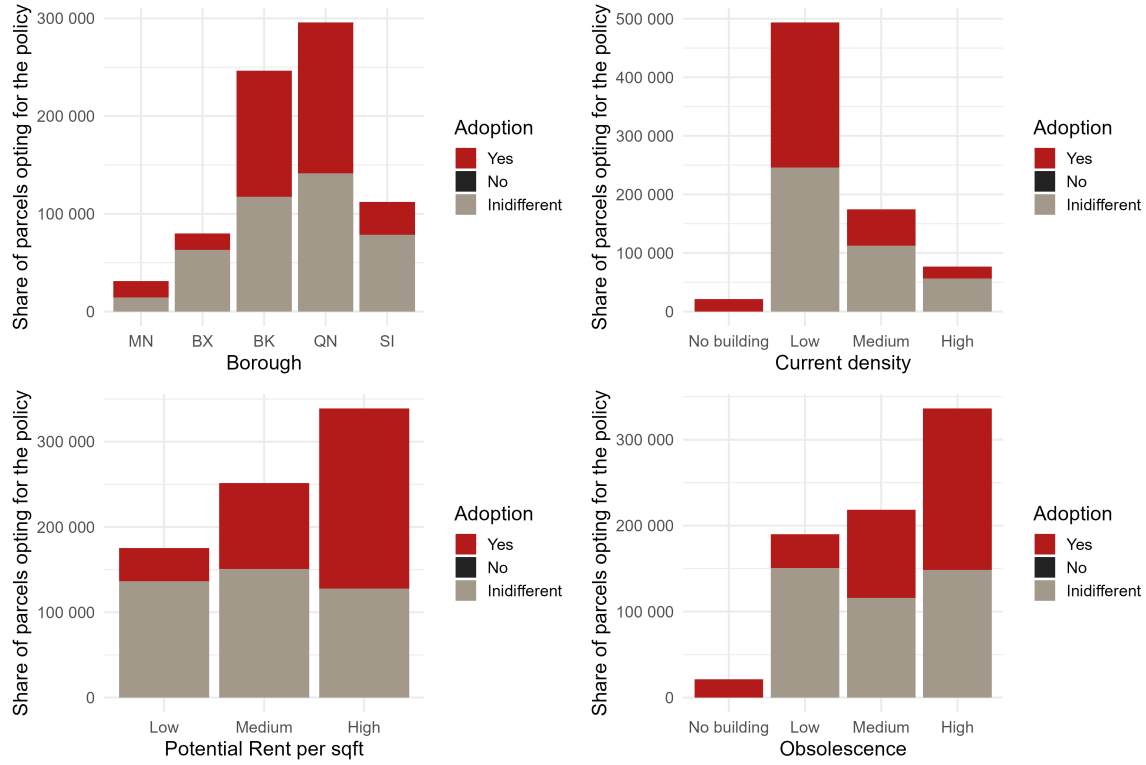
Table D1 reports the share of parcels that would opt into each policy. The model predicts that approximately 45% of parcels adopt either the voluntary Density Incentive Zoning or the Fiscal Incentive Zoning. About 12% of adopters differ across these two bundles, underscoring the importance of parcel-level heterogeneity in the adoption decision. When density and fiscal incentives are combined, adoption rises sharply to 73.7%, with 27.3% of parcels switching from non-adoption to adoption once the fiscal incentive is added to the density incentive zoning. Figures D5 and D6 decompose adoption rates by location and building characteristics. Participation is systematically higher among parcels with low current density, high potential rent, and those located in Queens and Brooklyn.

Figure D5: Voluntary Adoption of Density Incentive Policy



Note: These bar graphs show the share of parcels that opt-in ($\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$), opt-out of the policy $\mathcal{P}$, or are indifferent under the policy counterfactual that combine density bonuses ($\phi = 0.50$), and an affordable mandate ($\alpha = .30$), compared to the status-quo policies.

Figure D6: Voluntary Adoption of Fiscal Incentive Policy



Note: These bar graphs show the share of parcels that opt-in ($\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$), opt-out of the policy $\mathcal{P}$, or are indifferent under the policy counterfactual that combine tax exemptions ($\tau = 20$), and an affordable mandate ($\alpha = .30$), compared to the status-quo policies.

Table D1: Share of Parcels that Opt-in Voluntary Policy Interventions

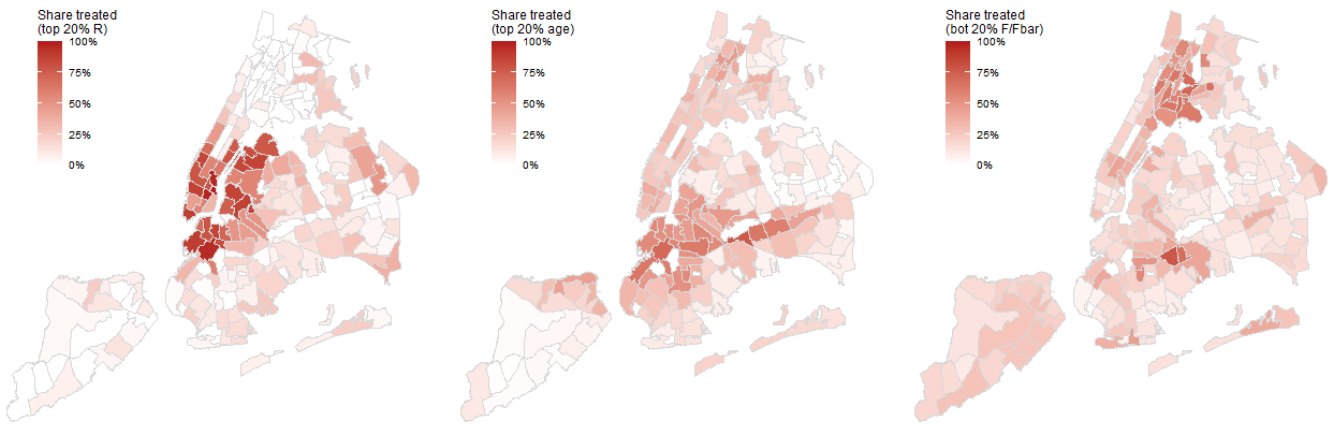
	Number of Parcels	Share (%)
All Parcels	765,830	100
(7) Density Incentive Zoning ($\alpha = 0.30$ & $\phi = 0.50$)	355,930	46.5
<i>but not in fiscal IZ</i>	94,390	12.3
(8) Fiscal Incentive Zoning ($\alpha = 0.30$ & $\tau = 20$)	350,894	45.8
<i>but not in density IZ</i>	89,354	11.7
(9) All Policies	564,752	73.7
<i>but not in density IZ (i.e., switchers with τ)</i>	208,822	27.3
<i>but not in fiscal IZ (i.e., switchers with ϕ)</i>	215,553	28.1

Notes: This table reports the number and the share of parcels that would opt-in the policy bundle $\mathcal{P}$ under voluntary adoptions. These parcels are defined as those that are better off under the new policy as in the baseline policy ($\pi_i(\mathcal{P}) \geq \pi_i(\mathcal{P}_0)$).

D.4 Targeted Counterfactuals

Given the importance of the built environment and the local demand in explaining redevelopment activity, we perform counterfactuals in which we target specific groups of parcels: (1) parcels with high underlying demand (the highest 20% R_i); (2) parcels with obsolete buildings (the lowest 20% $z_{B,i}$); and (3) parcels with high shares of undeveloped capacity (the lowest $F_i/\bar{F}_i$). Figure D7 shows the location of those targeted parcels. Those parcels are not the same. When focusing on high-rent potential parcels, we target mostly parcels in Manhattan, western Brooklyn, and Queens. Most obsolete buildings are located in Brooklyn, and high-unused-capacity parcels are located mostly in the Bronx. The lowest overlap is between the first and last groups, since it is rare for parcels to have unused capacity in high-demand areas.

Figure D7: Targeted Parcels in Targeted Counterfactuals



We reconduct our two unilateral policy counterfactuals on targeted parcels and present the results in Table D2. Targeting the 20% most valuable parcels delivers 61% of the citywide residential floor-space production under upzoning but only 44% under the tax exemption policy. The pattern reverses for the least-utilized parcels. Targeting those yields 77% of citywide production under tax exemption but only 48% under upzoning. This asymmetry reflects the mechanics of each instrument. Density bonuses are most effective where land values are high enough to make additional floor area profitable, whereas tax exemptions operate on the margin of development timing and are therefore more powerful on underbuilt parcels. Targeting obsolete parcels, by contrast, produces limited outcomes across all three policies.

Under all counterfactuals targeting high-value or obsolete parcels, expected citywide rents rise, consistent with the replacement margin effect dominating the supply effect. Targeting the least-utilized parcels, however, produces the lowest rent rise. Because least-utilized parcels include many vacant or near-vacant lots, redevelopment there involves little demolition of existing units, attenuating the replacement margin. In dense urban environments where new construction requires tearing down existing structures, supply expansions carry an inherent upward pressure on rents that is largely absent when development occurs on empty land.

Table D2: Targeted Counterfactual Results

Built Environment				Rent outcomes		Costs of Interventions		
$\Delta S(\mathcal{P}, h)$ (% chg.)	$\Delta \text{NetUnit}(\mathcal{P})$ (# of units)	$\Delta \text{Aff. Unit}(\mathcal{P})$ (# of units)	$\Delta \text{Rent}(\mathcal{P})$ (% chg.)	$\Delta \text{Shr. Q1}$ (pp)	$\Delta PV^{\text{Tax}}(\mathcal{P})$ (% chg.)	$\Delta PV(\mathcal{P})$ (% chg.)	Shr. PV-losers (%)	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
(0)	Baseline under $\mathcal{P}_0$	125,685	Baseline Outcomes 310 \$2,682.45	20.00	\$406.06 Bn.	\$1,574.13 Bn.	–	
(1)	All parcels	71,710	Upzoning policy — $\phi = 50\%$ 22 1.58	0.28	0.05	–1.20	0.87	
(1a)	Most expensive parcels	48,890	33 1.46	0.18	0.44	–0.64	0.89	
(1b)	Most obsolete parcels	20,095	9 0.60	0.05	0.02	–0.29	0.90	
(1c)	Least utilized parcels	32,577	–6 0.41	0.17	0.05	–0.60	0.88	
(2)	All parcels	89,532	Tax exemption — $\tau = 20$ 269 1.81	0.23	–3.71	–0.93	0.86	
(2a)	Most expensive parcels	45,424	262 1.68	0.13	–1.99	–0.15	0.89	
(2b)	Most obsolete parcels	25,266	10 0.82	0.06	–0.82	–0.27	0.89	
(2c)	Least utilized parcels	68,665	247 0.79	0.17	–3.11	–0.65	0.87	

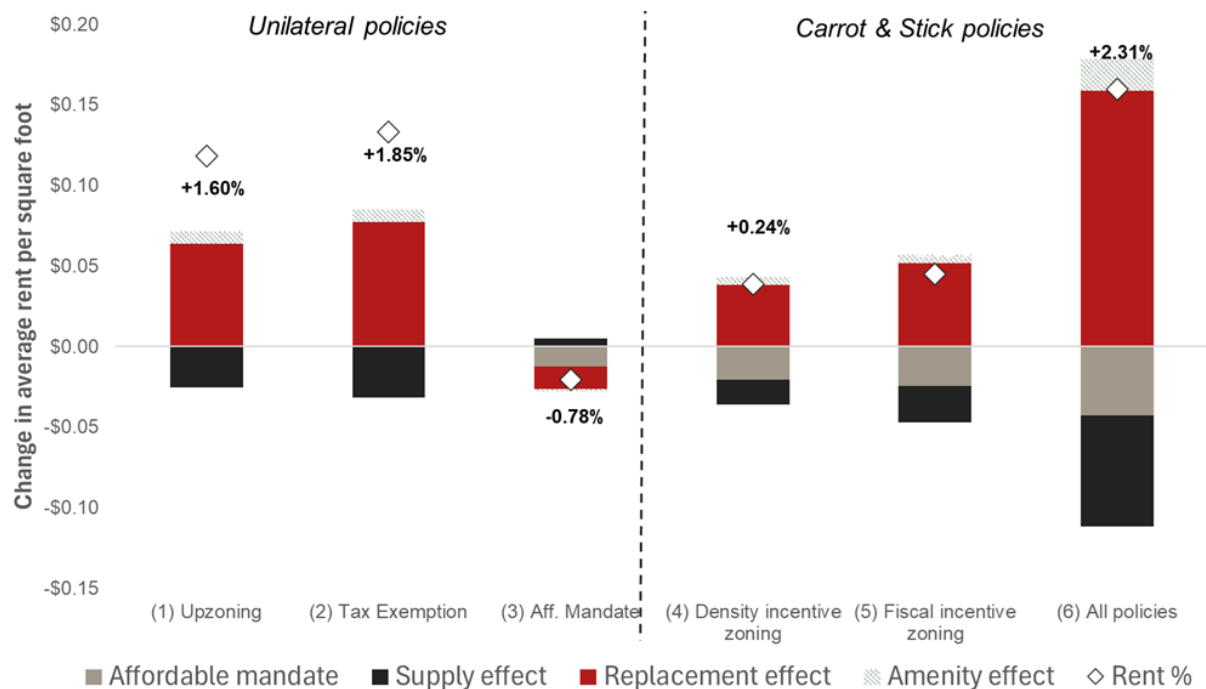
Notes: Row (0) reports baseline outcomes under the no-policy environment $\mathcal{P}_0$ in levels. Rows (1)–(2) report changes relative to the baseline. Each row reports outcomes under alternative regulatory environments $\mathcal{P}$, all measured over a 10-year horizon. Within each policy panel, All parcels applies the policy citywide; the remaining rows restrict the mandate and incentives to the top 20% most expensive parcels (by rent potential), the top 20% most obsolete parcels (by age), and the bottom 20% least utilized parcels (by F_0/F), respectively, with no policy applied elsewhere. Column (1) reports the percentage change in total expected supply in square feet. Column (2) reports the change in expected housing units constructed net of demolitions. Column (3) shows the change in affordable units out of this net change. Column (4) reports the percentage change in average expected citywide rent per housing unit. Column (5) depicts the change in percentage points in the share of units falling within the first quintile of the baseline rent distribution. Columns (6) and (7) report the percentage change in the present value of property tax revenues and total residential asset values, respectively. Column (8) shows the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$).

D.5 Size of Units and Main Results

The size of housing units (UnitSize_i) enters our model only as a reporting convention because the model is specified in per-square-foot terms and all supply and amenity spillovers are computed based on expected floor area. Presenting results in housing units may, however, raise concerns regarding the robustness of our conclusions to this normalization

Robustness to the square-footage basis. We first address the normalization concern directly by replicating the rent decomposition exercise in square-foot terms rather than housing units. Figure D8 shows that the same pattern emerges, if anything more strongly. The replacement effect dominates the supply effect across all six policy scenarios. This confirms that the replacement effect is not an artifact of the unit-size reporting assumption but reflects the underlying mechanism linking redevelopment decisions to rent outcomes.

Figure D8: Counterfactual Results — Expected Rents in Square Footage Basis

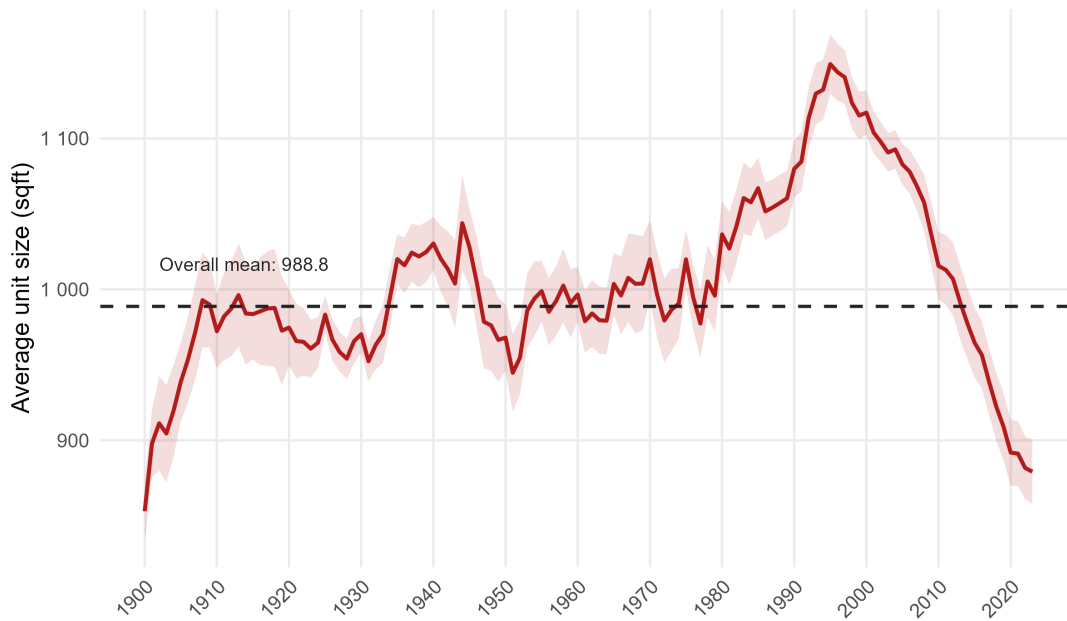


Note: The points report the average expected rent per square foot change under each counterfactual policy relative to the baseline at a 10-year horizon. The stacked bars decompose this change into four components. The Affordable Mandate component isolates the direct effect of the affordable share α on rental income, holding expected construction activity fixed. The Supply effect captures the change in expected rents attributable to additional citywide housing supply. The Amenity effect is computed analogously and reflects the impact of local supply increases on neighborhood rents. The Replacement effect is defined as the residual component and captures rent changes associated with replacing older, depreciated structures with new construction, including changes in redevelopment probabilities and endogenous updates to expected rental growth. Policies (1)–(3) correspond to unilateral interventions—upzoning, tax exemption, and affordability mandate—while policies (4)–(6) combine these instruments.

Result Sensitivity to change in unit size. To provide more evidence regarding the potential reporting bias related to the normalization, we plot in Figure D9 the rolling weighted average of residential unit size by year of construction, using PLUTO data. After several decades of growth, unit sizes have declined substantially since the early 2000s. Over the last decade, newly built units

were on average 7.0% smaller than the existing stock. Because our baseline results hold unit size fixed, this structural shift implies that our housing production figures may be biased downward if smaller units continue to be built over the horizon h .

Figure D9: Weighted Average Size of Constructed Unit



Note: This figure plots the rolling average of residential unit size (in square feet) by year of construction using information from PLUTO. The rolling averages series uses a ± 5 -year window. The shaded ribbons represent 95% confidence intervals around the rolling mean.

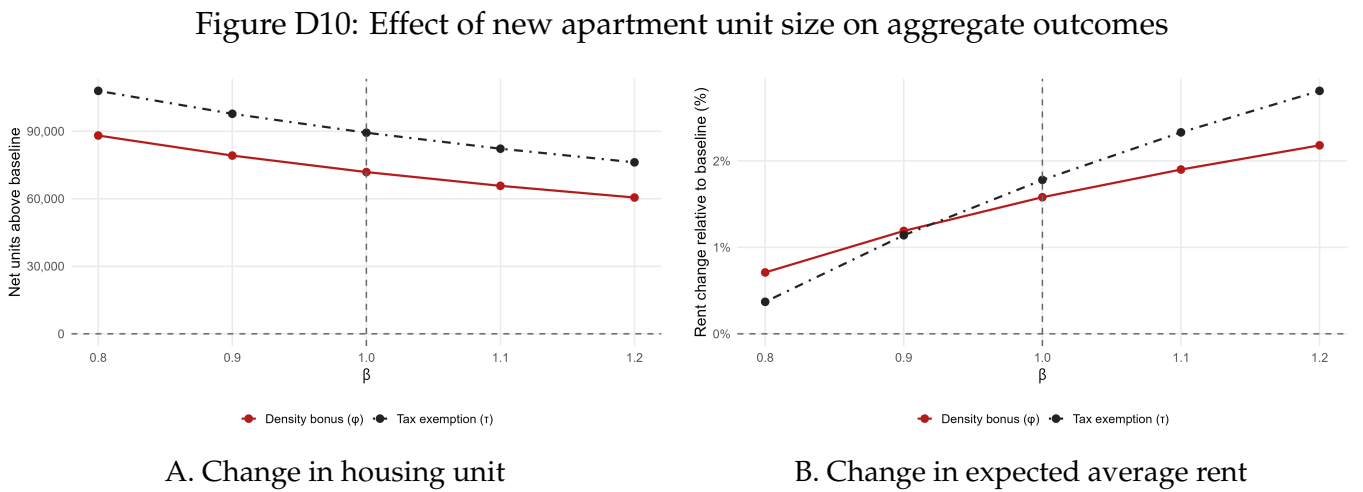
To quantify this bias, we introduce a unit-size adjustment parameter β , defined as the ratio of the new unit size to the existing stock's unit size. We re-conduct the main counterfactual analysis across a range of $\beta \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$. The expected number of units on parcel i under policy $\mathcal{P}$ becomes

$$\text{Unit}_i(\mathcal{P}, h; \beta) = \underbrace{\frac{(1 - \mathbf{P}_{ih}^*(\mathcal{P}))F_{B,i}L_i}{\text{UnitSize}_i}}_{\text{Exp. units with no development}} + \underbrace{\frac{\mathbf{P}_{ih}^*(\mathcal{P})F_i^*(\mathcal{P})L_i}{\beta \text{ UnitSize}_i}}_{\text{Exp. units with development}}.$$

so that $\beta < 1$ implies smaller new units and hence more units per square foot of new construction. We also adjust the supply feedback so that rent growth responds to changes in the number of housing units rather than floor area. Since rental income is unaffected by the normalization, rent per unit mechanically decreases with β to the extent that there is expected housing production ($F_i^* \mathbf{P}_i^* > 0$). Figure D10 reports how net unit production (Panel A) and expected rent per unit (Panel B) vary with β for upzoning ($\phi > 0$) and tax exemption ($\tau > 0$).

As β falls, a given amount of new floor area is divided into more units, raising the unit count gain from each policy and dampening rent per unit. The qualitative conclusions regarding average rents are fully robust. Rents rise under both policies for all values of β considered. Even at $\beta = 0.8$, which implies that new units are 20% smaller than the existing stock (a more extreme shift than observed historically), average expected rents still increase under both policies. Though the replacement effect is attenuated because the new units replacing older, larger structures are

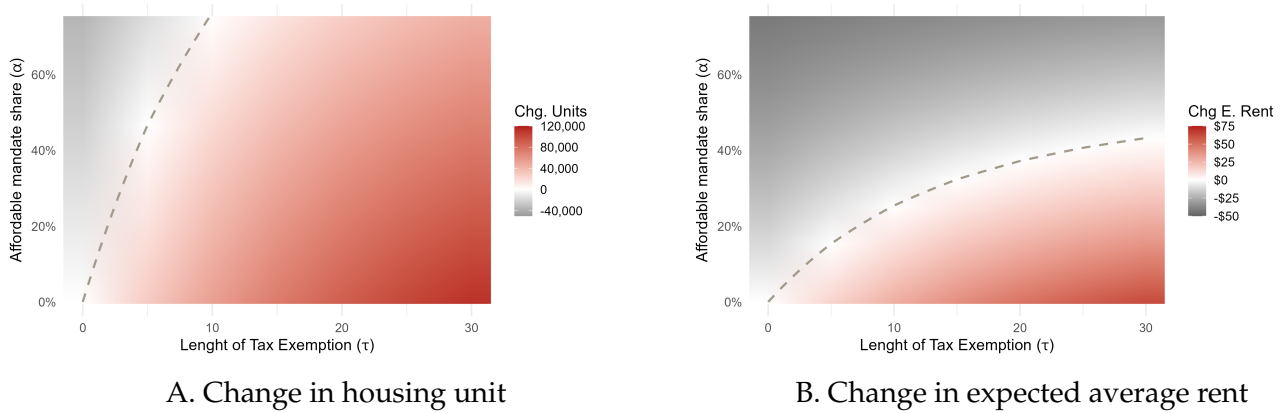
themselves smaller. Only a drastic and sustained reduction in unit size would be sufficient to generate rent decreases alongside increases in supply.



Note: These figures show the sensitivity of the main policy results to the assumption governing the size of newly built units relative to the existing stock, captured by the parameter β . Panel A reports the change in expected net units constructed relative to the no-policy baseline, for the density bonus (ϕ , solid red line) and the tax exemption (τ , dash-dot black line), as a function of β . Panel B reports the corresponding change in expected rent per unit relative to the no-policy baseline.

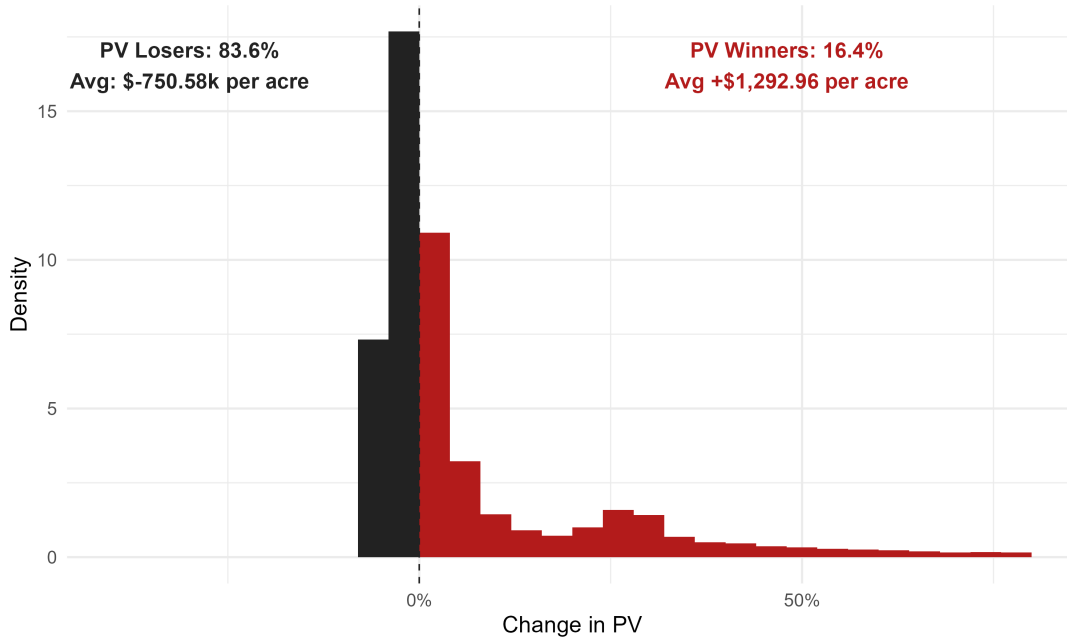
D.6 Other Supporting Figures and Tables

Figure D11: Continuous Policies Outcomes - Fiscal Incentive Zoning



Note: These heatmaps show the outcomes of a Fiscal Incentive Zoning program under continuous values of τ (x-axis) and α (y-axis). Changes are computed over the outcomes of the status-quo under policy P_0 . Panel A shows the net change in the number of housing units $E[AptNet|\mathcal{P}]$. Panel B shows the change in average expected rents per unit. The dotted lines show outcomes that equate the value under the status-quo policy P_0 .

Figure D12: Distribution of change of present values with combined incentive policy



Note: These bar graphs show the distribution of the percentage change in parcel present value $\Delta\pi_i(\mathcal{P})$ for the combined incentive zoning policy that combines density bonuses ($\phi = 0.50$), tax exemptions ($\tau = 20$), and an affordable mandate ($\alpha = 0.30$), relative to the baseline P_0 . Each bar represents the density of parcels experiencing a given percentage change in present value. Red bars denote parcels whose present value increases under the policy (winners, $\pi_i(\mathcal{P}) > \pi_i(\mathcal{P}_0)$), while black bars denote parcels whose present value decreases (losers, $\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$). Average changes are expressed per acre of land.

Table D3: Counterfactual Results

	Built Environment			Rent outcomes		Costs of Interventions		
	$\Delta S(h; \mathcal{P})$ (% chg.)	$\Delta \text{NetUnit}(\mathcal{P})$ (# of units)	$\Delta \text{AffUnit}(\mathcal{P})$ (# of units)	$\Delta \text{Rent}(\mathcal{P})$ (% chg.)	$\Delta \text{Shr. Q1}$ (pp)	$\Delta PV^{\text{Tax}}(\mathcal{P})$ (% chg.)	$\Delta PV(\mathcal{P})$ (% chg.)	Shr. PV-losers (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(0) <i>Baseline under $\mathcal{P}_0$</i>	3.75 Bn.	125, 685	<u>Baseline Outcomes</u> 310	\$2, 682.45	20.00	\$406.06 Bn.	\$1, 574.13 Bn.	–
(1) <i>Upzoning ($\phi = 50\%$)</i>	1.76	71, 710	<u>Unilateral Policy</u> 22	1.58	0.28	0.05	–1.20	86.7
(2) <i>Tax exemption ($\tau = 20$)</i>	2.18	89, 532	269	1.81	0.23	–3.71	–0.93	85.9
(3) <i>Affordability mandate ($\alpha = 30\%$)</i>	–0.37	–15, 124	41, 656	–0.77	0.01	–0.28	0.17	14.0
(4) <i>Density Incentive Zoning ($\alpha = 30\%$ & $\phi = 50\%$)</i>	1.08	<u>Mandatory Carrot & Stick Zoning</u>						
(5) <i>Fiscal Incentive Zoning ($\alpha = 30\%$ & $\tau = 20$)</i>	1.58	43, 552	67, 502	0.23	0.24	–0.40	–0.85	87.5
(6) <i>All ($\alpha = 30\%$, $\phi = 50\%$, & $\tau = 20$)</i>	4.80	65, 007	84, 479	0.31	0.22	–3.36	–0.75	87.3
		195, 457	144, 759	2.25	0.69	–5.35	–2.74	83.6
(7) <i>Density Incentive Zoning ($\alpha = 30\%$ & $\phi = 50\%$)</i>	1.04	<u>Voluntary Carrot & Stick Zoning</u>						
(8) <i>Fiscal Incentive Zoning ($\alpha = 30\%$ & $\tau = 20$)</i>	1.59	42, 338	34, 538	0.65	0.07	0.11	–0.46	88.4
(9) <i>All ($\alpha = 30\%$, $\phi = 50\%$, & $\tau = 20$)</i>	4.91	65, 834	84, 901	0.47	0.12	–3.06	–0.45	86.9
		200, 158	147, 028	2.71	0.39	–4.61	–1.98	83.2
(10) <i>Density Incentive Zoning ($\alpha = 30\%$ & $\phi = 50\%$)</i>	1.04	<u>Mandatory Carrot & Stick Zoning with no supply effect</u>						
(11) <i>Fiscal Incentive Zoning ($\alpha = 30\%$ & $\tau = 20$)</i>	1.65	42, 698	66, 662	0.53	–0.17	0.26	–0.02	12.6
(12) <i>All ($\alpha = 30\%$, $\phi = 50\%$, & $\tau = 20$)</i>	5.35	68, 393	86, 412	0.97	–0.32	–2.13	0.52	1.41
		219, 790	156, 370	4.41	–1.02	–1.68	0.97	0.1

Notes: Row (0) reports baseline outcomes under the no-policy environment $\mathcal{P}_0$ in levels. Rows (1)–(12) report changes relative to the baseline. Each row reports outcomes under alternative regulatory environments $\mathcal{P}$, all measured over a 10-year horizon. Rows (1)–(6) correspond to mandatory policies; rows (7)–(9) to the voluntary counterparts of policies (4)–(6); rows (10)–(12) to mandatory policies shutting down the supply feedback channel. Column (1) reports the percentage change in total expected supply in square feet. Column (2) reports the change in expected housing units constructed net of demolitions. Column (3) shows the change in affordable units out of this net change. Column (4) reports the percentage change in average expected citywide rent per housing unit. Column (5) depicts the change in percentage points in the share of units falling within the first quintile of the baseline rent distribution. Columns (6) and (7) report the percentage change in the present value of property tax revenues and total residential asset values, respectively. Column (8) shows the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$).

E Model Extension: Allowing for a renovation option

We extend the model and introduce renovation as an alternative to redevelopment. Renovation restores the quality of an existing structure, $z_{B,i} \rightarrow 1$, while leaving density unchanged. This extension isolates a quality-improving margin that leaves housing supply fixed, thereby providing a direct test of whether our main results hinge on the replacement of the existing stock rather than its incremental upgrading.

E.1 Model and Calibration Assumptions

Developers choose between two irreversible actions, $m \in \{\text{renov}, \text{rebuild}\}$, in addition to maintaining the existing structure. Parcel value satisfies

$$\pi(R_i, F_{B,i}, z_{B,i}, \alpha_{B,i}; \mathcal{P}, \Theta) = \max \left\{ \pi_i^{\text{renov}}(\dots; \mathcal{P}, \Theta), \pi_i^{\text{rebuild}}(\dots; \mathcal{P}, \Theta) \right\}.$$

Each intervention m has its own optimal trigger $R_i^{e*,m}$ and construction duration d^m . The developer chooses the option that yields the highest parcel value at time $t = 0$. If the rebuild dominates the renovation options in value at $t = 0$, the developer selects rebuild, and the relevant trigger is $R_i^{e*,\text{rebuild}}$, regardless of whether $R_i^{e*,\text{renov}}$ is lower or higher.

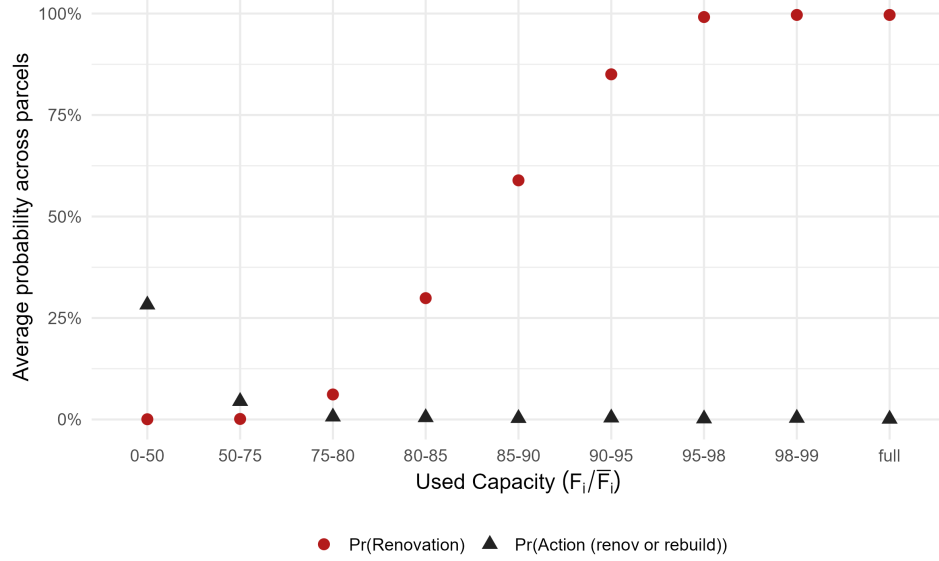
The renovation extension keeps the redevelopment structure and introduces four modifications that isolate a quality-improving, non-supply-expanding margin:

1. **No density gain:** Renovation does not increase building density, so $F_i^{*,\text{renov}} = F_{B,i}$.
2. **No demolition cost:** Demolition costs vanish under renovation, so $\gamma(F_{B,i}, L_i) = 0$.
3. **Shorter duration:** Renovation completes more quickly than redevelopment, with $d^{\text{renov}} < d$.
4. **Policies do not apply to renovation:** Policies do not apply to renovation, so $\mathcal{P}^{\text{renov}} = \mathcal{P}_0$.
Developers who renovate neither receive tax incentives nor face affordability mandates.

These features imply that renovation improves building quality without expanding housing supply. The renovation margin leaves the supply channel, captured by ζ_S , unchanged and raises rents through quality improvements and amenity spillovers captured by ζ_A . Because the policies do not apply to renovated buildings, developers can avoid the policy bundle $\mathcal{P}$ by selecting renovation. This introduces an opt-out margin that attenuates policy effects. This distinction matters for policy design because, in this context, redevelopment determines housing supply, while renovation changes only the quality of the existing stock.

The renovation option introduces a single new parameter: the renovation duration d^{renov} . We calibrate d^{renov} from permit-level data in the [New York City Department of City Planning Housing Database Project-Level Files](#), computing the average time between permit issuance and project completion for “Alteration” permits. This procedure yields $d^{\text{renov}} = 2.21$ years, which is 36% shorter than the redevelopment duration d , so renovation reduces exposure to foregone rental income.

Figure E1: Renovation Decision and Probability of Action by Used Capacity



Note: Each point represents the mean probability within a bin of the density used ratio $F_i / \bar{F}_i$, where F_i is the current floor-area ratio (FAR) and $\bar{F}_i$ is the zoning limit. Red circles show the probability of choosing the renovation option over redevelopment. Black triangles show the probability of any replacement action occurring within the 10-year horizon.

E.2 Results

E.2.1 Baseline results and diagnosis

Developers choose renovation only on parcels that lie close to their zoning limit, where redevelopment no longer increases density. Figure E1 plots, across bins of used capacity $F_i / \bar{F}_i$, the probability of renovation (red), and the probability of any action (renovation or redevelopment) within the 10-year horizon (black). When substantial capacity remains unused, renovation is negligible. Below 75% of capacity, virtually no parcel prefers renovation over redevelopment, consistent with evidence that developers build to their maximum allowed density (Brueckner et al., 2024; Büchler and Lutz, 2024). The renovation option becomes relevant only above 85% of capacity, and dominates above 95%. When $F_i \approx \bar{F}_i$, redevelopment yields no density gain, so renovation becomes the optimal intervention by model construction.

Renovation is the preferred option on many parcels, yet has limited aggregate implications for housing supply because it arises mainly on parcels that are unlikely to redevelop. The probability of any action is highest ($\approx 30\%$) for low-density parcels with substantial redevelopment upside, and then declines sharply as parcels approach their zoning limit. For built-up parcels, the foregone rental income during renovation rarely offsets the associated quality gains, even with the shorter renovation period. Allowing for renovation, therefore, leaves aggregate development activity largely unchanged, including in regions where renovation is the dominant intervention.

The first row of Table E1 reports baseline outcomes with the renovation option. Aggregate housing supply remains essentially unchanged. Total expected production equals 172,469 units, and 125,595 units net of demolitions. These figures are nearly identical to the baseline without the renovation option, which delivers 168,395 total units and 125,685 net. The larger difference

Table E1: Results with Renovation Option

	Built Environment			Rent outcomes			Costs of Interventions		
	$\Delta \text{Shr. Renov}$ (%)	$\Delta S(i; \mathcal{P})$ (% chg.)	$\Delta \text{NetUnit}$ # of Apt.	$\Delta \text{Aff. Net}$ # of Apt	ΔRent (%)	Shr. Q1 (pp)	ΔPV^{Tax} (%)	ΔPV (%)	Shr. PV-losers (%)
(0)	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(0) Baseline under $\mathcal{P}_0$	53.8%	3.75 Bn.	125, 595	Baseline Outcomes 312	\$2, 684.16	20.00	\$406.07 Bn.	\$1, 574.14 Bn.	-
(1) Upzoning ($\phi = 50\%$)	-53.48	1.79	72, 844	Unilateral Policy 26	1.81	0.04	0.64	-0.61	86.51
(2) Tax exemption ($\tau = 20$)	-49.24	2.22	91, 163	278	2.01	0.03	-3.25	-0.46	85.32
(3) Affordability mandate ($\alpha = 30\%$)	8.39	-0.37	-15, 219	43, 392	-0.64	-0.08	-0.04	0.41	13.57
(4) Density Incentive Zoning	-53.33	1.08	43, 845	Mandatory Carrot & Stick Zoning 67, 530	0.40	-0.01	0.10	-0.34	86.55
(5) Fiscal Incentive Zoning	-18.52	1.60	66, 171	85, 585	0.49	0.06	-2.95	-0.33	86.83
(6) Combined Incentive Zoning	-53.62	4.92	200, 661	147, 266	2.69	0.35	-4.54	-1.90	83.10

Notes: Row (0) reports baseline outcomes when the renovation option is available alongside redevelopment under the no-policy environment $\mathcal{P}_0$. Rows (1)–(6) report changes relative to this baseline under alternative regulatory environments $\mathcal{P}$, all measured over a 10-year horizon. Each row corresponds to a counterfactual regulatory environment $\mathcal{P}$, and all reported quantities reflect deviations from the baseline calibration $\mathcal{P}_0$ over a 10-year horizon. Column (0) reports the share of the parcels that opt-in the renovation option. Column (1) reports the percentage change in total expected supply in square feet. Column (2) reports the change in expected housing units constructed net of demolitions. Column (3) shows the change in affordable units out of this net change. Column (4) reports the percentage change in average expected citywide rent per housing unit. Column (5) depicts the change in percentage points in the share of units falling within the first quintile of the baseline rent distribution. Columns (6) and (7) report the percentage change in the present value of property tax revenues and total residential asset values, respectively. Column (8) shows the fraction of parcels whose present value declines under the policy ($\pi_i(\mathcal{P}) < \pi_i(\mathcal{P}_0)$).

between the two numbers indicates that, under this model, a greater share of the housing stock consists of renovated units. This result confirms that the aggregate housing supply implications of renovation are, however, modest. Expected citywide rent is marginally higher with renovation (\$2,684 vs. \$2,682), which reflects quality improvements in the existing stock rather than a reduction of housing supply.

E.2.2 Counterfactuals results

Renovation creates an escape margin from policy. Developers who choose renovation avoid the policy bundle $\mathcal{P}$, since $\mathcal{P}^{\text{renov}} = \mathcal{P}_0$. Table E1 reports the counterfactual results with the renovation option and shows that this escape margin is quantitatively important once policies include affordability mandates.¹⁸ Under the mandate alone, the share of parcels that prefer renovation increases by 8.4 percentage points, which reflects the incentive to avoid the α requirement. In contrast, policies that offer incentives tilt choices toward redevelopment, reduce the reliance on renovation, and increase gross housing production.

Expected average rent rises more across all counterfactuals when the renovation option is available. The share of units in the bottom quintile of the baseline rent distribution changes little relative to the model without renovation. This pattern reflects substitution toward renovation for low-rent parcels, which preserves the composition of the lower tail of the rent distribution. Renovation raises rents through quality improvements while leaving the supply channel ζ_S unchanged. The supply mechanism that would otherwise dampen rent growth, therefore, weakens, which makes it more difficult for policies to reduce rents through replacement of the existing stock.

Renovation also dampens policy incidence. Present value losses borne by landowners fall less when the renovation option is available, and the share of PV-losers is also reduced across all counterfactuals. The expanded set of developer choices drives this pattern. Renovation offers a lower-cost alternative to redevelopment that preserves parcel value on otherwise constrained parcels. Without renovation, parcels close to their zoning limit have no margin to respond to adverse policies other than waiting. Renovation introduces an adjustment margin that partially hedges this exposure.

E.3 Summary

Allowing for renovation does not alter the main aggregate results. Development activity continues to concentrate on parcels with unused capacity, where redevelopment dominates renovation. As a result, renovation has negligible effects on aggregate housing supply. At the margin, renovation improves quality and generates amenity spillovers without expanding the housing stock. When renovation is exempt from α and τ , it creates an escape margin from policy that weakens the effects of affordability mandates. These findings reinforce the central mechanism of the paper. Redevelopment decisions that replace the existing stock shape housing outcomes, whereas renovation operates mainly through incremental upgrading of that stock.

¹⁸To solve the fixed point, developers need to make predictions about which developers renovate and which prefer the rebuild option. Because this is a binary decision that can create an unstable equilibrium, we smooth these outcomes by transforming the discrete choice into a probability of choosing one option over the other, following the same procedure as the adoption algorithm described in Appendix A.8.